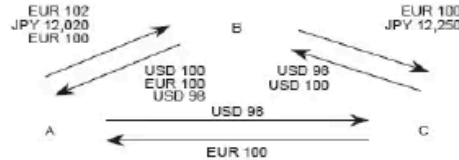


Examples of bilateral and multilateral netting

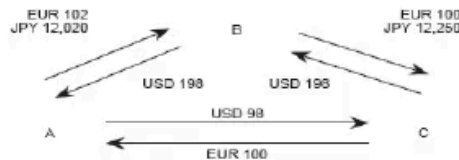
We take a situation where the following transactions between Banks A, B and C all need to be settled for the same value date:

Transaction N°	Bank	Pays	Receives	Counterparty
1	A	EUR 102	USD 100	B
2	B	EUR 100	USD 98	C
3	A	JPY 12,020	EUR 100	B
4	A	USD 98	EUR 100	C
5	A	EUR 100	USD 98	B
6	B	JPY 12,250	USD 100	C

Without netting, the payments to be made are as follows:

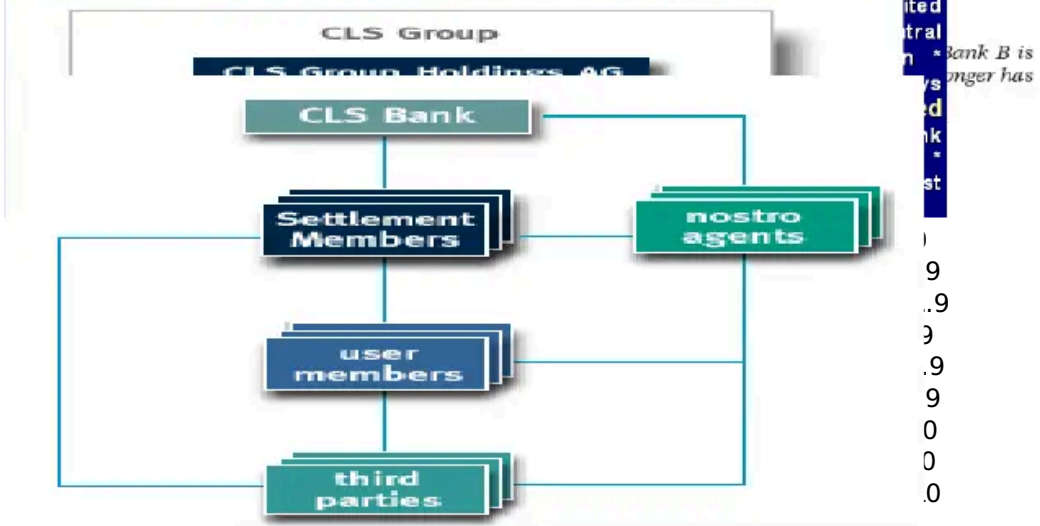


Under a bilateral netting system, there is at most one payment for each pair of counterparties and each currency, which corresponds to the net balance of the transactions in the currency pair between the pair of counterparties:



In this example, a single euro payment between Bank A and Bank B is substituted for the payments that would be required without netting of EUR 102 and EUR 100 from Bank A to Bank B and of EUR 100 from Bank B to

Australia - Australia & New Zealand Banking Group * Commonwealth Bank of Australia * National Australia Bank * Westpac Banking Corporation * **Belgium** - Fortis Bank * KBC Bank * **Canada** - Bank of Montreal * The Bank of Nova Scotia * CIBC * Royal Bank of Canada * The Toronto-Dominion Bank * **Denmark** - Danske Bank * Nordea * **France** - BNP Paribas * Caisse Nationale De Credit Agricole * Credit Lyonnais * Societe Generale * **Germany** - Bayerische Landesbank * Commerzbank * Deutsche Bank * DZ Bank * Dresdner Bank * Hypo Vereinsbank * Westdeutsche Landesbank * **Italy** - Banca Intesa * UniCredito * **Japan** - The Bank of Tokyo-Mitsubishi, Ltd * Mizuho Corporate Bank * Norinchukin Bank * Sumitomo Mitsui Banking Corporation * The Sumitomo Trust & Banking Co. Ltd * UFJ Bank * **Luxembourg** - Dexia Banque Internationale a Luxembourg * **Netherlands** - ABN AMRO Bank * ING Bank * Rabobank Nederland * **Norway** - Den



When a participant fails to make all its scheduled pay-ins for the 8:00 deadline, CLS sends it a pay-in call urging it to deposit a minimum sum on its account. The relatively small sum can be deposited in the currency of the participant's choice.

At 9:00 the CLS system rejects transactions that are not yet settled and involve a participant that has not complied with its pay-in schedule. This changes the projected positions of the counterparties to the failed transactions. Such counterparties could end up with a potentially larger negative projected position than before in some currencies. They could even swing from a positive projected position to a negative projected position. CLS then sends them a "pay-in call for settlement", urging them to transfer central bank funds to CLS Bank to enable immediate settlement of all of their transactions in the queue.

At the 10:00 deadline for Asian Pacific currencies and the 12:00 deadline for other currencies, CLS sends "pay-in calls for currency close" to the surviving participants, if needed. Even though they have complied with their pay-in schedules and met the pay-in calls for settlement, some of them may still have negative positions in some currencies on their accounts with CLS Bank.

At the same time, transactions involving the failed participant may have already been settled, even though the latter has not made any pay-ins, or has made only some of the pay-ins scheduled. In this case, CLS Bank does not have enough liquidity in the currencies concerned to make the pay-outs due to the counterparties in these transactions. On the other hand, since CLS Bank stopped the pay-outs to the failed participant, it has excess liquidity in the corresponding currencies. Therefore, it can exchange these currencies for the ones it needs to complete the pay-outs to the surviving participants. It swaps these currencies with the designated liquidity providers. The latter are banks, which may or may not be CLS participants, that have contracted to provide liquidity in eligible currencies under such circumstances.

The liquidity providers' commitments are set at a level such that, even if the liquidity provider with the largest commitment for a given currency fails, the other liquidity providers' commitments are adequate to cover the corresponding short position limit. Thus, in theory, CLS Bank will always be able to make the expected pay-outs, even if the participant with the largest debit position fails.

The measures intended to ensure settlement of transactions could turn out to be inadequate to cover the resulting short positions in specific currencies in the event of multiple failures, or if exchange rate variations exceed the market volatility haircuts. In this event, CLS Bank has planned loss-allocation arrangements for the surviving participants.

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13.17 Summary.....	19

13.4 About Continuous Linked Settlement

The CLS system was launched on 9 September 2002. It is a system which is jointly held by the large banks worldwide which are its shareholders (as of September 2007, there are 57 member banks and 1846 third party institutions that participate in this system). It is regulated by Federal Reserve Board of New York. CLS system acts as an intermediate for the risk-free cross border forex transactions.



Figure 13.1 Objectives of CLS

The CLS bank deals in the following fifteen currencies. (These are banks not currencies)



Figure 13.2 Central Banks in CLS

13.5 Conventional Forex Transaction Settlement

Prior to the formation of CLS bank in 2002 the process of forex transaction would expose participating banks to the settlement risk. We will see how with the help of an example. Two parties, Bank A and Bank B, have struck a deal for a forex transaction in which Bank A would sell Bank B 10 million US dollars at a rate of 1 dollar for 120 yen. To settle the transaction, Bank A has to deliver 10 million dollars to Bank B (also known as

the dollar leg) and receive 1200 million yen from Bank B in exchange (also known as the yen leg). And vice-versa, Bank B has to deliver 1200 million yen and receive 10 million dollars from Bank A in exchange.

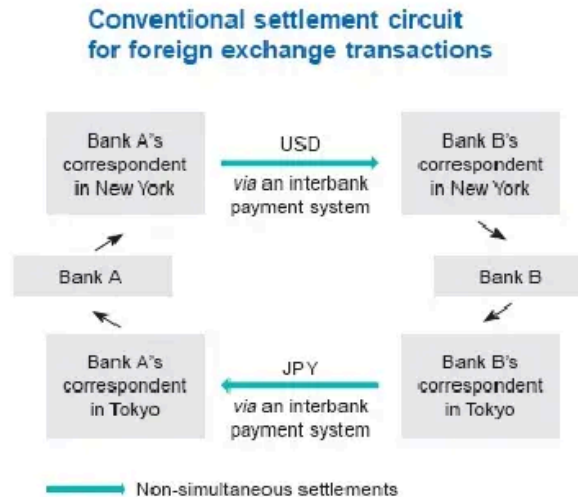


Figure 13.3 Dollar and Yen leg of settlement

The above **figure 13.3** shows that the transaction of both legs i.e. the transaction of a dollar and of a Yen are made through completely different channels that is through bank A's and B's respective corresponding banks and inter-bank payment networks. Hence the transaction here isn't simultaneous as the each bank is exposed to a settlement risk. That is the Bank A will make a final irrevocable payment of Dollar 10 million without knowing that when will it get the same amount of it in yen from the bank B and vice-versa.

The Allsopp report defines this risk i.e. the settlement risk as "one party to a foreign exchange transaction will pay the currency it sold but not receive the currency it bought".

The Allsopp Report contains analysis based on a survey of 80 banks from G10 countries conducted in 1994 and 1995. The survey showed that the average exposure to settlement risk in foreign exchange transactions extended over several days. This defied the common belief that the risk stemmed solely from time zone differences and only lasted a few hours at most, as well as the belief that the risk is incurred only by the counterparty that has the time zone difference working "against" it.

13.6 Purpose of CLS

- CLS is a means of settling foreign exchange transactions finally and irrevocably
- It eliminates settlement risk, improves liquidity management, improves operational efficiency and effectiveness
- It's an effective cross-currency settlement process. The average daily turnover in global FX transactions at almost US\$2 trillion. It's a process of effective risk free settlement FX market has long needed
- The CLS implemented the concept of simultaneous payments by applying multilateral netting of currencies and acting as the settlement agent

- Before the establishment of CLS, both parties of the trade paid separately. With taking time-zone differences into consideration, the risk of one party not meeting its obligation and hence defaulting accentuates manifolds. CLS eliminates the temporal risk originated out of different time zone and hence makes possible same day trading with finality
- Enhanced customer service by enabling them to deal with trading counterparties, reduce costly reconciliation, and exploit the real-time information on currency cycling and settlement that only CLS can provide.

A QUICK VIEW OF BENEFITS OF CLS:

- Payment costs lowered due to netting of payments.
- Real-time reporting through SWIFT
- Reduced settlement risks on FX trade flows enables Straight Through Processing of FX trades.

13.6.1 Herstatt Risk

The most well-known example of settlement risk is the failure of a small German bank, Bankhaus Herstatt in 1974. On 26th June 1974, the firm's banking license was withdrawn, and it was ordered into liquidation during the banking day; but after the close of the German interbank payments system (3:30pm local time). Some of Herstatt Bank's counterparties had irrevocably paid Deutschemarks to the bank during the day but before the banking license was withdrawn. They had done so in good faith, believing they would receive US dollars later in the same day in New York. But it was only 10:30 am in New York when Herstatt's banking business was terminated. Herstatt's New York correspondent bank suspended all outgoing US dollar payments from Herstatt's account; leaving its counterparties fully exposed to the value of the Deutschemarks they had paid the German bank earlier on in the day. This type of settlement risk, in which one party in a foreign exchange trade pays out the currency it sold but does not receive the currency it bought, is sometimes called Herstatt risk. It is however an inappropriate term since it has materialized in other cases and under differing circumstances. The collapse of US investment bank Drexel Burnham Lambert in 1990, Bank of Credit and Commerce International the following year and Barings in 1995 are all excellent case study material for 'Herstatt' risk. The more appropriate name for 'Herstatt' risk is foreign exchange settlement or cross-currency settlement risk. The amount at risk equals the full amount of currency purchased and lasts from the time that a payment instruction (for the currency sold) can no longer be cancelled unilaterally until the time the currency purchased is received with finality (irrevocable and unconditional).

Hence as it can be read that the Continuous Linked Settlement eliminates the settlement risk from the forex transaction and hence implements the concept of multicurrency PVP system that is payment versus payment which states Delivery-versus-payment means that the final transfer of one asset occurs if, and only if, the final transfer of an (other) asset(s) occurs. Assets could be monetary assets (such as foreign exchange), securities or other financial instruments. A multi-currency DVP system would thus eliminate Herstatt or cross-currency settlement risk.

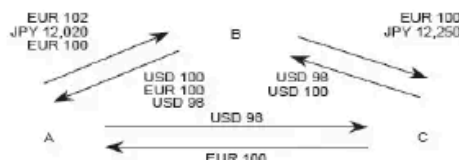
13.7 An Example of Multilateral Netting Settlement by CLS

Examples of bilateral and multilateral netting

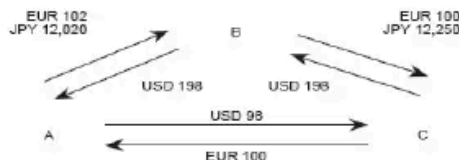
We take a situation where the following transactions between Banks A, B and C all need to be settled for the same value date:

Transaction N°	Bank	Pays	Receives	Counterparty
1	A	EUR 102	USD 100	B
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3	A	JPY 12,020	EUR 100	B
4	A	USD 98	EUR 100	C
5	A	EUR 100	USD 98	B
6	B	JPY 12,250	USD 100	C

Without netting, the payments to be made are as follows:

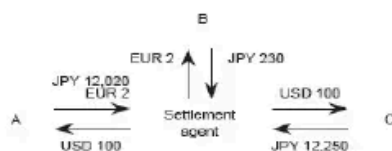


Under a bilateral netting system, there is at most one payment for each pair of counterparties and each currency, which corresponds to the net balance of the transactions in the currency pair between the pair of counterparties:



In this example, a single euro payment between Bank A and Bank B is substituted for the payments that would be required without netting of EUR 102 and EUR 100 from Bank A to Bank B and of EUR 100 from Bank B to Bank A. Furthermore, Bank C can meet its dollar obligations towards Bank B by making a single payment, instead of two. The same is true of the dollar obligations between Bank B and Bank A.

Under a multilateral netting system, each bank has to make or receive at most one payment in each currency. The amount of the payment is equal to the net balance of all of the participant's transactions in that currency with all its counterparties in the system. The payments are made through a settlement agent:



The dollar payment that Bank B is supposed to make to Bank A and the dollar payment that Bank B is supposed to receive from Bank C under the bilateral netting arrangement are netted off: Bank B no longer has to make or receive any dollar payments. A similar situation applies to Bank C's payments in euro.

13.8 The CLS shareholders

Australia - Australia & New Zealand Banking Group * Commonwealth Bank of Australia * National Australia Bank * Westpac Banking Corporation * **Belgium** - Fortis Bank * KBC Bank * **Canada** - Bank of Montreal * The Bank of Nova Scotia * CIBC * Royal Bank of Canada * The Toronto-Dominion Bank * **Denmark** - Danske Bank * Nordea * **France** - BNP Paribas * Caisse Nationale De Credit Agricole * Credit Lyonnais * Societe Generale * **Germany** - Bayerische Landesbank * Commerzbank * Deutsche Bank * DZ Bank * Dresdner Bank * Hypo Vereinsbank * Westdeutsche Landesbank * **Italy** - Banca Intesa * UniCredito * **Japan** - The Bank of Tokyo-Mitsubishi, Ltd * Mizuho Corporate Bank * Norinchukin Bank * Sumitomo Mitsui Banking Corporation * The Sumitomo Trust & Banking Co., Ltd. * UFJ Bank * **Luxembourg** - Dexia Banque Internationale a Luxembourg * **Netherlands** - ABN AMRO Bank * ING Bank * Rabobank Nederland * **Norway** - Den norske Bank * **Singapore** - DBS Bank * Oversea-Chinese Banking Corporation * United Overseas Bank * **Spain** - BBVA * Banco Popular Espanol * Banco Santander Central Hispano * **Sweden** - Skandinaviska Enskilda Banken * Svenska Handelsbanken * **Switzerland** - Credit Suisse * UBS * Zurcher Kantonalbank * **United Kingdom** - Barclays Bank * HSBC Holdings * Royal Bank of Scotland * Standard Chartered Bank * **United States** - American International Group * Bank of America * The Bank of New York * Bank One * Bear Stearns Securities * Citibank * The Goldman Sachs Group * JPMorgan Chase * Lehman Brothers * Mellon Bank * Merrill Lynch * Morgan Stanley & Co. * Northern Trust Corporation * State Street Bank and Trust Co.

Figure 13.4 Shareholders of CLS

13.9 The CLS Group of Companies Hierarchy



Figure 13.5 CLS Group

13.9.1 CLS Group Holdings AG (CLS Group Holdings)

CLS Group Holdings is the group holding company of CLS UK Intermediate Holdings Ltd, CLS Bank International (CLS Bank) and CLS Services Ltd. CLS Group Holdings is a company incorporated under the laws of Switzerland and is regulated by the Federal Reserve as a bank holding company in the United States.

13.9.2 CLS UK Intermediate Holdings Ltd

CLS UK Intermediate Holdings is the intermediate holding company of the CLS Group and is a limited company incorporated under the laws of England and Wales. CLS UK Intermediate Holdings is a 'shell' company from a governance perspective and its principal role is to provide certain corporate services to CLS Bank and its affiliated companies (i.e. Finance, Human Resources, Audit and Communications).

13.9.3 CLS Bank International

CLS Bank is a unique, independent financial institution that provides payment versus payment settlement for payment instructions arising from FX transactions in eligible currencies. It is a wholly owned subsidiary of CLS UK Intermediate Holdings. CLS Bank is an Edge corporation organized under the laws of the United States and regulated by the Federal Reserve Bank of New York.

13.9.4 CLS Services Ltd

CLS Services is a limited company incorporated under laws of England and Wales. The principal role of CLS Services is to provide effective operational and back-office support to CLS Bank and its affiliated companies.

13.10 Parties Involved in the CLS Settlement

CLS is only available through the unique and regulated relationship between CLS Bank, the central banks in whose currencies CLS settles, and members of CLS Bank.

The CLS process involves a number of different parties:

- shareholders
- Members - either Settlement Members or User Members
- Third parties.

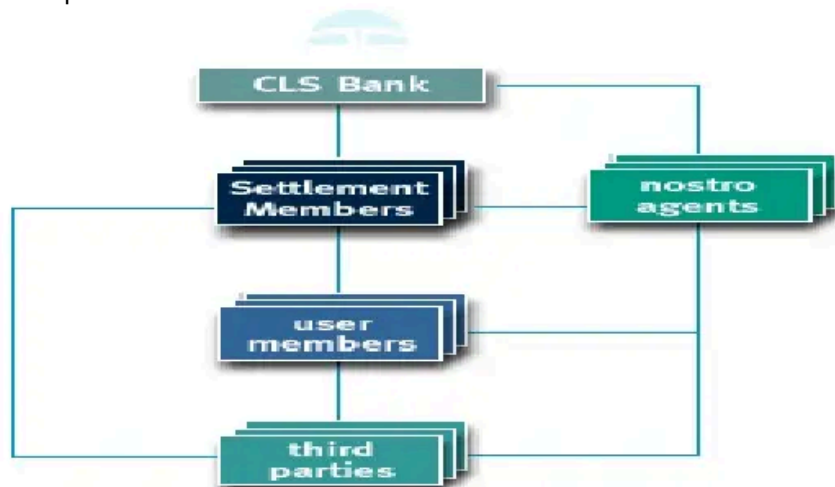


Figure 13.6 Parties of CLS Bank

13.10.1 Shareholders

CLS Bank is owned by 71 of the world's largest financial groups throughout the US, Europe and Asia Pacific. They are responsible for more than half the value transferred in the world's FX market. Five CLS shareholders alone represent over 44% of this market. Shareholders have invested in CLS to develop CLS settlement. Each has purchased an equal shareholding in the CLS Group of companies. Each shareholder has the exclusive right to become a CLS Bank Settlement Member with direct access to the CLS system.

13.10.2 Settlement Members

A Settlement Member must be a shareholder of CLS Group and must show that they have the financial and operational capability and sufficient liquidity to support their financial commitments to CLS. They can each submit settlement instructions directly to CLS Bank and receive information on the status of their instructions. Each Settlement Member has a multi-currency account with CLS Bank, with the ability to move funds. Settlement Members have direct access and input deals on their own behalf and on behalf of their customers. They can provide a branded CLS service to their third-party customers as part of their agreement with CLS Bank.

13.10.3 User Members

User Members can submit settlement instructions for themselves and their customers. However, User Members do not have an account with CLS Bank. Instead they are sponsored by a Settlement Member who acts on their behalf. Each instruction submitted by a user member must be authorized by a designated Settlement Member. The instruction is then eligible for settlement through the account Settlement Member's account.

13.10.4 Third parties

Third parties are customers of settlement and user members and have no direct access to CLS. Settlement or user members must handle all instructions and financial flows, which are consolidated in CLS. The terms on which members can act on behalf of third parties are governed by private arrangement. These do not directly involve CLS Bank and third parties do not have any relationship with CLS Bank. Members may provide a trademarked CLS service to their third-party customers.

13.10.5 Nostro agents

Nostro agents:

- receive payment instructions from Settlement Members
- may have multiple relationships with Settlement Members
- must provide time-sensitive fund transfers to Settlement Members' accounts at CLS Bank
- Receive funds from CLS Bank, User Members, third parties and others for credit to the Settlement Member account.

13.11 How CLS Works

CLS bank has several members or shareholder banks. Every member bank has its currency account with the CLS bank. The CLS bank also holds the RTGS settlement accounts with the respective central bank of member bank's nation. The CLS does all the settlement on the multilateral netting basis.



Figure 13.7 CLS bank is multi-currency bank

13.12 The concept of Pay-In and Pay-Out

Pay-In is an amount of a particular currency that a participating bank will supply into its CLS bank's multicurrency account. Hence we can say that after making a pay-in to the CLS bank the participating bank's position on that currency will get stronger. This means that participating bank has surplus of the currency, in which the pay-in is made, in its multicurrency bank account with CLS. For example bank A sells 100 dollars to bank B in order to buy 4500 Rupees, but it doesn't have 100 dollars in its dollar account with the CLS then it will have a negative position of -100 after the transaction hence at a specified time bank A will make a pay-in into its dollar account into the CLS bank.

In the same way pay-out is when the participant takes out the particular amount of currency from its CLS account which reduces its position and hence the bank will only do it when it has a positive position on that currency. The ability of CLS Bank to make pay-outs in a given currency depends on the funds available in that currency on its central bank account, which means it depends on the pay-ins received.

DID YOU KNOW?

On 13th November 2007, CLS bank International set a new record for volume of payment instructions settled in one day. CLS Bank settled 1,140,644 payment instructions with a gross value of US \$ 6.5 trillion.

Source : www.clsgroup.com

13.13 A Transaction in a CLS system

Following is an example for the transaction in a CLS system between two banks A and B.



Figure 13.8 Transaction in CLS system

The bank A wants to buy 102 euros from the bank B hence for that it has to supply an amount of 100 dollars to the bank B.

If the bank B accepts the transaction request from the bank A then following will be entries of the transfer of funds.

Bank A's multi-currency account:

- Debit: USD 100

- Credit: EUR 102

Bank B's multi-currency account:

- Debit: EUR 102

- Credit: USD 100

According to **figure 13.9** the settlement process works in three phases

- Submission of instructions
- Settlement and funding
- Execution of final trade.

13.13.1 Submission of instructions

Every bank by 06:30 submits its payment instructions to the CLS bank. Payment instructions to the CLS bank are sent in the message format MT300 specified by the SWIFT for the forex transactions. At 6:30, the system calculates the theoretical multilateral net positions in each currency that would result on the participants' account with CLS Bank after execution of all of the foreign exchange transactions submitted for settlement on that day. Participants have to make pay-ins for currencies in which their theoretical multilateral position is negative. For that purpose, CLS Bank sends each participant its pay-in schedule for the day. The pay-in deadlines are 8:00, 9:00 and 10:00 for Asian Pacific currencies (yen and Australian dollar at present) and 8:00, 9:00, 10:00, 11:00 and 12:00 for the other eligible currency.

13.13.2 Settlement and Funding

- Commences at 07.00 CET and is scheduled to complete at 09.00 CET.
- Risk Management Tests applied to both Settlement Members and no settlement is done if the bank fails in the test.
- Real-time exchange of currencies i.e. CLS Bank will not settle the trade unless both parties have the required funds available.
- Exchange is effected with finality & irrevocability for Settlement Members over their accounts with CLS Bank.

13.13.3 Execution of trade

The execution of trade begins as soon as the settlement members start paying their obligations and at the end of the settlement time frame window of 5 hrs that is from the 07:00 - 12:00 the pay-in and pay-out schedules are matched. That is the total value of the money paid should be equivalent to the total value of the money received. CLS doesn't allow for the credit cover hence not a single penny will be paid if the account balance of the participant becomes negative.

13.14 The mathematics behind the settlement in a CLS system

Table 13.1 Settlement under CLS System

Settlement under the CLS system

Our example is a simplified one involving three currencies and three banks named A, B and C. Market volatility haircuts are ignored. Unless otherwise indicated, all amounts are expressed in dollar equivalents. The aggregate short position limits (ASPLs) are 200 for Banks A and C and 150 for Bank B. The short position limit (SPL) for each currency is 100.

After the first pay-ins are made, the positions on the multi-currency accounts are as follows:

	EUR	GBP	USD	Total	Aggregate negative position	ASPL
A	—	700	100	800	—	200
B	100	—	200	300	—	150
C	500	—	400	900	—	200
SPL	100	100	100	—	—	—

The following transactions have been placed in the queue:

Transaction N°	Bank	Buys		Sells		From/to
		Initial amount	USD equivalent	Initial amount	USD equivalent	
1	A	USD 100	100	EUR 102	100	B
2	B	EUR 205	201	USD 200	200	C
3	B	EUR 318	312	GBP 200	312	C

After settling transaction 1, the positions are changed to:

	EUR	GBP	USD	Total	Aggregate negative position	ASPL
A	-100	700	200	800	-100	200
B	200	—	100	300	—	150
C	500	—	400	900	—	200
SPL	100	100	100	—	—	—

The only negative position is on Bank A's euro account, but it is both within Bank A's aggregate short position limit and within the euro short position limit. Furthermore, Bank A still shows a positive position across all currencies. Transaction 1 does not cause the position on Bank A's multi-currency account to exceed the limits set by the risk controls. Therefore it can be settled.

After settling transaction 2, the positions are changed to:

	EUR	GBP	USD	Total	Aggregate negative position	ASPL
A	-100	700	200	800	-100	200
B	401	—	-100	301	-100	150
C	299	—	600	899	—	200
SPL	100	100	100	—	—	—

A negative position is shown on Bank B's dollar account, but it is both within Bank B's aggregate short position limit and within its dollar short position limit. Furthermore, Bank B still shows a positive position across all currencies. Therefore, transaction 2 can be settled.

After settling transaction 3, the positions are changed to:

	EUR	GBP	USD	Total	Aggregate negative position	ASPL
A	-100	700	200	800	-100	200
B	713	-312	-100	301	-412	150
C	-13	312	600	899	—	200
SPL	100	100	100	—	—	—

Bank B's aggregate negative position now exceeds its aggregate short position limit. In addition, Bank B's negative position in sterling exceeds the short position limit for sterling. This means that transaction 3 does not meet the settlement criteria and it is put back in the queue for another try at settlement later.

(Source: www.banque-france.fr)

(Source: www.cls-groups.com)

The CLS settlement process

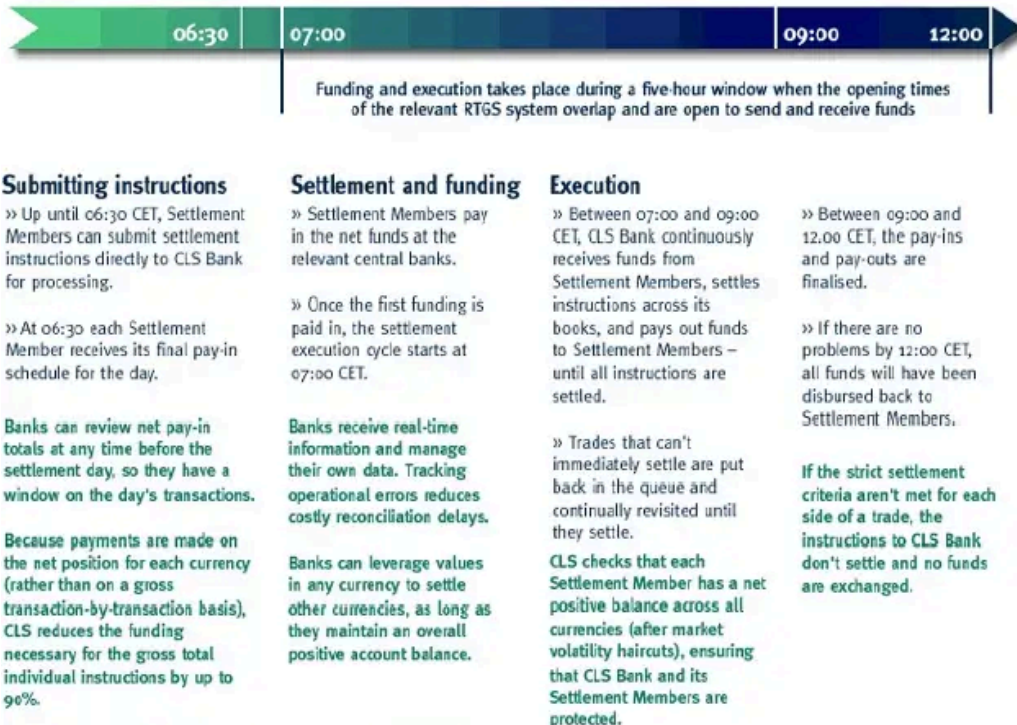
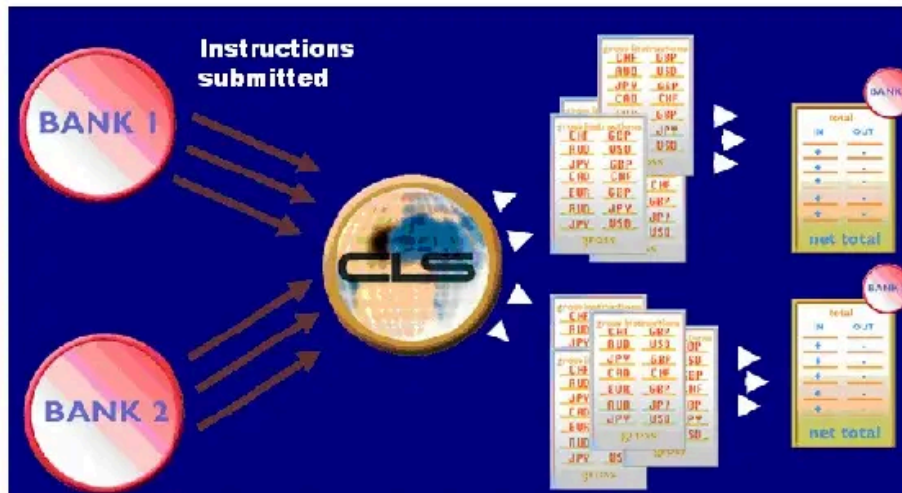


Figure 13.9 Processing in CLS



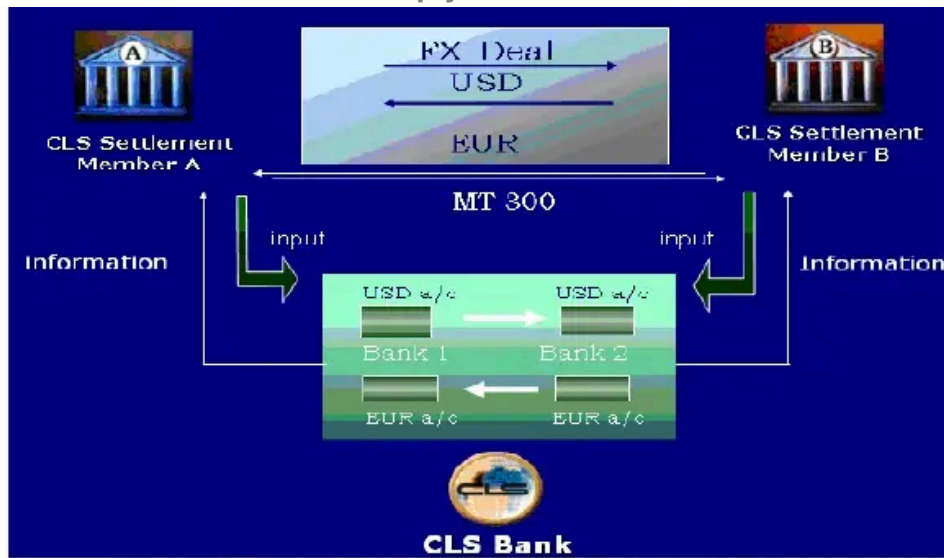
(Source: www.clsgroups.com)

Figure 13.10 Sequence of Instructions Flowing

Haircut (volatility margin) applied to Settlement Member's currency balances to minimise credit risk for CLS Bank-

After the submission of instructions the members start paying their respective netted obligations calculated by CLS bank. As soon as along with the payment of instruction the execution of settlement also starts.

13.15 Flow of Information and the payments



(Source: www.cls-groups.com)

Figure 13.11 Flow of Information in CLS

The above diagram makes it clear how the payment flows in a forex transaction through CLS. The bank A wants to have X number of euros from some other settlement bank B. So the information flows by sending a message through SWIFT networks. The message type used is MT300 meant for forex transactions. The CLS bank holds accounts for both of the banks and hence it will transfer the equivalent amount of dollars to the accounts of the bank B and puts the equivalent amount of euros to the account of the bank A.

13.16 Risk Management in CLS

Following are the three risk control measures employed by CLS:

- A participant's overall balance across all its currency sub-accounts must always be positive or equal to zero.
- A participant's negative position in a given currency must not exceed the limit called the "short position limit" (SPL),
- The sum of a participant's negative positions must not exceed the limit called the "aggregate short position limit" (ASPL).

These various balances, aggregates and limits are expressed in dollar equivalents. For this purpose, CLS updates the dollar rates of the currencies in real time using the average bid and offered rates of ten of the most active traders in the market. In view of the potential variations in these rates during the settlement process, a market volatility haircut is applied to the net positions. Each participant is assigned a specific aggregate short position limit (ASPL) that depends on its capital and its short-term rating. This

limit is aimed at managing counterparty risk by making sure that each participant's overall obligations are within limits at all times. It completes the system eligibility rules, which require a minimum rating for participants. The short position limit (SPL) is calculated for each currency. This limit ensures that the system can provide timely settlement even if the participant with the largest negative position in the currency concerned is unable to make all of its pay-ins. CLS Bank has signed contracts with a number of credit institutions called "liquidity providers" that undertake to provide the liquidity necessary to cover a shortfall up to the short position limit.

INSIDE / OUTSIDE SWAP service offered by CLS:

An I/O Swap comprises of two equal and opposite FX transactions that are agreed as an intraday swap. This reduces intraday cash flows while leaving institution's overall FX position unchanged. It exploits the likelihood that an institution with a large-short position in CLS will most certainly have one or more large long positions in CLS. I/O swaps can reduce these in CLS cash positions as well as liquidity position outside of CLS. CLS identifies potential I/O swaps notify the participants and implement I/O swaps effectively through the CLS system

SOURCE: www.Clsgroup.com



TATA CONSULTANCY SERVICES

Failure management procedures

As a general rule, CLS stops sending pay-outs to any participant that fails to meet its pay-in schedule. Pay-outs can resume as soon as the participant in question remedies its situation. Furthermore, any participant that misses a pay-in deadline is subject to a financial penalty.

When a participant fails to make all its scheduled pay-ins for the 8:00 deadline, CLS sends it a pay-in call urging it to deposit a minimum sum on its account. The relatively small sum can be deposited in the currency of the participant's choice.

At 9:00 the CLS system rejects transactions that are not yet settled and involve a participant that has not complied with its pay-in schedule. This changes the projected positions of the counterparties to the failed transactions. Such counterparties could end up with a potentially larger negative projected position than before in some currencies. They could even swing from a positive projected position to a negative projected position. CLS then sends them a 'pay-in call for settlement', urging them to transfer central bank funds to CLS Bank to enable immediate settlement of all of their transactions in the queue.

At the 10:00 deadline for Asian Pacific currencies and the 12:00 deadline for other currencies, CLS sends 'pay-in calls for currency close' to the surviving participants, if needed. Even though they have complied with their pay-in schedules and met the pay-in calls for settlement, some of them may still have negative positions in some currencies on their accounts with CLS Bank.

At the same time, transactions involving the failed participant may have already been settled, even though the latter has not made any pay-ins, or has made only some of the pay-ins scheduled. In this case, CLS Bank does not have enough liquidity in the currencies concerned to make the pay-outs due to the counterparties in these transactions. On the other hand, since CLS Bank stopped the pay-outs to the failed participant, it has excess liquidity in the corresponding currencies. Therefore, it can exchange these currencies for the ones it needs to complete the pay-outs to the surviving participants. It swaps these currencies with the designated liquidity providers. The latter are banks, which may or may not be CLS participants, that have contracted to provide liquidity in eligible currencies under such circumstances.

The liquidity providers' commitments are set at a level such that, even if the liquidity provider with the largest commitment for a given currency fails, the other liquidity providers' commitments are adequate to cover the corresponding short position limit. Thus, in theory, CLS Bank will always be able to make the expected pay-outs, even if the participant with the largest debit position fails.

The measures intended to ensure settlement of transactions could turn out to be inadequate to cover the resulting short positions in specific currencies in the event of multiple failures, or if exchange rate variations exceed the market volatility haircuts. In this event, CLS Bank has planned loss-allocation arrangements for the surviving participants.

Box 13.1 Failure Management Procedure (Source: www.banquefrance.fr)

13.17 Summary

- The CLS system was launched on 9 September 2002. It's a system which is jointly held by the large banks worldwide which are its shareholders.
- The CLS implemented the concept of simultaneous payments by applying multilateral netting of currencies and acting as the settlement agent.
- CLS is only available through the unique and regulated relationship between CLS Bank, the central banks in whose currencies CLS settles, and members of CLS Bank.
- The CLS process involves a number of different parties:
 - shareholders
 - Members - either Settlement Members or User Members
 - Third parties.
- CLS bank has several members or shareholder banks
- Every member bank has its currency account with the CLS bank
- The CLS bank also holds the RTGS settlement accounts with the respective central bank of member bank's nation
- The CLS does all the settlement on the multilateral netting basis
- Pay-In is an amount of a particular currency that a participating bank will supply into its CLS bank's multicurrency account
- Pay-out is when the participant takes out the particular amount of currency from its CLS account which reduces its position and hence the bank will only do it when it has a positive position on that currency
- Following are the three risk control measures employed by CLS:
 - A participant's overall balance across all its currency sub-accounts must always be positive or equal to zero.
 - A participant's negative position in a given currency must not exceed the limit called the "short position limit" (SPL),
 - The sum of a participant's negative positions must not exceed the limit called the "aggregate short position limit" (ASPL)
- Each participant is assigned a specific aggregate short position limit (ASPL) that depends on its capital and its short-term rating.

14.4. Introduction

The world today has turned into a global village, with faster means of communication, shortening of distances and with increased frequency of international trade, also called cross border trade.

With a growth rate of almost 100 percent in the past decade the total international trade has peaked to over \$10.5 trillion in the year 2005. Hence, in consideration of a trade in goods and services rendered, payments need to be made across nations involving different currencies.

Most cross-border trade payments are handled through correspondent banking relationships, whereby a series of banks and domestic payment systems are typically linked together to move funds.

Did you know?

Payments are big business. Revenues from the U.S. payments industry alone have grown at 6% per year since 1994, topping \$207 billion in 2004. In aggregate, the payments business generates more revenues than do the airline, personal computing, lodging, or entertainment industries.

In terms of volume, cross-border payments are estimated to represent approximately 8% of total payments. Although it is difficult to size exactly, one can indirectly estimate the relative magnitude of cross-border payment flows by analyzing the scope of international trade. During the past ten years, the world trade volume as measured by total imports has roughly doubled in dollar value from \$5.5 trillion in 1996 to \$10.6 trillion in 2005. Correspondingly, one can surmise that the cross-border payments related to international trade have doubled in size.

(Courtesy: Boston Consulting Group)

There are three major challenges that are incidental in implementing the mechanism of Cross-border payments:

1. Most payment systems are governed by local laws and practices within existing domestic banking and financial structures of the respective countries.
2. Dearth of a common global standard and variations between systems has diluted the ability of both bank and corporate treasury/enterprise systems to seamlessly pass data and communicate with each other.
3. Government regulations are changing how payments are made.
4. Payments are subject to domestic regulations which compound the challenges of cross-border payments because often rules vary between an originating and receiving country.

This we will discuss at the end of the session.

14.5. How Cross Border Payments Work Today

Most of the world's major banks maintain correspondent banking relationships with local banks in each of the important financial centres of the world. This two-way link between banks is one of many inter-bank relationships, such as nostro/vostro accounts and the selling of cash management and treasury services to other financial institutions. The institution *providing* the services is the "correspondent bank or upstream correspondent", while the institution *buying* the service is the "respondent bank or downstream correspondent". At least 80% of bank-to-bank cross-border payments currently take place through traditional correspondent banking arrangements or via intra-bank transactions.

Often banks do not separate domestic and cross-border payments, blurring the line of demarcation in a payment flow. Global financial institutions utilize their internal networks to clear and settle both domestic and cross-border payments. Often many payments are bundled in a single transfer, with both domestic and international transactions combined by currency.

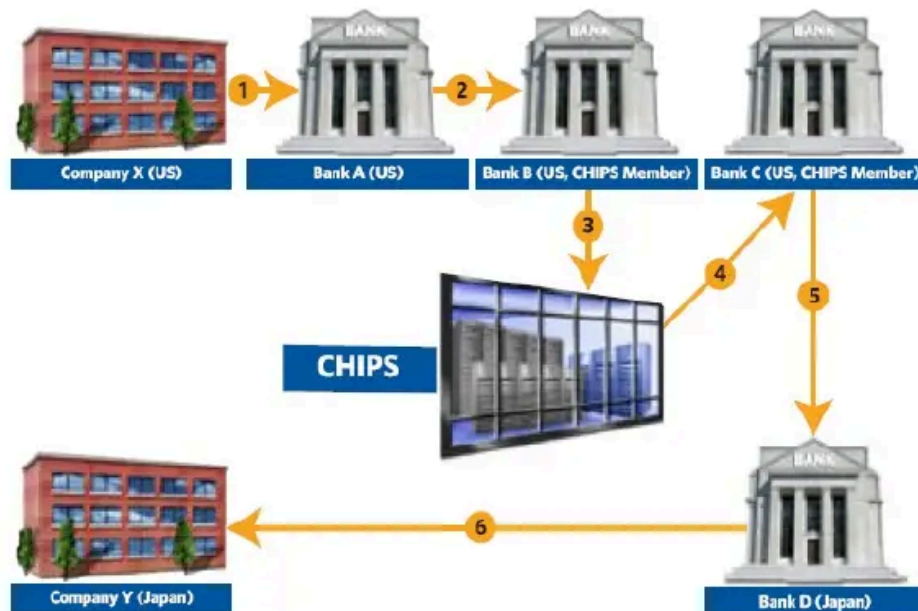
Many cross-border payments are actually settled in a specific country's domestic settlement system.

For example, a British company making a U.S. dollar payment to a Korean company transfers the necessary dollar amount from its U.S. correspondent bank to the Korean company's U.S. bank account in the U.S. If the Korean company does not maintain an account at a bank in the U.S., the funds are transferred to the Korean company bank's correspondent bank in the U.S.

An example of how cross-border payments work:

Company X in the United States needs to make a payment to Company Y in Japan. Company X requests its bank in the United States, Bank A, to send a U.S. dollar payment to Company Y. Since Bank A does not belong to CHIPS, (*Clearing House Inter bank Payment System (CHIPS) is a bank-owned, privately-operated, real-time, multilateral electronic payments system that transfers funds and settles transactions in U.S. dollars*) it requests its correspondent bank, Bank B, which is a member of CHIPS, to facilitate the transfer. Bank B sends the funds transfer via CHIPS to Bank C which is also a member. Bank C is the correspondent bank for Bank D which is where Company Y has an account to receive funds.

The **Figure 14.1** is the diagrammatical representation of the process:



- (1) Company (X) in the U.S. requests its U.S. bank (A) to send a dollar payment to its client (Y) in Japan.
- (2) Bank A asks its U.S. correspondent bank in the U.S. (B) to facilitate this transfer.
- (3) Bank (B), a member of CHIPS, sends the funds transfer command to CHIPS.
- (4) CHIPS executes the fund transfer by crediting the account of another U.S. CHIPS member bank C.
- (5) Bank (D) in Japan is bank C's correspondent bank.
- (6) Company Y has an account with Bank D.

Figure 14.1 a typical example of cross border payment process

A typical cross border transaction: In the above example we saw the transaction from the point of view of a corporate customer. Payments tend to be high-volume, lower-value payments – such as direct deposits of payroll, dividends, annuities, vendor and tax payments, and direct payment of utilities, loans and insurance premiums. These types of payments are supported by the national payment systems in many countries as a safe, reliable, efficient and cost-effective alternative to paper-based payments.

For many years, financial institutions have offered cross-border payment services to their customers, typically utilizing proprietary correspondent banking relationships. Over the last several years, there has been a marked increase in the amount of international trade due in part to the emergence of trading zones such as NAFTA, as well as the impact of the Internet, which has increased the need for a cost-effective, highly automated method for moving lower value payments across national borders. Additionally, the recipients of these payments are becoming increasingly mobile – working, traveling and retiring internationally. The Internet has also fueled the need for payments to travel across borders due to the increase in investment activities and the purchase of goods and services without respect to national boundaries. Without a cost-

effective and efficient payments mechanism, these types of activities may not reach their full potential.

An example of a typical cross-border consumer transaction would be a pension or annuity payment from an employer in the United States to a beneficiary who has relocated to the Canada. The US employer would originate a credit transaction denominated in US dollars.

This transaction would in turn be sent to the employer's financial institution in the United States, which would then forward the transaction to a financial intermediary (referred to as an Originating Gateway Operator or OGO). The OGO processes the transaction and forwards it to a Receiving Gateway Operator (RGO) in Canada. The foreign exchange is performed at this point by either the OGO or the RGO, depending upon their agreement, and the file is converted to the format of the receiving country. The RGO then forwards the transaction to the beneficiary's financial institution, which gives final credit to the beneficiary in Canadian dollars. If the transaction is going from Canada to the U.S. the cycle is reversed. A typical corporate cross-border transaction would be payment for goods or services purchased by a company in the U.S. from a company in Canada, or vice versa.

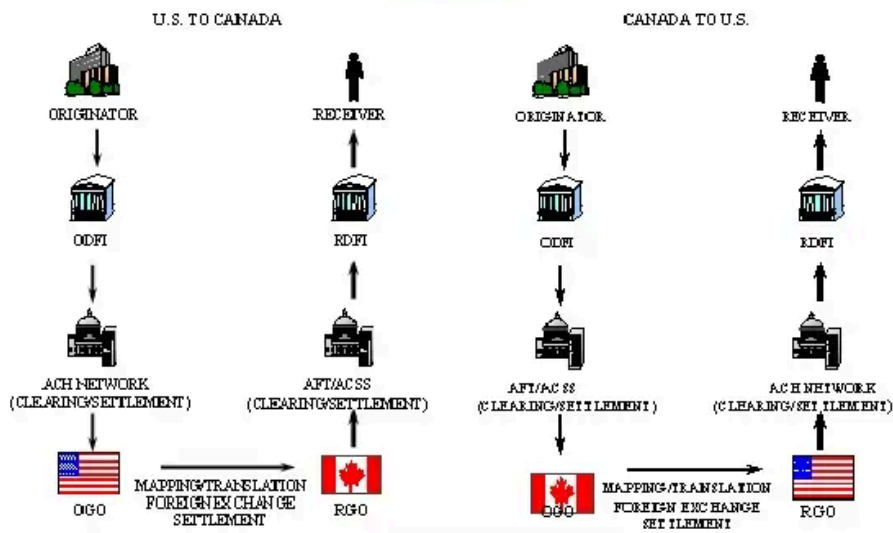


Figure 14.2 A Typical cross border payment process

The **Originator** is a company or individual with the need to make payments to or collect money from another company or individual. A contractual agreement or authorization is established between the payer and the payee prior to the first payment being made.

The Originator also establishes a contract for sending payment instructions with an Originating Depository Financial Institution.

An **Originating Depository Financial Institution (“ODFI”)** is the financial institution that initiates the payment instructions between financial institutions.

The next party in the transaction flow is the Originating Gateway Operator or OGO. The OGO is responsible for receiving payment instructions from the ODFI - with whom the OGO has an existing relationship - and forwarding the instructions in a timely fashion to the next participant in the payment flow, the Receiving Gateway Operator or RGO. The format mapping and translation, foreign exchange conversion and inter-gateway settlement for the entry occur in accordance with the agreement between the OGO and RGO.

A **Receiving Depository Financial Institution (“RDFI”)** is the financial institution that receives the payment instructions from the RGO. The RDFI is responsible for settlement and posting of the transaction to the Receiver. Its liability is the same for a cross-border transaction as it is for any other automated clearing house transaction processed within its national payment systems rules.

The **Receiver** is a company or individual to whom a transaction has been sent. The Receiver is bound by the rules and regulations of its national payment system as they apply to any domestic transaction.

14.6. Cross Border Payments Environment

14.6.1 Agreements

There must be an agreement in place between the gateway operators covering:

1. Adherence to the Cross Border Payment Operating Rules
2. Foreign exchange conversion
3. Technical and operational responsibilities
4. Settlement
5. Definition of a commercially reasonable time frame

Agreements are also required between the Originator and ODFI and the ODFI and the OGO.

14.6.2 Transmission of Payments

When payments are transmitted from the Originator to the ODFI, and from the ODFI to the OGO, they are subject to the requirements, rules and regulations of that country's national payment system. The foreign exchange conversion, format conversion and settlement that take place between the OGO and RGO are subject to the Cross Border Payment Operating Rules and the agreement in place between the two Gateway

Operators. The RDFI and Receiver are subject to the requirements, rules and regulations of that country's national payment system.

14.6.3 Settlement

Settlement takes place between each of the participants in the cross-border payment transaction and is governed by the national payments system rules of the participating countries. The Cross Border Payment Operating Rules place additional responsibility on the gateway operators. The OGO warrants to the RGO that settlement is final and the RGO warrants to the RDFI (in the receiving country) that settlement from the OGO is final. Gateway Operators assume additional risk if they transfer payments prior to settlement finality, as they will be liable for those payments in the event that settlement is revoked. Settlement is irrevocable between Gateway Operators once funding has occurred.

14.6.4 Foreign Exchange

The foreign exchange component of cross-border payments is stipulated in the agreements that are established between the various participants. Three foreign exchange scenarios exist:

- *Fixed to Variable*

Fixed origination currency amount to variable receiving currency amount – The Originator initiates a payment in their country's currency, the payment undergoes a foreign exchange conversion, and the Receiver's payment is in its country's currency.

- *Variable to Fixed*

Variable origination currency amount to fixed receiving currency amount – The Originator initiates a payment based on a specific foreign exchange rate for payment to the Receiver's account in their country's currency.

- *Fixed to Fixed*

Fixed origination currency amount to fixed receiving currency amount – Both the payment from the Originator and the payment to the Receiver are in the same currency.

14.6.5 Cross Border Payment Technical Standards

Cross-border payments should be transmitted using certain technical standards so that cross-border transactions are readily identified by financial institutions so that they may apply special handling requirements for cross-border payments, as appropriate. These formats also enable participants to transmit and receive detailed information that is unique to cross-border payments, such as:

1. Information relating to foreign exchange
2. Origination and destination currencies

3. Destination country
4. Identifiers for consumer and corporate payments
5. Identification of foreign receiving DFI
6. Foreign payment amount
7. Foreign trace number
8. Foreign Receiver's account number

14.6.6 Risk Management

Originators and ODFI's should be aware that the rights and privileges of the RDFI and Receiver in the receiving (foreign) country are applicable to the transaction. This may have an impact on authorizations, time frames allowed for returning items and other handling of exception items such as incorrect account numbers, payments made in error, etc. The Originator and the ODFI should fully understand the payment system rules environment of the countries to which they will be sending payments.

Foreign exchange on cross-border payments creates an area of risk not present in domestic payments. There may be foreign exchange fluctuation on return items, leading to a return amount that differs from the amount input. There may be items that are processed in error or duplicate items. The ODFI needs to understand the foreign exchange risks and should address how they will be handled in its agreement with the Originator.

The role of Gateway Operator brings a new participant to the typical flow of payments that stay within a country's domestic payment system. The Gateway Operator needs to be aware of the same foreign exchange impacts as the ODFI and should cover the handling of the associated risk in their agreements with each other and with the financial institutions for which they are processing.

14.7. Cross Border Payments Models

14.7.1 Origination Model

The Originator must utilize an ODFI for origination of cross-border payments. The ODFI may also be acting as the OGO. In the processing flow diagrams that follow, the ODFI and the OGO are shown as two separate entities. In the cases where the ODFI is also the OGO, items may be transferred cross-border with a reduced cycle time because the payments need not be submitted through the national payments system of the Originator's country.

It is assumed that all items are processed according to the rules, requirements and timing criteria as dictated by the originating country's national payments system and are formatted according to the specifications. Settlement finality for the entries in the

originating country's payments system is warranted by the OGO to the RGO upon funding of the entries to the RGO.

14.7.2 Receiving Model

Cross-border entries destined to a Receiver's account with an RDFI that is not also acting as the RGO flow through the receiving country's national payments system as depicted in the process flow diagrams. The entries are processed according to the rules, requirements and timing as dictated by the national payments system of the receiving country. When the RDFI is also functioning as the RGO, the cycle time for the processing of the payment may be reduced due to the entries not being submitted through the receiving country's national payments system.

14.8. Case Studies

14.8.1 Case Study 1- Credit Origination with Variable Amount and Return

(Variable to Fixed transaction)

In this case study, a US company is sending weekly payroll entries to its salaried employees in Canada. The payment amount must always be the same Canadian dollar value every week, so the US dollar amount will vary depending on the foreign exchange rate.

Cross-Border Payment Operational & Processing Flow Model
U.S.-to-Canada

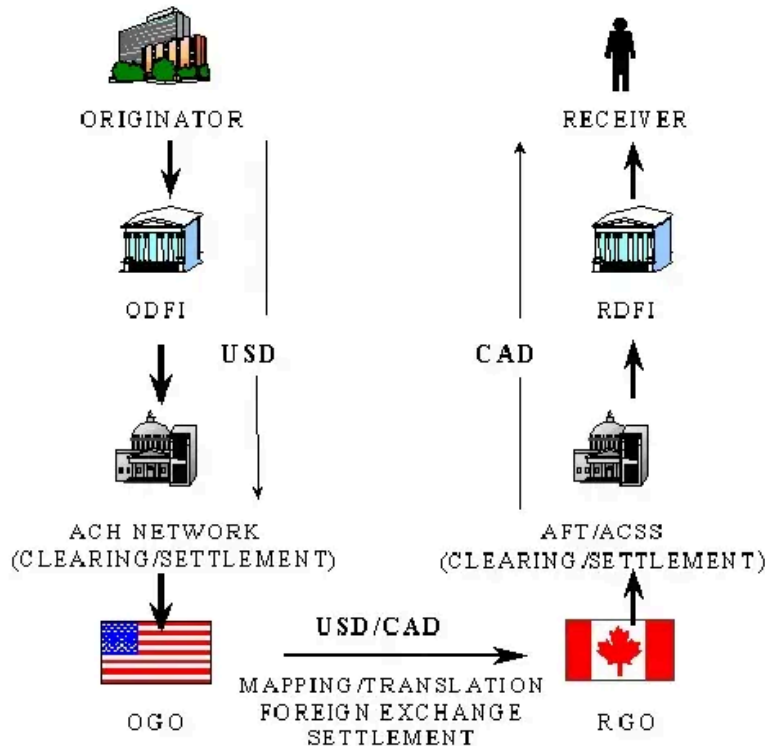


Figure 14.3 Process Flow Diagram

The company has decided to make cross-border payment to:

- 1) achieve a better foreign exchange rate for their employees by leveraging multiple payments rather than having each employee encash their checks individually,
- 2) reduce the delivery timeframe, and
- 3) eliminate the need for their employees to visit their financial institutions to cash their paychecks.

The payroll entry is originated as a variable-to-fixed credit transaction of CAD 1000 and the payment passes through the system. The employee's Canadian account is credited with CAD 1000 and the employer's USD account is debited for USD 670 based on a 0.67 CAD-USD exchange rate.

In this instance, the Canadian employee's bank account has been closed, and the RDFI returns the CAD 1000 payment to the Gateway Operator. The Gateway Operator then converts back to the standard format, performs the foreign exchange, and forwards the returned payment to the US Gateway Operator as USD 680 (because of exchange rate fluctuations, the amount actually credited back to the employer's account may be different than the original amount). Normally, two transactions will be originated back

to the employer's account, one for the exact original dollar amount (for easier reconciliation with the original payment), and a second entry to offset the foreign exchange fluctuation (whether credit or debit).

14.8.2 Case Study 2 - Debit Origination for Collections

(Fixed to Variable transaction)

In this example, a US insurance company collecting insurance premiums from Canada needs to have the funds credited to its US account without the hassle of managing paper check collections. As the Canadian customers will be paying their premiums in CAD, this will mean that a variable amount of USD will be credited to the insurance company's account.

A fixed-to-variable PBR debit entry is initiated for CAD 120 and passes through the system. Upon receipt, the RGO issues a domestic Canadian debit for CAD 120 to the customer's account, and the insurance company is credited with USD 80.40 – with the timing of that credit (i.e. before or after the debit is finalized in Canada) dependent upon the agreements between the various parties.

14.9. The Global Payment System

There is a growing interdependence of national payment systems, arising from the needs of international trade and finance, and increasingly evident in terms of foreign participation in domestic payment systems and in domestic financial markets generally. There is, in effect, a global payment system.

One of the best illustrations of these interdependencies is provided by the foreign exchange market. Unlike some financial markets, the foreign exchange market has no single location. Traders operate in different centers around the world, dealing with each other both within individual centers and between different centers. Settlement of a foreign exchange deal (two business days after the deal date in the case of a 'spot' deal) will involve two payments, one in each of the currencies being traded. Thus, using the example in figure above, to settle a US dollar/Yen deal agreed by two banks in London requires that each of the banks send a message (usually via the SWIFT- an international telecommunications network) to their correspondent in the country where they have to deliver the yen or the dollars, instructing those correspondents to arrange delivery of the relevant currency. The influence of the world-wide foreign market on turnover in high-value payment systems is substantial: for the most heavily traded currencies, payments related to foreign exchange settlement can account for as much as 50% by value of the daily turnover. Large fluctuations in market activity will thus feed through (after a two-day lag in the case of spot trades) to the relevant payment systems, and are

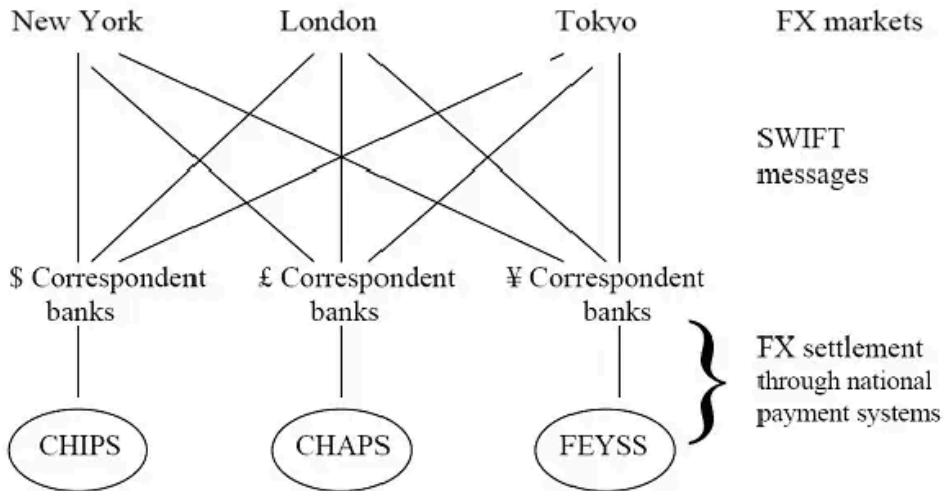


Figure 14.4 Global Payment Systems

capable of putting pressure on the operational and liquidity management capabilities of those systems. By the same token, a serious disruption in a national payment system, which resulted in the failure of trades in overseas market centers to settle, could have a serious impact on confidence in those markets.

14.10. Developing the Business Case for Cross-Border Payments Origination

14.10.1 Major Drivers of Decision

The business case for origination of cross-border payments is driven by the market demand for the application. On the corporate side the globalization of the economy is leading to greater foreign trade activity and increasing the need for companies to make more - or to begin to make more - international corporate payments. On the consumer side, as more employees and pensioners are located in (or retire to) foreign countries, they are expecting to receive their payments in an efficient manner and with certainty of payment.

The primary features a cross-border payment system must have to be an attractive option to the Originator are:

- 1) It needs to be less costly than their current method or other available methods
- 2) There needs to be certainty and finality of payment
- 3) It must be easy to use
- 4) It must be timely.

14.11. Cross Border Payment Challenges

- 1) **Inadequate domestic infrastructure** – For past few decades many nations have developed their own large and small value funds transfer systems in accordance with their own nation specific requirements, hence these systems are very independent and there is a clear lack of standardization and automation in inter-bank and intra-bank networks. This adversely affects banks and businesses alike and results often in manual intervention to collect and repair data. The current methods of funds transfer across the border are very inefficient and costly in which the funds are transferred bilaterally, the use of non-standard customer interfaces, incompatible formats between domestic and foreign banks, and the low degree of automation in banks' internal systems.
- 2) **No common message standards** – Currently the rates of Straight through Processing (STP) is very high means the manual intervention and the paper processing is very high in the current systems. For the reduction of that manual intervention more and more payment instructions and settlement should be done electronically with common protocols which are internationally recognized.
- 3) **Impact of regulatory requirements** – The complex governance structures of these disparate payment systems – some public, some private, some operated as industry associations – only add to the challenge. Achieving coordinated change at an industry level is nearly impossible without government mandates. However, when government mandates occur, they tend to focus more on responding to crises (or preventing crises) than on promoting efficiency.

14.12. Benefits of the Cross-Border Payment System

This system drives access to cross-border e-payments universal and offers several other advantages:

- * Financial institutions of all sizes may offer cross-border payment origination services to their business customers
- * Batches can contain any mixture of domestic and cross-border transactions, allowing processes to remain streamlined
- * Participating companies benefit from the improved reliability, stability and efficiency of the existing national payments systems that are part of the process.

14.13. Trends in Cross-Border Payments

- **Growth of transnational payment systems** – Apart from the domestic payment system now there is a profound growth and newer developments in the transnational payments systems such as CLS (Continuous Linked

Settlement) and TARGET. Moreover in the card systems there are giants like Visa and MasterCard which are global. Another thing that can be observed is in countries like Switzerland and Hong Kong, new arrangements have been developed for the settlement of local payments in foreign currency. These arrangements neither fit perfectly in the traditional category of “correspondent banking” or in that of “payment systems”. The main common characteristic of these arrangements or systems is that they do not settle in central bank money but across accounts held with a commercial bank and that they are based on clearly defined and transparent rules for payment activities. There are other transnational systems like Swiss Euro Clearing Bank (SECB) developed by Swiss financial institutions established as a cross-border solution in order to facilitate their cash management in euros. In Hong Kong, the U.S. dollar and Euro clearing systems, USD CHATS (Clearing House Automated Transfer System) and Euro CHATS, were introduced in 2000 and 2003, respectively. They enhance the safety and efficiency of settling these foreign currencies in the local time zone. The growth in transnational systems can improve the efficiency of cross-border payments by reducing clearing and settlement times, minimizing float. Better visibility of funds flows supports improved cash forecasting. Finally, standardized formats will reduce costly errors and repairs.

- **Government-led initiatives and mandates are increasing** – To prevent the issues like money laundering and financial terrorism the central banks and government is taking up new initiatives. One of the initiatives is Single Euro Payments Area (SEPA). Government-led initiatives are focusing on the reduction of costs to the end-users, adoption of common payment standards, and reducing the ability of payment systems to be used for illegal means. Ultimately, this will translate into higher costs for banks that provide cross-border services. However, this leads to revenue opportunities for those banks that provide services to other banks.
- **Risk and liquidity usage are being closely managed** – More premium is now being put on striking a balance between the above two. The exposure to credit risk is increased if payments are delayed but also for the payer its liquidity costs decrease but on the other hand when the payments are made immediately with more liquidity in the system the risks are reduced but cost of making the payment increases for the payer.
- **Multinational banks and corporations are expanding** – The trend of consolidation in the banking sector is becoming a great force. Mergers and

Acquisitions are becoming big headlines. For example, we have witnessed the emergence of mega banks such as the merging of Bank of America and Nations Bank, as well as J P Morgan Chase merging with Chase Manhattan Bank.

- **Operational efficiencies are being sought through outsourcing** - The financial institutions like banks are concentrating more on customers and the marketing functions. Their paradigm is becoming service oriented and hence the back office tasks are being outsourced now. Banks have increasing recourse to such entities, allowing banks to specialize in the “sales function” (covering direct relations with clients, including account holding) while outsourcing “production function” such as the processing of payments and securities.

14.14. Summary

- Many cross-border payments are actually settled in a specific country's domestic settlement system.
- Cross-border payments should be transmitted using certain technical standards so that cross-border transactions are readily identified by financial institutions so that they may apply special handling requirements for cross-border payments, as appropriate.
- There must be an agreement in place between the gateway operators in cross-border payment covering:
 - o Adherence to the Cross Border Payment Operating Rules
 - o Foreign exchange conversion
 - o Technical and operational responsibilities
 - o Settlement
 - o Definition of a commercially reasonable time frame.

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15.4. Introduction

As economies have grown and incomes have increased, consumers and companies have demanded more convenient electronic ways of paying for goods at the point of sale as well as settling bills remotely from home. To provide customers with a more convenient way to make payments, SEPA [Single European Payments Area] has been introduced where the people will be able to make payments in Euros whether in Europe or outside the national boundaries under the same basic conditions, rights and obligations, regardless of their location.

SEPA is currently defined to consist of the EU 25 Member States plus Iceland, Liechtenstein, Norway and Switzerland. The very essence of SEPA is to eliminate these borders and create a Single Euro Payments Area.

15.5. Vision of SEPA

SEPA project dates back to 1990 when European Commission published report "Making Payments in the Internal Market" which outlined a vision of a single payments area stating "the full benefits of the single market will only be achieved if it is possible for business and individuals to transfer money as rapidly, reliably and cheaply from one part of the community to another as now is the case with (in) most Member States". In the year 1999, Economic & Monetary Union (EMU) developed the concept of integrated European markets for goods and services which could be realized through SEPA. The EPC's [European Payments Council] vision of SEPA reflects those of the EC [European Commission] and the ECB [European Central Bank] and is summarized as follows:

"SEPA will be the area where citizens, companies and other economic actors will be able to make and receive payments in Euro, within Europe, whether between or within

national boundaries under the same basic conditions, rights and obligations, regardless of their location.”

SEPA will have following impact

- All electronic payments will be impacted and as a result, credit transfers, direct Debits and card payments will migrate to common interoperable formats and process
- The system will be more efficient with tangible benefits for the economy and society as a whole.
- Impact on the fragmented national payments instruments with the implementation of new, common business rules and technical standards
- Euro will be systemically strengthened as a currency by being underpinned with an integrated payments environment
- It will also generate through harmonization more efficient payment systems with tangible benefits for the economy and society as a whole.

15.6. Scope of SEPA

There is a priority implementation focus on the Euro area, currently 12 countries (13 from January 2007), and the change programmed will radically impact their whole domestic payments environment. Within Europe, outside the Euro area, there will be opportunities to participate in Euro Payment Systems, and communities will be able to adopt SEPA standards and practices to contribute to the *Single Market for Payment Services*. The vast majority of banks throughout SEPA active in making and receiving Euro payments are expected to participate in the SEPA Schemes and issue.

15.7. Payments landscape after SEPA

The following **Figure 15.1** shows the difference between present and future after implementation of SEPA

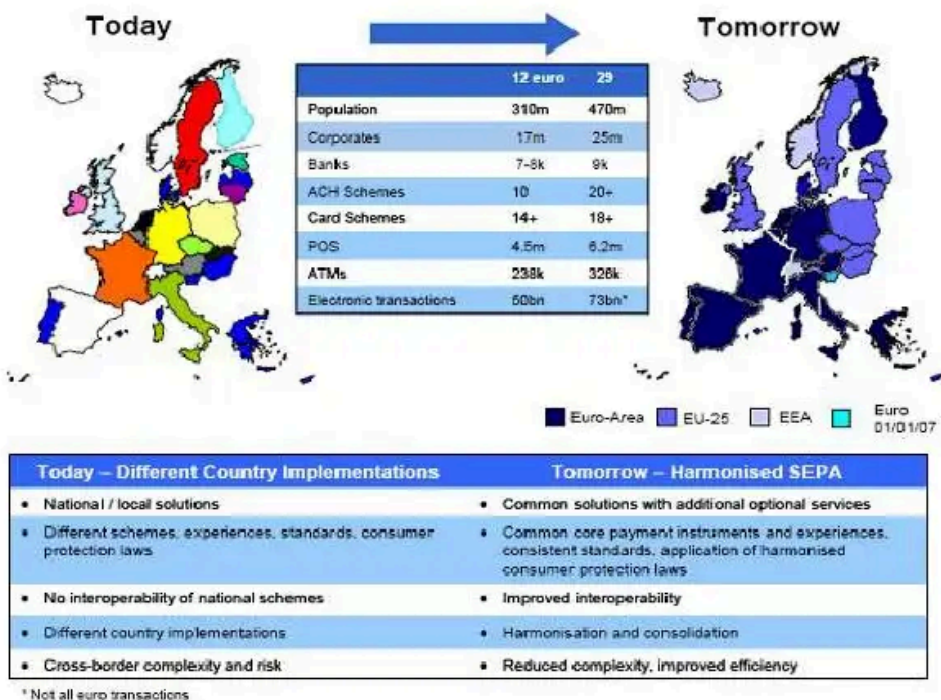


Figure 15.1 Future after SEPA (source: European Payment Council)

The Euro area alone currently processes some 50 billion electronic retail transactions and between two to four times in cash each year. This volume is generated by 310 million citizens, 16–18 million large and small corporate, 7,000–8,000 banks, 4.5 million points of sale and 240,000 ATMs.

Given the size of the market, the costs of bank migration will be very substantial; however long-term efficiency gains will eventually more than offset the initial outlay. SEPA will impact euro payments made within its geographic area. Currently there is a priority implementation focus on euro area (13 countries) and the change will radically impact the whole domestic payment environment.

15.8. SEPA Programme

The SEPA programme has been initiated by 3 main European bodies – the European Commission, the European Central Bank, and the European Payment Council.

The three European Bodies as shown in the **Figure 15.2**, laid down the programme of SEPA

a) The European Commission (EC): The overall initiative is steered by the European Commission (EC). European Commission published Payment Services Directive (PSD) in December 2005 which was designed to harmonise and remove legal barriers for payments throughout the European Union (not just the Euro area).



Figure 15.2 Pillars of SEPA (Source: European Payments Council)

The reason was that EC recognized that a single payments market would only be possible within a common legal framework that would remove the local anomalies and differences.

b) European Central Bank (ECB): ECB has developed guidance on SEPA requirements and set the implementation timelines to ensure an efficient and orderly payments market in the Euro area. The ECB has issued several reports between 1999 and 2003 to guide the creation of a common Euro area payments market. The ECB fully expects the launch of SEPA instruments in 2008 and considers that most of the SEPA objectives can be implemented by the (end of) 2010.

c) European Payment Council: EPC founded in 2002 has responsibility for designing the new SEPA payment instruments. The EPC has the role of designing and specifying the core common services which will operate within a single European payments market place. It also provides guidance and co-ordination to enable the development of SEPA standards; it identifies and synthesizes best practice in the payments industry and supports and monitors the implementation of SEPA.

15.9. SEPA-Rulebooks, Implementation and Migration

For SEPA development, EPC has covered two different approaches, which are complementary to each other.

1. For Electronic Transfer Schemes (ETS) a “replacement” strategy has been chosen with new common credit transfer and direct debit schemes for the overall SEPA.

2. For the highly complex cards business, the strategy has been that of “adaptation” of existing schemes to a new set of business and technical standards.

These two approaches have core feature of different infrastructure for the scheme they follow.

Here the Scheme could be defined as Credit transfer, debit transfer and card strategy whereas, Infrastructure could be, the layer comprising of different networks, clearing and settlement houses. Rulebooks for each of the scheme have been defined comprising of different standards, rules and obligations.

15.9.1 Timeline & Implementation of SEPA

The EPC, working with the ECB, has drawn up a timeline based upon three deliverable phases:

1. Design and preparation;
2. Implementation and deployment and
3. Co-existence and gradual migration.

From 2008, the three SEPA payment instruments (credit transfers, direct debits ,cards) will operate alongside existing national processes, with full migration achieved from the end of 2010 onwards. After the evolution of credit transfer, debit transfer and card framework no national credit or debit or card framework will be used. SEPA will be successfully implemented on a solid organizational structure within which all the players and stakeholders know their duties and can execute their responsibilities.

The Implementation of SEPA will be in following phases

Phase 1 – Design and Preparation involves the design of the two new ETS and the Cards Framework during 2005. 2006 will focus on the development of standards and the specification of the technical detail and security.

Phase 2 – Implementation and Deployment overlaps standards and specification with implementation, piloting and launch by the end of 2007 for the two new ETS.

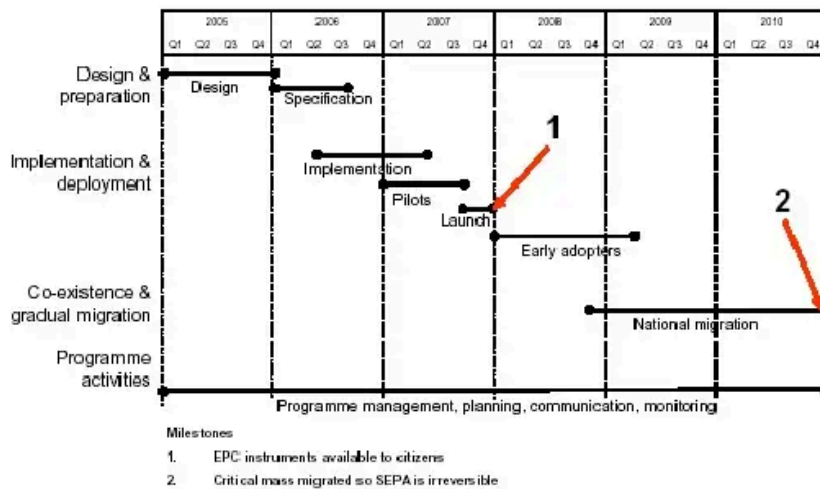


Figure 15.3 Timeline of Implementation (Source: European Payment Council)

Phase 3 – Co-existence and Migration will be a transitional period in which there will be a co-existence of national and SEPA schemes and a gradual migration to the latter from January 2008 to end 2010 and beyond.

15.10. SEPA End-User Experience

SEPA will have an impact on the users who make their payments in Europe or are the citizen of Europe. Implementation of SEPA involves a set of complex changes to payments, commercial practices and infrastructures. Its impact on the day-to-day lives of consumers, merchants and corporate within SEPA and the EU is discussed below with practical applications.

15.10.1 Citizen Europe

- All banks will be able to offer accounts that are usable in all SEPA countries.
- When spending in other countries, citizens can feel more secure, carry less cash and be less reliant on local ATMs.
- Home country payment card will be accepted for payments in any SEPA country and they will receive full details of any merchant currency conversion charges across SEPA.
- Sending money within SEPA to family and friends will be simplified with uniform processes, rules and account codes and transparent fee structures with no deductions from the transferred amount.
- Harmonization of Direct debits within SEPA and common processes adopted for mandate set up, first payments and standing data changes, improving the service for regular bills payment.

- Salary payments and credits to the current account will have predictable posting dates.
- New and common legal framework applied for refunds, disputes and complaints.

The difference experienced by the citizen could be seen in **Table 15.1**.

15.10.2 European Mobile Citizens

People participate in programs to obtain work experience and to study in other countries. Usually these people often find it difficult to use a home based bank to support payments in other country. Same is the case with home country payment cards. Money transfers direct to family and relatives at home, or to home banks, is expensive and takes time to reach their destination. The **Table 15.2** below describes the additional benefits SEPA will offer to the “European Mobile Citizen”.

15.10.3 Small and Medium sized European merchant

SEPA will impact small and medium European merchant’s approach to card acceptance and the services they receive from their banks.

- With the introduction of SEPA, domestic market specific practices will become more consistent. Card schemes will move to more standardised approaches.
- Card acceptance-net will be widened, which will enable domestic only payment brands to be supported by all terminals across the EU.
- All SEPA approved cards will be chip based (i.e. the magnetic stripe will not be SEPA compliant) and will be authenticated using PIN.
- By moving to common standards SEPA will open up the market for acquiring services leading to increased choice of providers and the development of many new products and services.

15.10.4 Large European merchants

- Large retailers will experience significant benefits through lower processing costs as a result of a common Euro area wide cash repositioning strategy.
- Large merchants will see significant savings in terminal costs and POS {Point of Sale} processes and infrastructures as a result of a single SEPA software application, the removal of multiple terminals and the introduction of common terminal to host standards.
- Under SEPA Europe’s largest merchants with multi-country operations will also obtain additional improvements over and above those of smaller players.
- Cross-border business expansion will no longer be constrained.

Table 15.1 Today/Tomorrow-SEPA Citizen of Europe

Today – Pre SEPA (2006)	Tomorrow – SEPA
Payments limited to home market only with separate services for foreign payments	Payments made from home bank to any of the SEPA countries using common standards
In other countries: National payment cards often not accepted by merchants Over reliance on ATMs for cash High cash holding - security concerns Lack of transparency of merchant currency conversion fees Different experiences at POS terminals in different nations	In other countries: Home payment card enabled for all SEPA-based merchants Fewer visits to ATMs required for cash withdrawal Increased card usage – less cash in wallet Full transparency of charges, fees and exchange rates in all EU nations Common experience and processes at POS terminals
Telephone or internet purchases restricted to home market for some	Purchases made in any country with a common legal framework for consumer protection and redress
Variable service sending money or making payments outside home market, delays in transferring value	Competitively priced, improved service for all payments outside home market with new clearer account codes, no value dating and no deduction of fees from the principal amount of payment
Inconsistent direct debit processes. Mandate set up and change processes complex in some markets	Standard Direct Debit Rules for all euro-area countries. Common mandate processes for set up, first payment and standing data change
In some markets extended periods to process salaries and other payments	Predictable transfer timeframes offered by banks with no regression from current service levels
Complex bank charges for payments services	Full disclosure of charges, fees and exchange rates for payments
Widely different consumer redress processes in every country	Common EU-wide process for consumer redress and refunds
In some markets a limited choice of banks from which to purchase payment facilities	Increased choice and ability to shop around for lowest cost or highest functionality accounts in SEPA countries
Common processes for transferring pensions when resident outside native/home country	Common processes for pension transfers within predictable time frames

(Source: European Payment Council)

Table 15.2 Today/Tomorrow- SEPA Europe Mobile Citizen

Today - Pre-SEPA (2006)	Tomorrow - SEPA
Multiple payment relationships – one for each working or studying country	One bank in home country accessible throughout SEPA
Few simple processes to pay rent deposits	Low cost and predictable SEPA cross-border transfer services from home account
No facilities for cross-border direct debits for rent and utility bills	Home bank accepts direct debits from any SEPA working or studying country
Limited facilities for cross-border salary payments from working or studying country	SEPA credit transfers can be posted from country of working to home account
High use of cash for all types of purchases	Home bank issued payment card used for most higher value purchases in euro-area
Money transfers to home bank or to family relatively complex	Simple, common transfer service for all SEPA countries
Limited choice of banks with full multi-country services	Increased choice - new products and services with more competitive offerings

(Source: European Payment Council)

15.10.5 Small and medium sized European corporate

- SEPA will also impact small and medium sized corporate enterprises, particularly those that trade cross-border.
- Under SEPA small businesses will have greater confidence when trading cross-border as a result of the implementation of Payment Services Directive (PSD) as its consistent legal framework for payments will introduce much improved certainty and clarity.
- SEPA wide transfers to pay for sales and purchases will become more efficient through the implementation of a harmonized timeframe, reaching any account within SEPA.
- A common SEPA wide full reach direct debit service will be introduced with a common IBAN/BIC account codes and an electronic mandate processing feature with the potential to dematerialise paper records thus reducing back office costs.
- SEPA will open up the market for payments, offering smaller enterprises greater choice of payment products, encourage innovation and new product offerings.

15.10.6 Large European Corporate and Public Administrations

Large corporations that operate across Europe want business to business (B2B) payment instruments to collect revenues from customers and to pay suppliers for goods and services. These companies work closely with their banks to build highly efficient payments processes in their domestic markets.

- A major benefit will be one single process for all incoming and outgoing payments, both national and cross-border.
- SEPA will offer opportunity for harmonised, guaranteed, and secure remittance information across Europe will be a major benefit for both invoicing and reconciliation. Standard, harmonised Scheme Rulebooks for direct debits and credit transfer, plus consistent PSD legal framework for exception processing, will improve efficiency and reduce back office reconciliation costs.
- Corporate with operations in several countries will be able to create (and receive) a single aggregated file for all payments (domestic and cross-border) and submit these to a single institution within standard clearing and settlement timeframes. There will no longer be a need for multiple files prepared to different standards. Common file standards will apply to all payments (and the use of IBAN [International Bank Account number]) resulting in fewer exceptions and higher STP.
- The number of banking relationships for effecting and receiving payments can be consolidated. Banks will offer extended reach and support SEPA-wide operations, reducing administration costs and improving efficiency.
- Entities with large volumes of direct debits and credit transfers will be able to shop around for banks able to clear and settle at best service, and lowest costs, inclusive of additional bank services, in any country.
- The new standard for pan-European direct debits will offer a much-improved service for corporate wishing to receive payments from customers working or living across Europe. The new standard for SEPA Euro credit transfers will enable predictable transfer times throughout SEPA, again improving the service to corporate customers.
- SEPA will deliver similar benefits to governments and public administrations, utilities and other organisations that make substantial payments within national and pan-European markets.
- Services to citizens will be enhanced and pensions, social security and other benefits paid within a clear timeframe to a best practice standard.

15.10.7 Large European Merchants

Europe's largest merchants face significant problems in developing common internal systems and processes for their operations in European markets. Many large super-market chains have internal systems that are specific to each country and, in addition, cannot benefit from economies of scale and the pooling of transactions. As shown in **Figure 15.4** largest merchants of Europe with multi-country operations will also obtain additional improvements over and above those of smaller players under SEPA.

- The most significant is the use of a single internal ePOS [Electronic Point of Sale]/EftPos [Electronic funds transfer Point of Sale] platform which will support payments processing for all countries and enable common systems for features such as mobile top-up (MTU), dynamic currency conversion (DCC) and bills payment (BP). This will eliminate multiple country specific systems.
- Cross-border business expansion will no longer be constrained. Multiple acquirers banking relations can be reduced, because SEPA will enable combined payment card acquiring for Euro transactions.

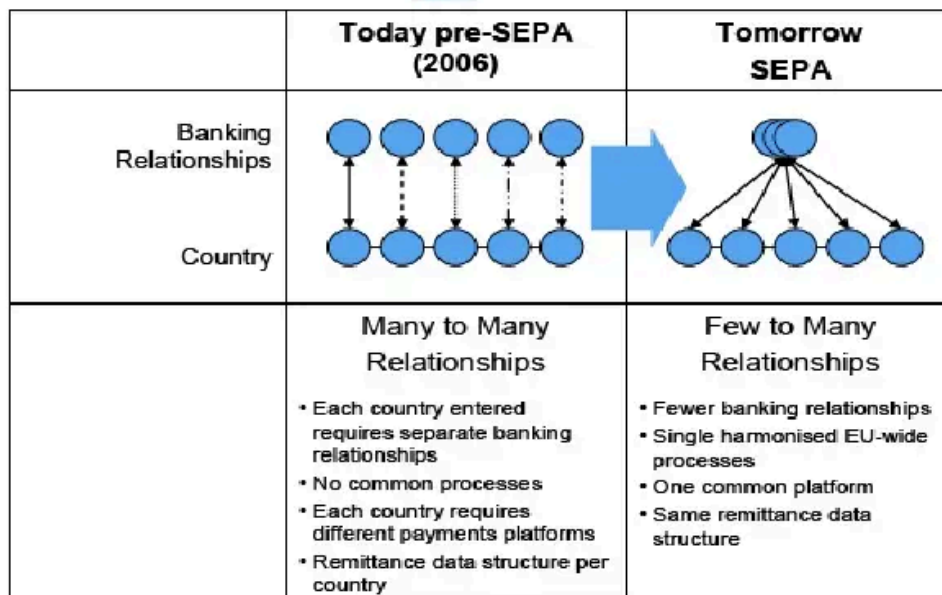


Figure 15.4 Benefits to Corporate

(Source: European Payment Council)

15.10.8 Owning or renting a home in another country

Many European citizens have properties outside their home country, which is let out, intend for retirement. Similarly, many rent properties in other countries for extended time periods. However today the transactions associated with renting or buying a home in another country, paying the agent, solicitor and taxes is time consuming. They have to open a bank account in the same country as the property because their domestic

account cannot easily support national transactions in another country. Similarly, there are currently limited facilities for receiving cross-border direct debits to pay rented or holiday home utility bills, taxes and other services. Finally, the process of transferring salaries, pensions and letting and rental fees to either the home country account or the holiday or rented country account, is complex and subject to delays.

The holiday home owner or renter will benefit from the introduction of SEPA. There will no longer need for accounts in two countries. The process of making payments for deposits, legal or agent's fees, can be rapidly conducted from the existing home account. Direct debits for rents, utility bills and taxes can be rapidly directed cross-border to the home account. Credit transfers for letting fees can be similarly processed. Finally, SEPA may also increase the number of banks offering products and services that support second home owners or those living in other countries. The choice of banking services and new banking products will therefore increase.

Table 15.3 SEPA for Non Residential Europeans

Today - Pre SEPA (2006)	Tomorrow - SEPA
• Sometimes complex payments processes for property purchase or rent, legal, agent and refurbishment costs	• Predictable SEPA-wide transfer process for payments anywhere in SEPA
• The usual remedy is to open an additional account in second home country	• Home bank may serve second home country
• No facilities for cross-border bills payment	• Second home utility or tax bills direct debited to home country account
• Extended times for transferring salaries, pensions and letting fees to holiday country account	• Standard transfer time to home country account using common account codes
• Limited choice of home banks offering services tailored to holiday home owners and renters	• Many new home banks offering new targeted SEPA products and services

(Source: European Payment Council)

15.11. Designing Components of SEPA

The features and design criteria used in developing two new SEPA Credit Transfer [SCT] and SEPA Direct Debit [SDD] Electronic Transfer Schemes and the SEPA Cards and Cash Frameworks are as follows.

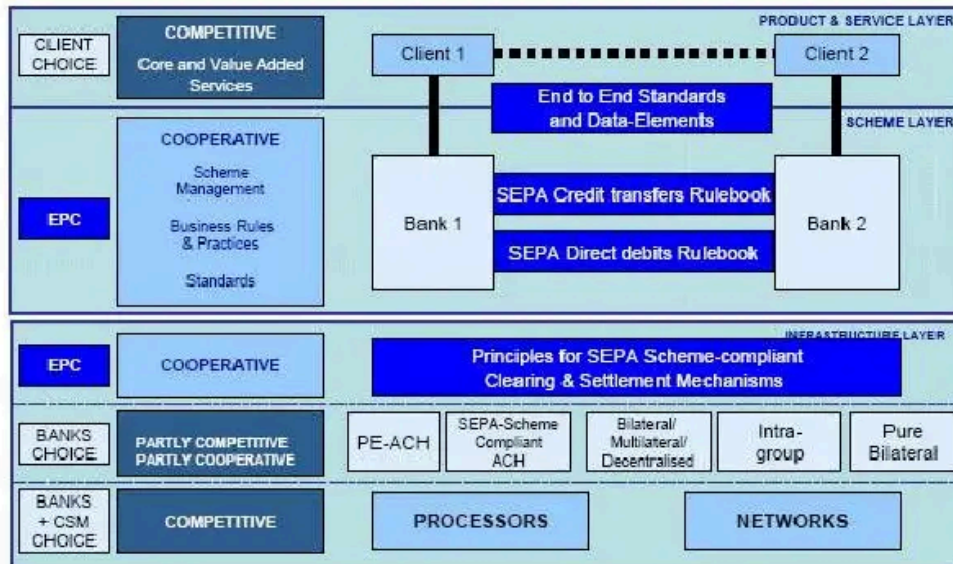
The architectural design of the SEPA deliverables is based on different layers of activity. First there will be the SEPA products and services of banks which the customer directly experiences, uses and pays for. The second layer is the scheme layer which defines the basis on which banks co-operate to provide standards, rules, and interoperability. The third layer is the processing infrastructure between banks and providers of payment services. The **figure 18.5** shows business architecture of the SEPA deliverables which is based on different layers of activity.

First layer is competitive bank layer {The SEPA products and services of banks will be defined, delivered and described by individual banking institutions on the basis of competition}. The second layer is related to the *scheme* co-operation. The third layer is related to the *processing infrastructure*.

Key differences between the scheme and infrastructure are given below in **Table 15.4**.

Table 15.4 Scheme and Processing Infrastructure

A Scheme	Processing Infrastructure
• A set of interbank rules, practices and standards for the execution of euro payment instruments within SEPA • A commercial framework which enables the authorisation, clearing and settlement of payment transactions • A brand, logo or trademark to enable widespread acceptance, where applicable • Rules which define the roles and responsibilities of banks (originators, beneficiaries, issuers, acquirers) and the relationships with each other, consumers and corporates, merchants and public administrations • Rules which define the operational service quality, service levels, settlement times, exception item processing • Standards which define the basis on which transactions and payment instruments inter-operate in a secure manner	• The core operational service provided to enable authorisation, clearing and settlement • The technology delivery system and network entities which support a payment scheme • The technology platform on which the clearing and settlement process operate • The customer support operations which support banks, corporates, merchants and public administrations • The development and maintenance functions that support the technology applications, IT platform and networks authorisation • Functions include clearing and settlement systems, card transaction switches and networks, merchant accounting and back end processes as operated by banks, interbank companies and commercial processors and central banks (settlement)

**Figure 15.5** Business Architecture of SEPA

Source: European Payment Council

15.12. Designing two new Electronic Transfer Schemes (ETS)

The two new Electronic Transfer Schemes (ETS) viz. SEPA Credit Transfer (SCT) and SEPA Direct Debit (SDD) are the deliverables of the European Payment Council [EPC] who developed them from 2004 through to 2006. These schemes are designed to give core information to customers, banks, and infrastructure based on the data accumulated from bank's day to day contact with their customers. A different methodology has been adopted for the design of these schemes, compared to that used for the cards arena which is shown in **Figure 15.6** below.

It should be noted that some parts (e.g. clearing and settlement functions) of the [Pan-European-Automated Clearing House/Clearing and Settlement Mechanism] PE-ACH/CSM infrastructures and card processing infrastructures could be executed by the same organization and/or infrastructure.

15.12.1 The SCT (SEPA Credit Transfer) and SDD (SEPA Direct Debit)

Three key domains of activity were identified during the design to ensure an optimal balance between competition and co-operation amongst banks, namely:

1) Enable banks to offer their own products and services on the basis of competition.

Banks will provide different types of payment services based on the core functionality of schemes and compete on factors such as pricing, service level, and optional services. They are free to add advanced features and practices and integrate their payment services into the broader range of banking services provided to their customer.

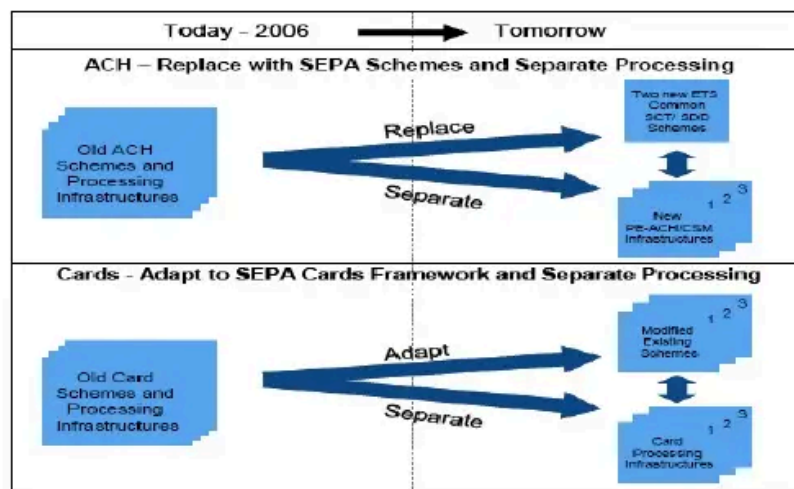


Figure 15.6 Two Electronic Transfer Schemes (Source: European Payment Council)

2) Create common inter-bank schemes.

In the second domain of 'scheme' it became clear that it is necessary to undertake a full scheme replacement strategy because various national schemes need to be replaced by one common business rulebook, data sets and standards. These schemes with common business rules and standards will enable banks to compete for clients and develop market products and services across SEPA, rather than just within the home markets, as at present.

The following paragraphs provide a summary of what the SEPA Credit Transfer and SEPA Direct Debit Schemes will enable banks to deliver to users and how they are structured in terms of the 'four-corner' model.

For Credit Transfer

- The originator (payer) completes a credit transfer instruction and forwards it to the Originator's (payer's) bank by any agreed means [2].
- The originator's bank receives and checks this, and rejects erroneous instructions, then the originator account is debited and the credit transfer is sent to the clearing and settlement mechanism (CSM) [3].
- The CSM forwards the credit transfer message to the beneficiary bank and settles the amount of the transfer [4].
- The beneficiary's bank receives the credit transfer message, checks the credit transfer message and credits the account of the beneficiary [5].

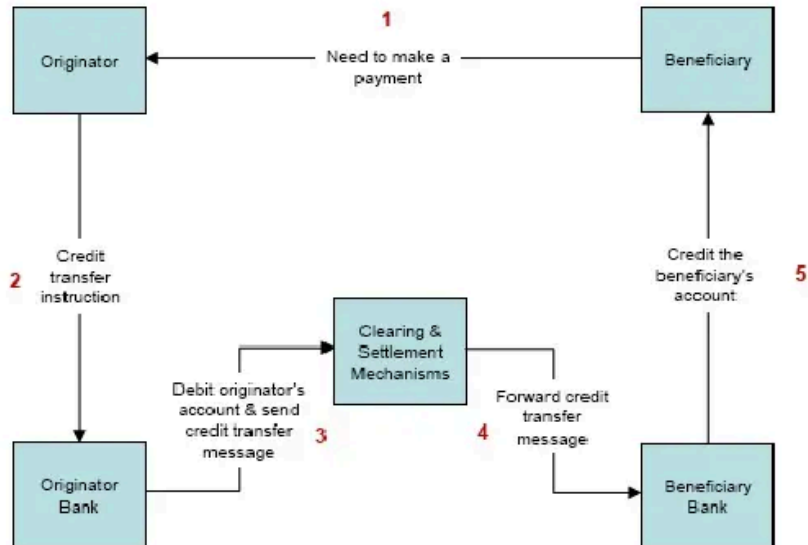
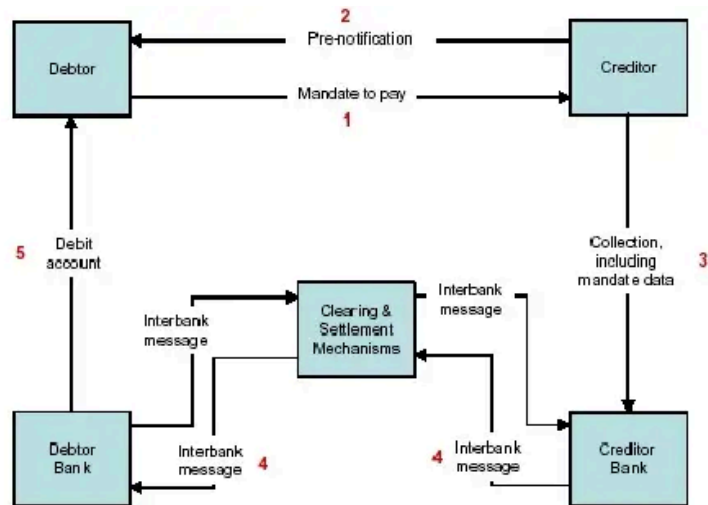


Figure 15.7 SEPA Credit Transfer

(Source: European Payment Council)

**Figure 15.8** SEPA Direct Debits

(Source: European Payment Council)

For direct debits

- A mandate is given by the debtor to authorize the creditor to initiate direct debit payments (called collections) [1] and allows the debtor bank to pay those collections. (Debtors are, however, entitled to request banks not to accept any direct debit collections on their accounts). A mandate can be a paper document or an electronic document created and signed in a secured environment. A mandate, after being signed by the debtor, must be sent to the creditor.
- After receiving the signed mandate, the creditor may start to initiate collections. Before a collection the creditor must send a pre-notification to the debtor [2], unless otherwise agreed between the two parties.
- The signed mandate must be stored by the creditor as long as the mandate is valid. The Mandated data are transmitted in electronic form along with each collection [3].
- The debtor bank may reject a collection for technical reasons prior to settlement. On the due date itself, the debtor bank must debit the debtor's account if the account status allows this, if not a return is generated. The debtor is entitled to obtain a refund from the creditor if he/she disagrees with the collection for reasons covered by the legal requirements defined in the PSD, for which he must send the request to the debtor bank within a period of six weeks. This refund does not relieve the debtor of its responsibility to resolve the disputed collection with the creditor.

Banks will provide further information on the detailed operation of services based on these electronic transfer schemes before the launch of the new SEPA instruments.

3) Competitive processing infrastructure

In SEPA, SCT (SEPA Credit Transfer) and SDD (SEPA Direct Debit) Rulebook, describes Clearing and Settlement Mechanisms (CSM) as the third domain of processing infrastructure. In the new SEPA environment the market can elect various optional CSM models, all of which must be SEPA scheme compliant namely:

PE-ACH- An ACH that is or is part of a Pan-European ACH, a SEPA-wide, country-neutral clearing organisation, providing reach to all banks in the SEPA Schemes, and which banks from anywhere within SEPA can elect to use on the basis of price and service.

SEPA Scheme Compliant ACH- An ACH capable of processing SEPA Scheme transactions within a defined market and which may or may not (yet) be in transition to a PEACH. **See Figure 15.8**

Multilateral CSM- A decentralized form of multilateral clearing and settlement (not an ACH structure) capable of processing SEPA Scheme transactions within a defined market

Bilateral- A decentralized form of bilateral clearing or settlement (e.g. correspondent banking).

IntraBank/IntraGroup. An intrabank and/or intragroup clearing and settlement arrangement, where both the originator/creditor and beneficiary/debtor have their accounts within the same bank or group. This is a competitive domain operating within a set of principles, as between infrastructures within which banks may cooperate to operate a particular infrastructure that suits their needs.

Table 15.5 Policy Framework of PE-ACH

Policy Framework for a PE-ACH
- Country neutral in terms of access, governance, rules - Guaranteed open and transparent access criteria for all SEPA banks in SEPA - SEPA-wide reachability to all SEPA banks, either directly or indirectly - Full interoperability and clearing and settlement services for both SCT and SDD - Clear statement of services, no barriers to entry - Open governance structure - Pricing fair, transparent, non discriminatory, scaled – national and SEPA payments the same - Robust operational rules for risk management, clearing, settlement, execution, security

(Source: European Payment Council)

To summarise, the SCT and SDD principles have enabled the creation of a common set of core rules and processes for the two new ETS schemes. The new structure has also

been designed to fit into the three layer model which will ensure competition by banks for customers and by the processing sector for services and networks. This framework is summarised in the **Figure 15.5**.

As discussed the business architecture of the SEPA deliverables is based on several different layers of activity. First there will be the competitive bank layer in which banks provide SEPA products and services for customer use. The second layer relates to the scheme co-operation. This defines the basis on which banks co-operate to provide standards, rules and interoperability. The third layer relates to the processing infrastructure. This layer is primarily competitive as between various competing channels, although communities of banks can and do cooperate to meet common needs.

Traditionally commercial aspects of the “payment scheme” are entangled within the rules for the operational company that delivers interbank payment processing. Under SEPA the new schemes and their rules, will be separated from the operational inter-bank service provider.

Each common scheme Rulebook details core and basic SCT and SDD services to enable a common level of service to be delivered to users for all ETS services within SEPA. The new Rulebooks will be implemented by banks and processors (PE-ACH or other CSM's) and services delivered on this basis across Europe. Rulebooks have been designed on following criteria.

Table 15.6 Design Criteria for SEPA Schemes

SEPA Schemes - Design Criteria
• Separates scheme governance from underlying processes • Encourages competition among banks in products and services • Consists of one common “all countries” framework • Enables SEPA-wide interoperability and universal reach • Builds highly automated, efficient, no exception, services to common standards • Stimulates processor and network provider choice and market consolidation • Enables highly secure, robust and resilient infrastructures • Removes disputes, inhibitors and barriers in national markets • Defines the roles and responsibilities of PE-ACH and other CSMs • References the Payment Services Directive • Enables implementation by launch date of 2008 and full replacement of existing solutions from 2010

(Source: European Payment Council)

15.13. What the New Electronic Transfer Schemes deliver

A summary of the major components and features of the two new schemes is

provided below.

Table 15.7 New ETS Scheme

SEPA Credit Transfer	SEPA Direct Debit
• Full SEPA-wide reach • Transfers in euro initiated from customer accounts • Basic credit non-urgent transfers	• Full SEPA-wide reach • Recurrent and one-off payments in euro • Both parties (Creditor and Debtor) hold accounts with SEPA Scheme Participants
• Uses IBAN and BIC as account codes • Based on UNIFI (ISO 20022) XML standards • Common maximum 3-day time cycle (subject to PSD provisions) • Accommodates secure customer reference data • No deductions from principal amount – separate fees • No value limit at scheme level (sending banks set own limits) • Receiving banks accept payments any value	• Uses IBAN and BIC as account codes • Based on UNIFI (ISO 20022) XML standards • 2-day time cycle unless it is a first payment when 5 days are necessary for set up and first payment • Accommodates secure customer reference data • Limit will be set for C2B payments. • Specific rules for “due date” - “settlement date” - “debit date” • Clear specification of mandate data and contractual terms • Dematerialises mandate – transmission to debtor bank
• Reflects FATF SR7 regulation • Common rules for clearing and settlement • Comprehensive rules for rejects and returns • Ability to provide additional optional services	• Common rules for clearing and settlement • Comprehensive rules for rejects, refunds and returns • Ability to provide additional optional services

(Source: European Payment Council)

15.14. SEPA Cards Framework

The SEPA Cards Framework (SCF) was developed by the EPC, working closely with banks as issuers and acquirers as well as the international and domestic card schemes (card schemes mean VISA, MASTERCARD etc).

Given the complexity of the cards business, widely differing approaches adopted in the national card scheme of each country and the presence of the International Card Schemes (ICS), EPC decided that it should not for many practical and cost reasons, build a new common card scheme and Rulebook unlike the new ETS schemes and it developed a policy document called the “SEPA Cards Framework” which would state

how issuers and acquirers and card schemes and operators must adapt their current operations to comply with the SEPA principles for card payments in Euro.

The card experts developed a three layer model to enable competition between all of Europe's card schemes, banks, processors and network providers but within a consistent framework that defined how the parties and players should interact. Thus international, national and new schemes will compete for members, banks will compete for customers and sell products and processors/networks will compete to service banks within a three layer model as shown in **Figure 15.9**. Three key areas for card scheme adaptation identified are as follows.

1) Improving Choice.

Each country operates a different national payment card infrastructure and applies different commercial frameworks standards, formats and protocols, which can sometimes limit competition and transparency. The SEPA Cards Framework [SCF] policy will ensure that national schemes adapt their existing commercial frameworks, standards and processes to "best practice" guidance and would also ensure that markets are accessible and competition strengthened.

2) Scheme Inter-operability

Through the international standards developed by Visa, MasterCard, American Express and Diners Club for credit cards, international interoperability is well advanced and are accepted in many countries worldwide. National payment cards often lack this interoperability and are not accepted by most of Europe's merchants outside their domestic markets hence SCF policy statement had to ensure interoperability of all cards by merchants in any country within the SEPA zone, as well as defining the need for new common standards and processes.

3) Competitive Processing Infrastructure

Most of the national payment card and ATM schemes are embedded within the payment card infrastructure's operating companies. EPC decided to separate card and ATM scheme from processing with the objective of increasing both competition on the card scheme layer and competition on the infrastructure layer.

15.14.1 The Deliverables from SEPA Cards Framework (SCF)

The Framework defines the environment within which payment card transactions are carried out. A summary of the major policy scope and features described within the SCF is provided below.

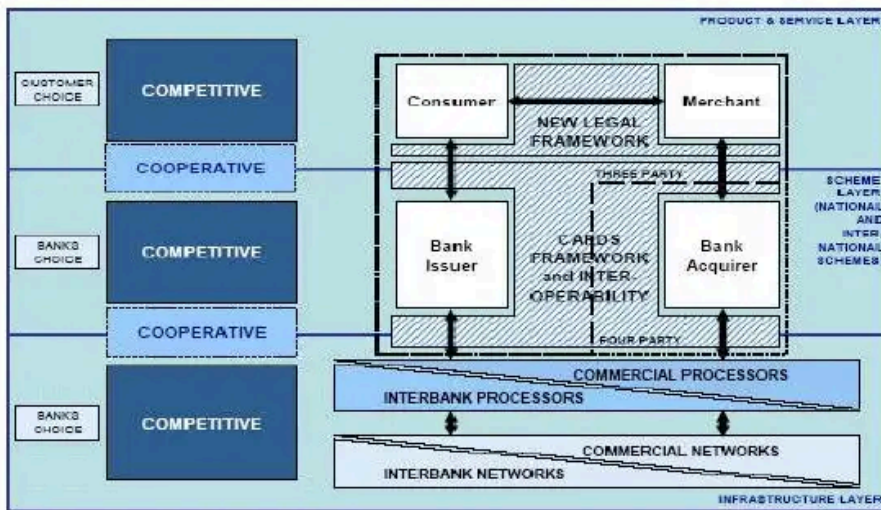


Figure 15.9 SEPA Cards Framework

(Source: European Payment System)

The SCF's objective is that banks, as owners of each of the national card schemes, will commit to the SCF's policy statements and decide how they will adapt to meet the SCF principles.

The Framework covers the use of "general purpose" cards for payments and cash acquisition within SEPA. The Framework defines the environment within which payment card transactions are carried out.

15.14.2 Various Roles within the Card Framework

1. SEPA Acquirer's role.

A SEPA compliant acquirer is one that acquires or and processes general purpose SEPA compliant Euro payment card transactions within the SEPA. Under SEPA acquirers are expected to perform the following:

- Enable a consistent cardholder experience at the POS (Point of Sale) following the transaction flow defined by the relevant scheme.
- Communicate the benefits of, and offer to, their merchants (from 1st January 2008) the acceptance at their terminals of transactions generated by one or more SEPA compliant card schemes.

Table 15.8 SEPA card Framework

The SEPA Cards “Adaptation” Framework – Policy Scope
• Separates scheme governance from underlying processing • Removal of legal, technical and commercial barriers and limitations, both national and EU wide • Open non-discriminatory scheme membership • Single, pan-European licence • Open and transparent pricing policies • EMV implementation: Chip and PIN implementation shift for all cards (national and ICS) by end 2010 • A common approach to fraud and reporting • Standardised processes and procedures for card acceptance and certification • Convergence of scheme rules for common non-competitive aspects • References the Payment Services Directive • Common, consistent experience for all cardholders at POS and ATM • Cardholder choice of card type accepted • Merchant choice of acquirer, scheme and terminal • Interoperability and all merchant SEPA card acceptance

(Source: European Payment Council)

- Remove any barriers that may limit their merchants accepting multiple schemes and co-branded schemes at their terminals.
- If several payment applications are contained in the same card and are supported by their terminal, enable cardholder choice of which payment application they use.
- Promote to their merchants the benefits of migrating to EMV (Europay, Mastercard, Visa) certified terminals with PIN pads.
- Complete the migration to EMV by the end of 2010 (with the exception of the environment in the Netherlands, where existing contractual arrangements with the Dutch merchant community supersede this).
- Implement the SEPA technical standards, including cardholder to terminal interface, card to terminal (EMV), terminal to acquirer interface, acquirer to issuer interface.

2. SEPA Card Issuer Roles. A SEPA card issuer is one that issues general-purpose SEPA compliant cards capable of being used for payments and/or cash acquisition transactions in Euro. SEPA compliant card issuers are expected to perform the following:

- Offer their customers (from 1st January 2008) SEPA compliant payment cards.
- Complete the migration to EMV by 2010.
- By the end of 2010 enable all their general purpose cards to become SEPA compliant.
- Communicate to their cardholders the benefits of SEPA compliant cards.
- Implement SEPA technical standards, including cardholder to terminal interface, card to terminal (EMV), and acquirer to issuer interface.

3. SEPA ATM Owner's Roles. A SEPA compliant ATM owner is one that acquires (or otherwise processes) general purpose SEPA compliant Euro cash withdrawal transactions at ATMs within the SEPA. SEPA compliant ATM owners are expected to perform the following:

- Enable a consistent cardholder experience at the ATM following the transaction flow defined by the relevant scheme.
- Ensure their ATMs accept cards of all schemes of which they are participants.
- Complete the migration to EMV by the end of 2010.
- Enable their ATMs to offer, as a minimum, the national language(s) and English.
- Where several cash withdrawal applications are contained in the same card and are supported by the ATM, enable cardholder choice of which application they will use.
- Implement SEPA technical standards, including cardholder to ATM interface; card to ATM (EMV); ATM owner to issuer interface.

It should be recognised that the acceptance of a card at any given terminal is ultimately dependent on the decision of a merchant to accept that particular card.

15.15. The Single Euro Cash Area Framework (SECA)

The Euro system Bank Note Committee, banks and other key players developed the plans for Single Euro Cash Area. The objective of SECA is to create, with the Euro system, a level playing field whereby the basic cash functions performed by each of the National Central Banks (NCB) in the Euro area are interchangeable i.e. there is a common level of service and common processes are followed by all Euro-area NCB's. SECA seeks to ensure the implementation of harmonized costs of distribution within the SEPA area. The effect will be to create a common infrastructure for cash (banknotes and coins) in all Euro countries. These changes will have a good impact on NCB's, banks, Cash-in-transit [CIT] companies, merchants and corporate. The key design

principles are summarized below. EPC also has work-plan to develop a framework for cash repositioning. This strategy will have the objective of encouraging consumers and merchants to migrate from cash to payment cards and other electronic payment instruments.

Table 15.9 SECA Design Principles

Single Cash Area – Design Principles
• Consistent customer experience at all euro-area ATMs. Homogenous note, coin quality and features • Consistent and homogenous wholesale and retail cash handling standards and processes in all national markets • Improved services to merchants and increased competition among cash-in-transit companies (CIT) • Removal of legal barriers and harmonisation of NCB conditions • Improving the efficiencies of the cash supply chain – reducing number of steps • Improving competitiveness through the use of alternative cash distribution models • Access to any NCB in any national market and a cash supply network that ensures optimal reach • Effective and efficient cash distribution management information systems for NCB's, banks and cash-in-transit providers • Harmonised NCB responsibilities and service level agreements for debiting or crediting, deposits and withdrawals, opening hours, authenticity and quality control. Common rules for NCB bank fees and charges • Stability of services and operational conditions to enable long term investment

(Source: European Payment Council)

15.16. Roles and Responsibilities for SEPA Implementation

SEPA will be successfully implemented based on a solid organisational structure within which all the players and stakeholders understand their roles and can cooperate and interact.

15.16.1 The Role of the EC (European Commission)

- As the political driver of the SEPA concept, the EC, have the critical role of providing the foundations for SEPA by ensuring the approval of the PSD by the European Parliament and European Council.
- The EC holds responsibility for the overall economic integration of the EU countries.
- The EC is also expected to provide “SEPA thought leadership” and encouragement to all the parties as the concept is developed and implemented.

15.16.2 The Role of the ECB [European Central Bank] and the Eurosystem

As a co-initiator of SEPA, the ECB and the Eurosystem focus on the Euro-area. The Eurosystem, coordinated by the ECB, has set the SEPA objectives and the high level statement of requirements.

- It consults with, and draws on, input from the SEPA stakeholder groups.
- The ECB also has the role of monitoring the EPC's progress in designing and specifying SEPA and attends EPC meetings.
- The Eurosystem produces an annual review of progress.
- Through the Eurosystem, the ECB also has the task of stimulating each central bank's role in coordinating national implementation of SEPA.
- The ECB also has a responsibility for SEPA leadership with a special focus on supporting successful standardisation.
- ECB has responsibility for delivery of TARGET2 starting November 2007 (first window), which will be an essential building block for settlement of SEPA payments and for other Euro payments (money market, forex, securities).
- In each national market the national central bank is expected to draw together the banking industry, governments and public authorities and stimulate and support the start-up of national SEPA implementation plans and organisations.
- The Eurosystem is also expected to promote the proposition that European and national public authorities (and their agencies) become "early adopters" of the new payment instruments.
- Each national central bank will also have a pivotal national role in helping resolve misunderstandings, facilitating the simplification of processes and orchestrating the comments of banks, merchants and corporate.

15.16.3 Role of the EPC

- The EPC is the third co-owner of the SEPA concept, with the brief to focus on design, specification and policy relating to SCT, SDD, cards and cash.
- In order to accomplish the ambitious deliverables timetable the four EPC Payment Instrument Working Groups (PIWGs) each have responsibility.
- The EPC has developed a formal SEPA-level programme management function, which will monitor the overall programme and interdependencies, provide a risk and issues management framework and produce consolidated progress reporting.
- The EPC also has a mandate to support the start-up of national SEPA implementation programmes in conjunction with the Eurosystem (see below), national associations and European associations.

- The EPC has also established a Roll-Out Committee (ROC) with the brief to prepare, plan and support the specific roll out of the SCT and the SDD schemes.
- The objectives of the ROC are: to provide guidance to ensure consistent implementation; to build and operate a scheme Rulebook change process; to define processes for the scheme service providers (i.e. CSM's and PE-ACH); resolve any scheme Rulebook or implementation disputes.
- The EPC has developed a communication plan to communicate the SEPA message and programme, and to help co-ordinate and align decision makers and implementers across the SEPA zone. However, it must be emphasised the EPC manages on the principle of self-regulation; it has no brief to be directive.

15.16.4 Role of banks

- Banks are committing to deliver SEPA payment services based on the two new Scheme Rulebooks and on the Cards Framework.
- Banks are responsible for developing their SEPA value propositions (products and services). There should be timely decision making by each bank's executive, plus the allocation of resources and the appointment of SEPA project directors and teams to carry out the essential marketing, product, operational and IT changes.
- Banks will also strongly encourage their national associations and payment scheme organisations to ensure the creation of an over-arching national SEPA implementation structure.
- Internally bank staff and relationship managers are already buying into the SEPA vision and are implementing external communication plans.
- Constructive engagement with customers and lobby groups, governments and other stakeholders, supported by appropriate communication programmes, will continue throughout the next four years.

15.16.5 Role of public authorities

Governments and public authorities will have to provide legislative and regulatory support to SEPA implementation by helping in the removal of blockages and barriers that may restrict the effectiveness of SEPA or delay implementation.

Also, as 'buyers' of payment services, they have a role as "early adopters" by encouraging the implementation of SEPA payment instruments within the government departments, revenue, tax benefits and within public utilities.

15.16.6 Role of users of payment services

- Customers must be persuaded that the new SEPA payment instruments bring value and convenience. Clearly banks must communicate the SEPA message to their customers to encourage them to migrate to IBAN (International Bank Account Number) and BIC (Bank Identification Code) usage.
- Public administrations, large corporate and merchants will have to interface to new systems and for many, build additional infrastructure components. For SME's the majority of the change will be prepared by their banks.
- Merchants in all nations will face changes to their terminals, with new card acceptance applications and protocols.
- There is potential for the creation of partnerships with financial institutions to promote adoption of the SEPA payment instruments.

15.16.7 Role of Payment Infrastructure Providers

- Both payment (card) schemes and clearing houses (ACHs) and operators will see their roles change over the next four years and it is clear that they are essential facilitators to make SEPA a reality.
- For ACH's, the role will be that of migration from a one country only scheme to a Euro-area common scheme.
- Many will need to change their strategies and structures and enable the separation of scheme from the infrastructure.
- Processing operators will see their roles change as they adjust to becoming a PE-ACH or SEPA scheme compliant CSM and servicing national and European customers.
- Similarly, payment (card) schemes and inter-bank processors face changes in their roles, responsibilities and business opportunities.
- These changes can only be effected by working closely with scheme members and owning banks to ensure the necessary changes are implemented smoothly and do not disrupt the implementation of SEPA.

15.16.8 Role of the payments supplier sector

- SEPA implementation is likely to cost the banking sector and corporate and public administrations a very substantial sum in changes to marketing communication, product development, operational processes and IT systems. This enormous change program cannot be achieved without the strong support and involvement of the supplier sector. The EPC expects software vendors, processors, systems integrators and consultancies to support the development

of new cost efficient products and services that will facilitate rapid and low risk conversion and migration to SEPA.

- This will be no easy task for the cards sector in particular, for standards and specifications will take time to be finalised and implemented.
- For the SEPA ETS Schemes, the UNIFI (ISO 20022) XML standards have been specified and the implementation guidelines are in preparation.
- However, suppliers can pro-actively assist the process by supporting their customers, by contributing to the standards-setting process and communicating the SEPA vision and concept to their product and services developers.

15.17. Potential Impacts and Opportunities for Banks

SEPA will create many strategic challenges and opportunities for bank to offer products and services so that it can build new businesses model. SEPA will also deliver benefits to customers, merchants, public administrations and corporate.

For Banks, SEPA is a strategic opportunity, and not just compliance. Individual banks and other payment industry players across SEPA will have to grasp the scale of the opportunity. All banks dealing in retail Euro payments will be impacted whether inside the Euro-zone or not. It is important that banks see SEPA as a business opportunity. Banks need to understand SEPA and the PSD, assess the impact on their revenues, review the impact of competitor's offers and analyse their strengths and weaknesses in the future payments market.

Change will bring many strategic opportunities for banks to innovate, to develop new products, to replace ageing systems and to improve operational efficiencies. Competition amongst banks is likely to increase. Larger banks can offer high volume payments processing products. Smaller banks can compete with larger ones, because one home account can serve customers in multiple countries (private clients, students, pensioners etc). Large and smaller banks can also specialise and develop and deliver products that will service niche sectors.

With change comes both threats and opportunities. Innovative market offerings linked to the benefits of improved efficiencies in service delivery will be essential requirements if all banks are to compete.

With the introduction of SEPA, geography and national markets will no longer be a barrier. Even smaller national and regional banks will have the opportunity to compete on a pan-EU basis – physical presence alone will no longer necessarily be a differentiator. Banks will have to provide new cross-border products and services to Pan-EU corporate and retailers to take advantage of the efficiencies delivered by SEPA. Banks will provide

scale processing offers to Europe's largest corporate. One common account processing engine can be developed to serve the whole SEPA area, reducing the costs of country specific platforms and enabling mergers and acquisitions.

Similarly, acquirers can offer large merchants much improved, multiple nation Pan-EU acquiring services for all payment cards. Common in-house platforms will enable large processors to develop new pan-EU services at lower cost.

Payment sector software suppliers can innovate and develop new interfaces and converters to enable SEPA migration. In addition, the costs of products and services will decline, as vendors develop common EU designs rather than national market specific, reducing bank's operational costs.

All banks will have to assess the impact of SEPA and the (PSD) on their revenue streams and re-evaluate their pricing strategies in the new market context. Increased transparency will reduce cross subsidisation and as a result, payment product prices can be more closely linked to cost. Transparency will enable simpler structures, improve customer clarity, as well as meeting the requirements of regulators and consumer groups.

15.18. Summary

- "SEPA will be the area where citizens, companies and other economic actors will be able to make and receive payments in Euro, within Europe, whether between or within national boundaries under the same basic conditions, rights and obligations, regardless of their location."
- There is a priority implementation focus on the Euro area, currently 12 countries (13 from January 2007), and the change programmed will radically impact their whole domestic payments environment.
- For SEPA development, EPC has covered two different approaches, which are complementary to each other. 1. For Electronic Transfer Schemes (ETS) a "replacement" strategy has been chosen with new common credit transfer and direct debit schemes for the overall SEPA. 2. For the highly complex cards business, the strategy has been that of "adaptation" of existing schemes to a new set of business and technical standards.
- The EPC, working with the ECB, has drawn up a timeline based upon three deliverable phases:
 1. Design and preparation;
 2. Implementation and deployment and
 3. Co-existence and gradual migration.

- The two new Electronic Transfer Schemes (ETS) viz. SEPA Credit Transfer (SCT) and SEPA Direct Debit (SDD) are the deliverables of the European Payment Council [EPC] who developed them from 2004 through to 2006. These schemes are designed to give core information to customers, banks, and infrastructure based on the data accumulated from bank's day to day contact with their customers.
 - The objective of SECA is to create, with the Euro system, a level playing field whereby the basic cash functions performed by each of the National Central Banks (NCB) in the Euro area are interchangeable i.e. there is a common level of service and common processes are followed by all Euro-area NCB's.
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Chapter-16 TARGET

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for associates

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Chapter-16 TARGET

16.1. Introduction

This session is a peek into upcoming European Union-wide Real Time Gross Payment systems, which is expected to change the payment scenario for all the participating European nations. This payment system is christened TARGET. This session also deals with the core functioning of TARGET, its components, its accounts and various modules.

16.2. Learning Objectives

After completing this session you would know

- What is TARGET1& TARGET2?
- What are the new features of TARGET2?
- Various modules OF TARGET2
- Detailed concept of SSP (Single Shared Platform).

16.3. Topics Covered

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16.4. TARGET - An Introduction

The EU-wide real-time gross settlement (RTGS) system, The **T**rans-European **A**utomated **R**real-time **G**ross settlement **E**xpress **T**ransfer system (TARGET) is one among the leading three largest wholesale payment systems in the world, which contributes to the smooth execution of monetary policy and singleness of money market within the EU region.

TARGET came into existence in January 1999, at the same time as the start of European Monetary Union (EMU). TARGET facilitated the conduct of the single monetary policy and the creation of a unified money market in the Euro area. TARGET is owned by the Euro system, which consists of the European Central Bank (ECB) and the 12 national central banks (NCBs) of the Euro area. TARGET has a decentralized structure linking together 17 national real-time gross settlement (RTGS) systems (Some RTGS systems are linked through other RTGS systems) and the ECB payment mechanism. TARGET is an essential initiative for the implementation of the monetary policy for the Eurosystem, and has contributed a lot to create a single money market within the Euro area.

Interesting Facts:

In 2002, TARGET held 85 percent of the EEA's high-value payments market share in terms of value and 59 percent of the volume. Notably, cross border payment volumes in TARGET expanded 19 percent 2002 over 2001 due to an increase in commercial customer payments, which have migrated from correspondent banks to interbank networks. However, the share of cross border payments relative to domestic payments declined from 39 percent to 31.3 percent 2002 over 2001, due to faster growth in domestic payments.



Figure 16.1 Members of TARGET

16.5. Purpose of TARGET

TARGET has been developed to achieve three main objectives:

- To provide a safe and reliable mechanism for the settlement of cross-border payments on an RTGS basis;
- To increase the efficiency of intra-EU cross-border payments; and, most importantly,
- To serve the needs of the ESCB's monetary policy.

The introduction of the Euro enabled internationally-oriented financial and non-financial enterprises in the EU to centralise their treasury operations, which were spread over a number of currencies. By using one currency instead of several, it is possible for all enterprises involved in cross-border activity to make considerable savings.

One pre-condition for optimizing these savings was that the payment systems themselves must be integrated.

All prevalent national currency areas had an integrated payment system at their disposal; it was essential for the Euro area to have the same facility. The successful implementation of the single monetary policy is reflected in a uniform money market interest rate.

The EMU-wide interbank market required, first, that credit institutions had both the incentive and the capability to manage their liquidity positions efficiently and, second, that arbitrage operations can be executed easily and swiftly throughout the Euro area.

This in turn presupposed the existence of an integrated EMU-wide payment system to ensure that liquidity could be transferred from one participant to another in a safe, easy and timely fashion within the new monetary area.

16.6. Principles of TARGET

Market Principle

The use of TARGET is only mandatory for payments directly relating to monetary policy operations (in which the ESCB is involved either on the receiving or on the sending side). The use of TARGET is not mandatory for either interbank payments or commercial payments. However, in order to contain the systemic risk inherent in large-value net settlement systems, all such systems operating in Euro are required to settle through TARGET. TARGET is mandatory for payments directly relating to monetary policy operations.

Irrevocability

The national rules of each RTGS system in TARGET stipulate that payment orders are irrevocable at the latest from the moment when the RTGS account of the sending participant is debited by the sending NCB/ECB for the amount of the payment order.

The moment when the RTGS account of the sending participant is debited by the sending NCB/ECB for the amount of the payment order, the payment becomes irrevocable.

Finality

TARGET provides a firm foundation for the management of payment system risks. It gives participants the possibility of settling payments in central bank money with immediate finality, thus eliminating the settlement risk between participants which is inherent in other payment mechanisms.

16.7. Structure of TARGET

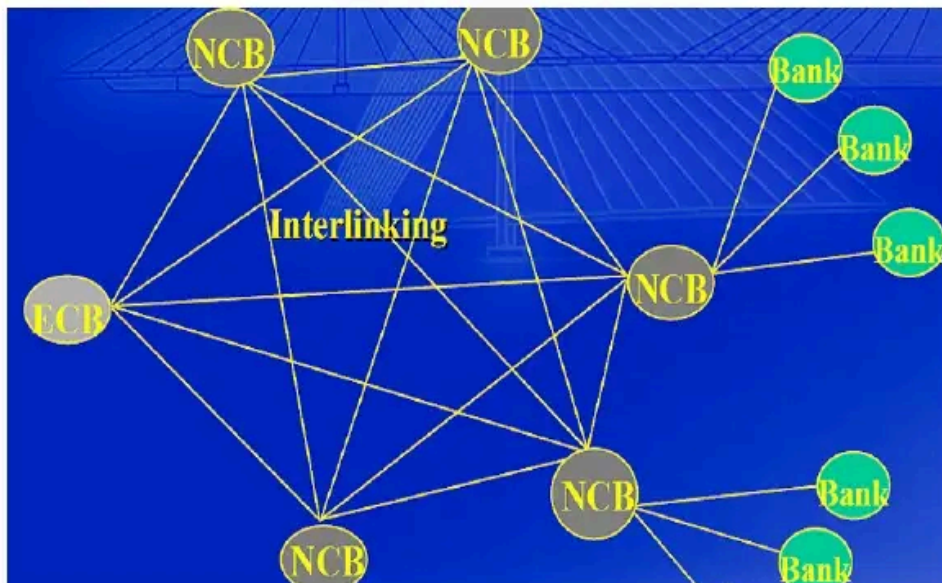
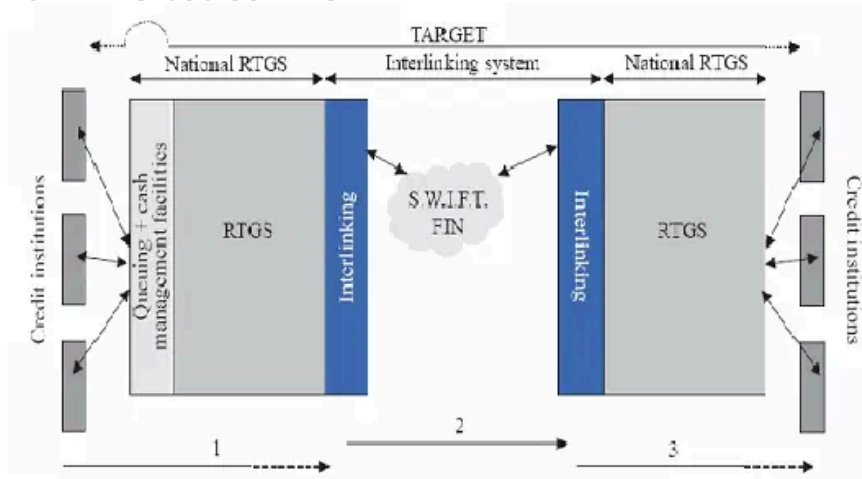


Figure 16.2 Interlinking of RTGSs

TARGET is a decentralised payment system, consisting of national RTGS systems and the ECB payment mechanism (EPM), which are connected to each other by the Interlinking system. The TARGET system makes use of the infrastructures in place in the EU Member States. RTGS system participants' settlement accounts are held at national central banks (NCBs). Individual payment messages are exchanged bilaterally and directly between the two central banks concerned, using reciprocal accounts for debiting and crediting. Only a few overall centralised functions are undertaken by the ECB (e.g. TARGET coordination and end-of-day procedures).

National RTGS systems have retained their specific features to the extent that they are compatible with both the singleness of the Eurosystem's monetary policy and the level playing-field for credit institutions. In addition, they must comply with the "Minimum common performance features of RTGS systems within TARGET", the "Interlinking Specifications" and the "TARGET Security Requirements". To enable the different RTGS systems to process and settle cross-border Euro payments, a number of common features have been agreed upon and implemented in the TARGET Interlinking.

Over and above these technical features, harmonisation is concentrated in four key areas:

- the provision of intraday liquidity for euro area RTGS system participants;
- operating times;
- TARGET cross-border holidays; and
- Pricing policies for TARGET payments.

The RTGS system in each country is represented by the NCB concerned.

The TARGET system is composed of:

- One national RTGS system in each of the EU countries that have adopted the euro. RTGS systems in those EU Member States which were already members of the EU at the start of Stage Three, but which have not yet adopted the single currency are allowed to be connected to TARGET, provided that they are able to process the euro currency alongside the national currency.
- The ECB payment mechanism (EPM).
- The Interlinking system, which interconnects the RTGS systems and the ECB payment mechanism. It consists of an IT system which provides inter-NCB accounts for recording mutual claims and liabilities between NCBs stemming from payment transfers, and a telecommunications network for the real-time transmission of interlinking data.

Each national RTGS system and the EPM is composed of:

- An IT system which provides final and irrevocable debiting and crediting functions along with some optional features such as queue management, gridlock resolution, provision of debit or credit advice; and
 - Telecommunications facilities for the real-time transmission of payment orders and additional information between RTGS system participants and the NCBs/ECB.
- Only NCBs and the ECB – with regard to the EPM – may be members of the TARGET Interlinking.

16.8. Participants in TARGET

National provisions regulate the relations and operations between NCBs and their RTGS system participants.

Credit institutions access TARGET via RTGS systems participating in or connected to TARGET. The interface with credit institutions is determined by the NCB concerned. In order to send and receive cross-border euro payments, credit institutions must either be:

- Direct participants in an RTGS system or be represented by a direct participant (in the latter case the institution is sometimes referred to as an “indirect participant”); or
- They must be customers of a participant or of an NCB connected to TARGET.

Credit institutions which are direct participants in an RTGS system maintain settlement accounts at the NCB responsible for this system. There are other participants in a variety of legal and technical situations where credit institutions which are not direct participants but which may hold an account at the NCB (albeit not an RTGS settlement account) can nevertheless access and receive funds from TARGET. They may choose to participate in their local RTGS system and/ or to participate as a remote access participant in another RTGS system. All TARGET RTGS system participants must be identified by a unique published Bank Identifier Code (BIC).

TARGET promotes STP. To ensure that validation takes place early on in the payment process, it was necessary to define a common reference which could be used by all TARGET RTGS systems and also by banks using TARGET.

For this reason it was decided to use BIC addresses published by S.W.I.F.T. as the means of identifying the sender bank, receiving bank and, where appropriate, the intermediary bank. Although the BIC is used to identify banks, it does not imply that the banks have to use S.W.I.F.T. as their message carrier.

BICs are listed in the widely available "SWIFT BIC Directory" of published codes, which is available both in electronic and in paper form.

In cooperation with S.W.I.F.T., the ECB has published the TARGET Directory, which lists those credit institutions which are addressable through TARGET. They are listed alphabetically by name of institution and are also sorted by BIC address.

16.9. Role of SWIFT

The logical and physical technical platform for the TARGET Interlinking is based on the SWIFTNet FIN network. S.W.I.F.T. has been selected as the preliminary network provider for the TARGET Interlinking. The national RTGS system user interface may also be SWIFTNet based or may use another network provider.

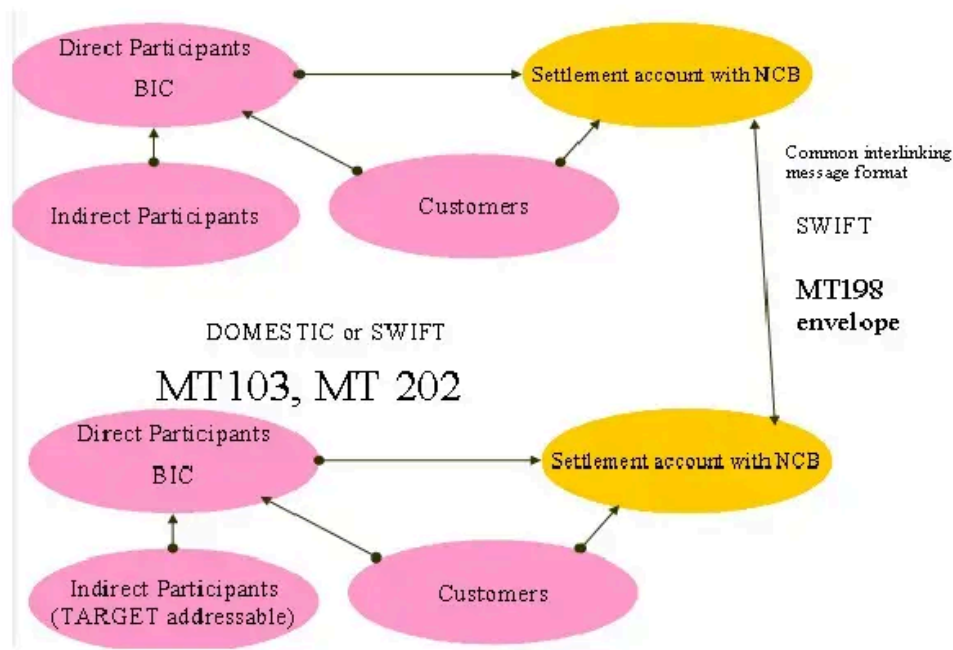


Figure 16.3 Message flow via SWIFT

Participant banks send payment instructions using domestically agreed message formats to their RTGS system. NCBs exchange messages through the Interlinking using the Common Interlinking Message formats.

Payment messages are mapped to an MT103 – standard or straight-through processing (STP) – or an MT202 Interlinking specification message. The messages are enveloped in an MT198 proprietary message format by the sending NCB for sending via the Interlinking. They are then unwrapped by the receiving NCB and forwarded to the final beneficiary.

16.10. Payments in TARGET

TARGET processes customer and inter-bank cross-border payments in Euro. Data in TARGET payments are transmitted without any alteration of the content from end to end, although the order in which data appear may be altered slightly, because payment messages are mapped from domestic formats to TARGET Interlinking formats and vice versa.

If, for any reason, a beneficiary is unable to apply a payment order that it has received via TARGET, there is a mechanism which provides information so that the payment can be returned by the receiving RTGS system participant to the TARGET participant which introduced it into TARGET.

TARGET has been designed to support STP to the greatest possible extent. Thus, TARGET participants must address cross-border payments directly to the receiving bank in its RTGS system (the receiving RTGS system). It means payments to be made to TARGET participants should include the published BIC of the credit institution concerned in the first credit field of a SWIFT payment message (and not the BIC of the NCB through the RTGS system of which the participant accesses TARGET).

16.10.1 Customer Payments

Customer payments are defined as payments in the SWIFT MT103 format, standard or STP, or equivalent national message formats. Customer payments can be transmitted via TARGET from 07.00 until 17.00. The 17.00 cut-off time applies to both domestic and cross-border customer payments.

The participants send their payment orders to the Euro RTGS system in which they participate using the SWIFT MT103 formats (standard or STP), or the equivalent national message format for TARGET payments, which is converted by the RTGS system into the common Interlinking message format.

The Interlinking uses SWIFT proprietary MT198 messages as its message framework. MT198 messages are “envelope” messages which contain free space that can be formatted by the users (the NCBs/ECB in TARGET) according to rules agreed bilaterally.

S.W.I.F.T. does not validate the content of the envelope, except to verify that the characters used belong to the SWIFT character set, that the length of each line does not exceed 78 characters and that the total number of characters in each message does not exceed 10,000.

16.10.2 Inter-bank Payments

Interbank payments are defined as payment messages in the SWIFT MT202 format or equivalent national message formats for TARGET payments. This type of message is sent by or on behalf of the ordering institution either directly or through any correspondent(s) to the financial institution of the beneficiary institution.

Interbank payments are payments, such as the payment leg of money market, foreign exchange and derivatives transactions, which take place between credit institutions or between NCBs/ ECB and credit institutions. Interbank payments can be transmitted via TARGET from 07.00 until 18.00.

The MT198 sub 202 General Financial Institution Transfer is used to transmit an interbank payment message through the interlinking. The sending and receiving RTGS systems process interbank payments in the same manner as that outlined for customer payments.

16.10.3 Validation of Customer and inter-bank Payments

The sending RTGS system checks the syntax of the data according to the appropriate standard, the value date of the payment order (the only value date for the TARGET system is "today") and the availability of the receiving NCB/ECB.

If syntax errors or other grounds for rejection are detected, the sending NCB/ECB handles the data according to domestic rules and returns the error information in the agreed domestic format.

The receiving RTGS system checks all those parts which are necessary to successfully credit the account of the beneficiary institution in line with its own national provisions. If the crediting of the beneficiary institution in the RTGS system is successfully completed, the receiving NCB/ECB sends a positive acknowledgement to the sending NCB/ECB. In the event that an error occurs during the processing and it cannot be completed, a negative acknowledgement is sent to the sending NCB/ECB stating the reason.

These reasons can be either the absence of any information by which to identify the receiving institution or an invalid format. The sending NCB/ECB then handles the data according to domestic rules.

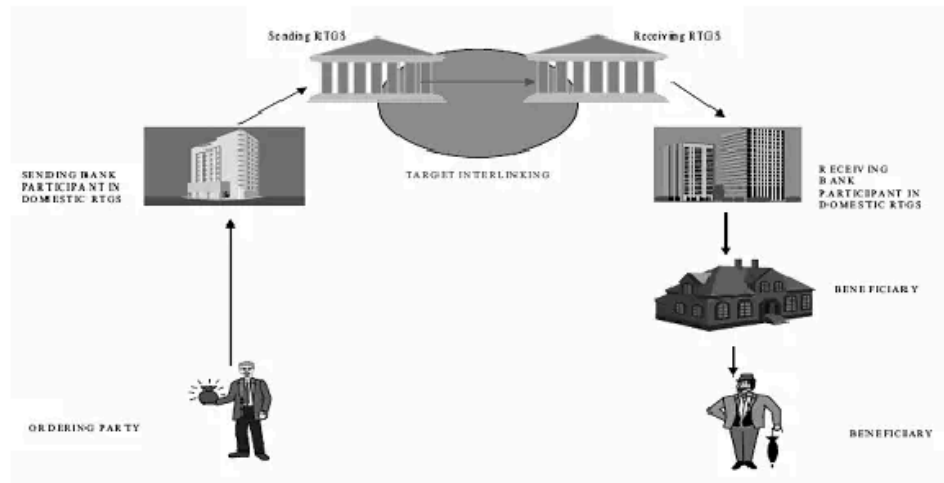


Figure 16.4 Cross-border Euro customer payment through TARGET

16.11. From TARGET1 to TARGET 2

TARGET has a number of shortcomings that stem from its heterogeneous technical design. In view of this, and because of developments such as the future enlargement of the Euro area, the Eurosystem is currently building the TARGET of tomorrow i.e. TARGET2. The new system will achieve even higher levels of safety and efficiency through a harmonized structure based on a common technical platform.

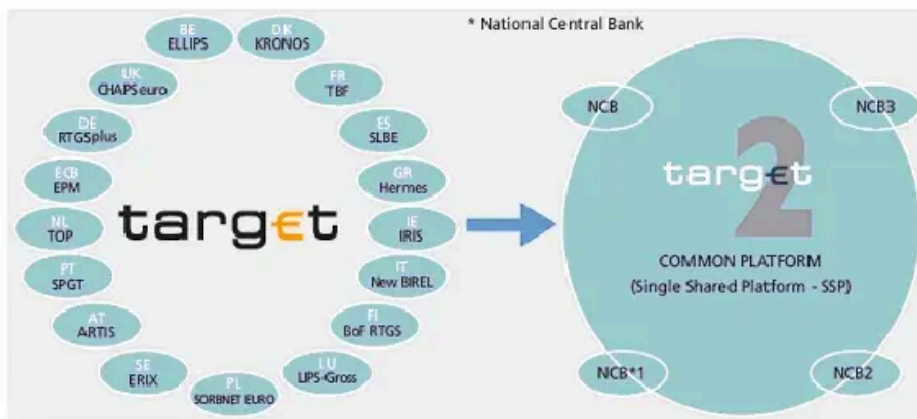


Figure 16.5 From star (TARGET) to wheel and spoke (TARGET2) topology

On 17 June 2005, the Governing Council of the ECB communicated to the market that the period of extensive user consultation had concluded and that the go-live date for the first migration window will be 19 November 2007.

The Euro system has agreed to split the migration into four waves (with the last wave being reserved for contingency purposes only) and has decided on the composition of the migration groups.

Group 1	Group 2	Group 3	Group 4
19 Nov. 2007	18 Feb. 2008	19 May 2008	15 Sep. 2008
Austria	Belgium	Denmark	<i>Reserved for contingency</i>
Cyprus	Finland	Estonia	
Germany	France	ECB	
Latvia	Ireland	Greece	
Lithuania	Netherlands	Italy	
Luxembourg	Portugal	Poland	
Malta	Spain	United Kingdom	
Slovenia			

16.12. Concept of TARGET2

In October 2002, the ECB announced the move to the next generation of TARGET, TARGET2, which is to be based on the EU's most efficient RTGS systems. It declared that any system that does not meet TARGET2's cost recovery requirements in 4 years will be closed. TARGET2 is to improve customer needs by offering core service levels, which will be defined in close cooperation with the user community, and cost efficiencies that will arise from a less fragmented IT infrastructure. Individual central banks will be able to offer specific, national services in addition to the core service levels. TARGET2 will consist of national components and a shared IT platform component that central banks can use on a voluntary basis. Although plans are for a single price structure for euro payments, volume based pricing or pricing based on timing of payments may be also offered. It is estimated that TARGET2 will be operational in the second half of this decade. A number of user groups meet regularly and discuss the requirements for the current and future TARGET system.

Thus, TARGET2 is the future real-time gross settlement system of the Euro system. It is based on a single platform infrastructure, i.e. the entire application is based on an integrated central technical infrastructure (so-called "**Single Shared Platform**" (SSP). Banca d'Italia, Banque de France and Deutsche Bundesbank are co-operating on the development of the new payment system.

In order to allow for a smooth migration from TARGET to SSP it is foreseen to support the current Interlinking logic also at SSP level, until the last migration window is closed, in order to allow a gradual migration of countries in several waves. Thus, Central Banks' will not have to migrate to the new environment "in one go". After the migration period is finished the Interlinking functionality will be removed.

From a user's perspective, TARGET2 offers a broad range of features and services to meet the requirements of all users (European banking industry, NCBs and ECB). TARGET-2 has following features (some of them new and some old with improvement):

- Homogenous Technical Design/ A Single Technical Platform
- TARGET-Wide Flexible Liquidity Management Services
- Support For Payments With a Debit Time Indicator
- Pooling of Intraday Liquidity
- Interaction With Ancillary Systems
- Strengthened Business Continuity Measures
- Information & Control Module
- TARGET2 Directory
- Operational Day.

16.13. Objectives of TARGET 2:

With the TARGET2 project the Eurosystem is pursuing the following main objectives:

1. Fulfilling the European banking industry's user requirements (eg Europe-wide liquidity management).
2. Compliance with the neutrality principle. TARGET2 should lead to a level playing field for banks and market infrastructures in the respective countries.
3. Provision of a harmonised level of service on the basis of a common technical platform.
4. Applying a single price structure that is applicable to both domestic and cross-border payments.
5. Safeguarding the principle of decentralisation within the ESCB. The participating central banks will retain responsibility for conducting business with their customers.
6. High degree of flexibility through a modular approach in order to accommodate the various interests of the participating central banks.
7. High level of standardization.
8. Benchmark-setting provisions for business continuity.

16.14. Features of TARGET-2 Explained:

1. A single technical platform -Probably the most important innovation is the consolidation of the technical infrastructure. TARGET2 will replace the decentralised technical structure of the current TARGET system with a single technical platform, known as the Single Shared Platform (SSP). This was also the overriding user requirement to emerge from the public consultation on TARGET2 held in early 2003.

2. TARGET-wide flexible liquidity management services- Payment system users are increasingly demanding enhanced liquidity management services and liquidity-efficient systems that combine the advantages of both gross and net settlement systems. TARGET2 will respond to these demands by offering both state-of-the-art liquidity management tools and liquidity-saving features.

3. Strengthened business continuity measures- TARGET2 will respond to the new threats in the post-September 11 world by offering the highest possible level of reliability and resilience, as well as sophisticated business contingency arrangements which are commensurate with the systemic importance of the TARGET infrastructure.

4. Integration with ancillary systems- TARGET2 will provide cash settlement services in central bank money for all kinds of ancillary systems (ASs), including retail payment systems, large-value payment systems, foreign exchange settlement systems, money market systems, clearing houses and securities settlement systems (SSSs). The main advantage of TARGET2 for ASs is that they will be able to access any account on the SSP via a standardised interface. TARGET2 will offer six generic procedures for the settlement of ASs (two real-time procedures and four batch procedures), which represents a substantial harmonisation of current practices.

5. Support for debit time indicator- TARGET2 will take into account the increased time-criticality of payments, particularly in the context of CLS, by making it possible to submit transactions with a debit time indicator.

6. Pooling of Intraday Liquidity- Liquidity pooling will be achieved by grouping a number of accounts. TARGET2 will offer two variants for liquidity pooling: i) the virtual account; and ii) consolidated information. In the virtual account option, a payment order submitted by a participant belonging to a group of accounts will be settled if the payment amount is smaller than or equal to the sum of the liquidity available on all accounts (including any credit lines) in the group. Otherwise the payment order will be queued. The consolidated information option is an information tool: it will give comprehensive information to the participant subscribing to the service about the liquidity position of all of the entities of the group at any given moment. Such information will also be provided in the virtual account option. However, payment amounts will be checked only against the liquidity available on the individual RTGS account of the sending participant. The liquidity available on other accounts in the group will not be used to settle the payment. In the event of insufficient liquidity on the sending bank's account, money will need to be transferred to that account.

Only credit institutions established in the European Economic Area (EEA) and directly participating in the system will be able to use the consolidated information option. Furthermore, owing to business and legal constraints, the virtual account option will

only be available for accounts of euro area banks held with euro area central banks. It will only be possible to establish a group of accounts (for the consolidated information or virtual account options) for credit institutions that belong to the same group.

7. Strengthened Business Continuity measures- TARGET2 will offer the highest possible level of reliability and resilience, as well as sophisticated business contingency arrangements commensurate with the systemic importance of the TARGET2 infrastructure.

The business continuity concept of TARGET2 consists of a multi-region/multi-site architecture. There will be two regions. In each region, there will be two sites some distance from each other. This will be combined with the principle of region rotation in order to ensure the presence of experienced staff in both regions.

8. Information and Control Module- TARGET2 users will have access, via the information and control module (ICM), to comprehensive online information and easy-to-use liquidity control measures appropriate to their business needs. Users of the ICM will be able to choose what information they receive and when. Urgent messages (e.g. system broadcasts from central banks and warnings concerning payments with a debit time indicator) will be displayed automatically on the screen. Through the ICM, TARGET2 users will have access to the payments module (PM) and the static data (management) module. Depending on the decision of the relevant central bank with regard to the use of the optional modules offered by the SSP, participants may also have access to the home accounting facility of the central banks and the applications for reserve management and standing facilities.

9. TARGET2 Directory- The TARGET2 directory will contain information on each institution that can be addressed in the TARGET2 system, and will be updated on a weekly basis to support system participants in their routing of payment instructions. The directory will use TARGET2-specific information provided by TARGET users during the SSP registration process in combination with SWIFT-related information. The TARGET-2 directory will be an electronic product/service provided to the direct participants by the Eurosystem.

10. Operational Day- In order to better meet users' business needs, the operational day in TARGET2 will be longer than that of the current TARGET system. TARGET2 will start the new business day on the evening of the previous day. The night-time window² will be available from 7.30 p.m. to 6.45 a.m. the next day, with a technical maintenance period of three hours between 10 p.m. and 1 a.m. The night-time window will facilitate the night-time settlement of the different ancillary systems in central bank money with finality, and will also support cross-system settlement during the night. Settlement of ASs will take place in dedicated accounts. During the night-time window, liquidity

transfers via the ICM between RTGS accounts and the dedicated sub-accounts will be possible. Ancillary systems and their participants will be able to choose whether or not to enable this liquidity transfer functionality, or to limit the functionality. Banks may alternatively decide not to participate in night-time settlement. The Eurosystem believes that the night-time window will generally increase the efficiency of night-time settlement and will favour initiatives such as cross-system delivery versus payment.

16.15. Single shared Platform

The following modules are implemented on SSP:

- Contingency Module (CM)
- Home Accounting Module (HAM)
- Information and Control Module (ICM)
- Payments Module (PM, including the interface for ancillary systems)
- Reserve Management (Module) (RM)
- Standing Facilities (Module) (SF)
- Static Data (Management) Module (SM).

We will see functionality of each module in subsequent sections.

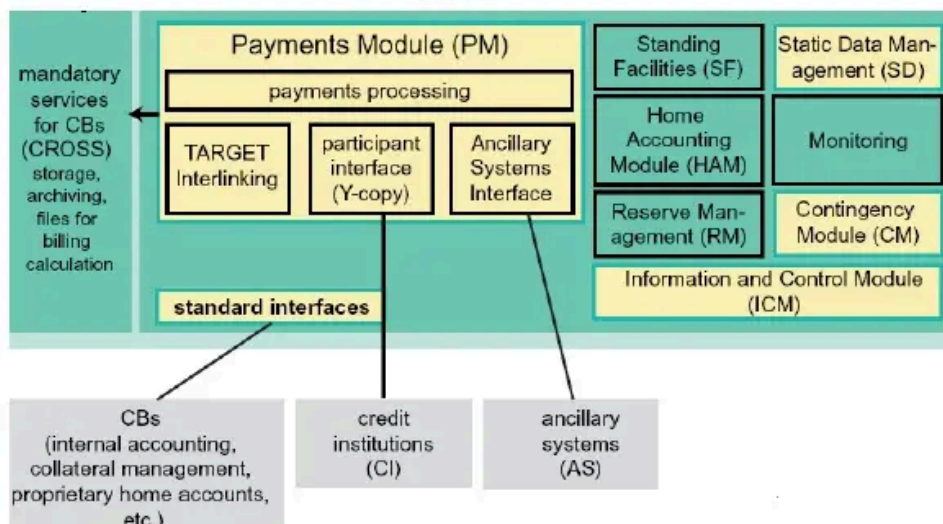


Figure 16.6 Single Shared Platform (SSP) depicting modules

16.16. Participants of TARGET 2

Direct participant

Direct participants have

- 1) direct access to the Payments Module (PM)
- 2) hold an RTGS account in the PM
- 3) access to real-time information and control measures.

The following institutions are eligible to be direct participants in TARGET2:

- Supervised credit institutions
- Treasury departments
- Public sector firms
- Investment firms
- Organisations providing clearing or settlement services
- Central banks.

Indirect participant

- are registered in the PM through participants with direct access
- are directly linked to one direct participant only (which may be located in another country)
- can be indirectly addressed in the PM
- have no own RTGS account within the PM
- must have published BIC.



Table No. 16.1 Direct and Indirect Participants

Feature	Direct participant	Indirect participant
sending and receiving payments	- directly using an own SWIFT interface - directly via a service bureau	via direct participant
own RTGS account	yes	no
liquidity provisioning	on its RTGS account	by direct participant
liquidity control	by itself	by direct participant
access to ICM	yes	no
addressability	directly	by direct participant
BIC	8 or 11-digit published SWIFT-BIC	8 or 11-digit published SWIFT-BIC or non-SWIFT-BIC
publication in BIC/TARGET2 directory	as a direct participant (TGT)	as indirect participant (TG+)

Multi-addressee access

In the TARGET2 system, direct participants will be able to authorize their branches and credit institutions belonging to their group, located in EEA (European Economic Area) countries, to channel payments through the direct participant's main account without its involvement by submitting/receiving payments directly to/from the system. This offers affiliate banks or a group of banks efficient features for liquidity management and payments business.

The payments will be settled on the main account of the direct participant.

Table No. 16.2 Direct and Multi-addressee Participants

Feature	Direct participant	Multi-addressee access
sending and receiving payments	- directly using an own SWIFT interface - directly via a service bureau	- directly using an own SWIFT interface - directly via a service bureau
own RTGS account	yes	no
liquidity provisioning	on its RTGS account	by direct participant
liquidity control	by itself	by direct participant
access to ICM	yes	no
addressability	directly	directly
BIC	8 or 11-digit published SWIFT-BIC	8 or 11-digit published SWIFT-BIC
publication in BIC/TARGET2 directory	as a direct participant (TGT)	-

Difference between indirect participant and Correspondent BIC:

Following not available to Correspondent BIC

- Settlement Finality Directive
- Compensation scheme.

16.17. Directories of the participants

TARGET-2 directory

1. The directory of TARGET2 contains the needed routing information for TARGET2 participants
2. Directory is updated weekly.
3. In the TARGET2 directory , details of BIC to be used in SWIFT header for receiver is also included
4. All the related Information on non-migrated participants is available in the directory
5. The routing logic between SSP users and a SSP user to a non-migrated participant is exactly the same. [Y- copy]
6. In case of a non-migrated participant, the TARGET2 directory delivers the *SSP Interlinking BIC*, which must be addressed in the SWIFT header instead of the receiver BIC .

BIC directory

1. BIC directory is updated monthly.
2. The BIC directory shows all global SWIFT participants and the payment system (s) to which they are connected.
3. For indicating direct and indirect SSP participation worldwide the respective TARGET service code (TGT or TG+) is mentioned for each SSP participant.

16.18. TARGET2 & its Modules

16.18.1 Payments Module (PM)

The Payments Module (PM) gathers all services related to the processing of payments. All direct participants (e.g. credit institutions, market infrastructures, other participants and CBs) have to maintain an account managed within the PM (called as RTGS account). All transactions submitted to and processed by the PM are settled on these RTGS accounts.

In addition to the services related to the processing of payments (settlement including optimization procedures, queue management, etc.), the PM offers advanced services for liquidity management (limits, reservation of liquidity, etc.), for the communication with participants (participant interface for credit institutions based on the Y-flow value-added service) and ancillary systems (ancillary system Interface with six generic settlement models based on SWIFT Net XML standards).

16.18.2 Information and Control Module (ICM)

The Information and Control Module (ICM) allows participants to access all information related to their accounts and use control measures through a comprehensive on-line

information tool. In particular, the participants benefit from a "single window access" to the Payments Module (PM), Static Data (Management) Module (SD) and depending upon whether the relevant central bank has decided to use optional services, to the Home Accounting Module (HAM), the Reserve Management (Module) (RM) and Standing Facilities (Module) (SF).

The ICM enables direct participants to control and manage actively their liquidity and payment flows (visibility of incoming and outgoing payment queue).

The ICM allows access to data of the current business day only, which are available through "pull mode". This mode gives each user the flexibility to decide what information should be updated and at what time.

Finally, the ICM can be accessed either through application-to-application (A2A) mode or user-to-application (U2A) mode.

16.18.3 Reserve Management (RM) and Standing Facilities (SF)

With regard to Reserve Management (RM) and Standing Facilities (SF), the choice to adopt standardised SSP modules or to manage them locally is up to the individual CB. For the local management, specific applications have to be in place at CB level. CBs who opt for these standardised SSP modules can offer the following services to their users:

Reserve Management Module (RM)	Standing Facilities Module (SF)
• Receiving end-of-day balances • Monitoring of the running average in the current reserve period • Calculation and settlement of interest to be paid (for minimum reserves) or penalties that banks have to pay (non-fulfilment of reserve requirement)	• Management of – Overnight deposit accounts – Marginal lending accounts • Transfer of liquidity to the overnight deposit account • Granting of overnight credit either – "on request" – automatically, if intraday credit remains at the end-of-day. • Calculation and settlement of interest to be paid by the CB (overnight deposit facility) or by the banks (marginal lending facility) • Automatic repayment of the deposit or the overnight credit

16.18.4 Home Accounting Module (HAM)

Need for Home Accounting

Direct TARGET2 participants will have to maintain an RTGS account in the Payments Module (PM). Nevertheless, each CB is free to maintain so-called additional "home accounts". This "home accounting" functionality could be used for the following reasons:

- Some banks may not be interested to participate directly in the RTGS system, but are nevertheless subject to minimum reserve requirement. In addition, they may wish to directly manage cash withdrawals etc. Therefore, they need an account with the CB outside the RTGS system.

- In some cases, depending on the specific situation in each country, it may be preferable to have a second set of accounts. This could be used to specific operations of direct participants, which already have an RTGS account. Some ancillary systems might, for example, decide not to migrate from the start to the SSP, but maintain - for the time being - a local infrastructure.

Each CB is fully responsible for the execution of its home accounting business.

In this context, each CB is also free to choose:

- Either to offer proprietary home accounting services or to rely only on TARGET2, if there is a need to offer such services.
- For what type of business the home accounting application is used.

Proprietary Home Accounting Module-

In this case, the service offered to its banks contains a limited number of well-defined transactions, but never fully-fledged payment services.

Home Accounting Module

The SSP also offers a standardised Home Accounting Module (HAM). This module is not intended to offer real payment services. These activities must be performed through a direct PM participant.

This is to accommodate the needs of those central banks wishing to use commonly shared resources to a larger extent. The HAM allows to manage accounts for financial institutions as well as accounts for CB customers not allowed, according to the TARGET Guideline, to open accounts in the PM (hereafter referred to as "CB customer's accounts").

The following standardised account services to its customers may be offered through the HAM by central banks that opt for the use of this module:

- Liquidity holding (e.g. maintaining reserve requirement either through RM or proprietary application) on an account with the CB
- Interbank transfers between accounts in the HAM held by the same CB
- Interbank transfers between the HAM accounts and RTGS accounts of direct participants
- Cash withdrawals with the respective CB
- Access to standing facilities either through the SF or through a proprietary application managed by the CB.

There is also the possibility that the HAM account is managed by a PM participant (the so-called co-manager). The aim of the co-management function is to allow small banks to manage directly their reserve requirement, but to delegate cash flow management to other banks.

The co-management is also available on a cross-border basis. The HAM is technically integrated in the SSP and can be accessed by customers via SWIFTNet.

16.18.5 Static Data Management

The Static Data Management Module provides the SSP operator, CBs and SSP participants with a homogeneous set of data thanks to:

- A unique point for the creation, modification and deletion of static data.
- Daily data transmission for all the SSP modules including the provision of a coherent procedure according to which the data are synchronously loaded into all other modules at the same point in time.
- Exceptional procedures for urgent updates of all the modules in case of emergency.
- For audit trail purpose the central repository keeps the "tracking" for all data updates (date and reference).
- Static Data module is not meant to manage the past versions of static data.
- Possibility to enter changes that become effective at a future date, including versioning facilities.

Users

Users of the Static Data Module are:

- CBs for consultation and updates
- Credit institutions for consultation and updates
- Ancillary Systems for consultation only
- SSP operators for consultation and updates

Controls and responsibilities for data

The entity in charge of data modifications is the one which is responsible for the data:

- CBs for all data related to the participant's structure and AS
- SSP operator for production data such as opening/closing time, calendar, etc.
- Credit institutions for specific credit institution data.

This respective entity is also responsible for the manual controls to be carried out in context with possible changes. Controls consist of both automated and manual procedures. Automated controls on the data format have to be strictly applied to the data before it is used in the production environment.

When information has to be checked against "subjective" elements, manual controls are carried out. In order to safeguard the integrity of data, manual controls and changes cannot bypass automated controls.

Four eyes principle can be enforced for any static data creation and updates. Therefore each entity (in most cases CB) in charge of static data management will be able to

define which data creation/updates are critical and should therefore be subjected to four eyes principle.

Critical static data are all data needed for processing payments in the PM and HAM. It includes participant's public data, participant's private data, ancillary system data, specific CIs data and SSP data.

Static Data Description

Data managed via ICM are represented by:

- the participant's structure (credit institutions and central banks)
- the AS (Ancillary Systems)
- specific data for SSP, such as TARGET calendar, monitoring data, etc.
- specific credit institution's data, such as default limits, standing orders and direct debit

According to the general principle regarding responsibilities on data ("The entity in charge of data modifications is the one which is responsible for the data"), CBs are responsible for all data related to the participants structure and ASs which are in their area of competence. They are in charge of the update of data, integrity and controls.

In specific circumstances, SSP operators are able to act on behalf of a CB. That's why SSP operators have full rights to update all data of the participants' structure.

Similarly, SSP operators are in charge of the update of production data.

For specific credit institution data, only CIs (Credit institutions) are responsible for their data, but SSP operators and CBs are able to update those, on behalf of a CI.

16.18.6 Contingency Module

The Contingency Module (CM) is a common mandatory tool for each CB joining the SSP. The CM runs in the non-active region. It is an independent module which includes all the functions needed to access the SWIFTNet services. Considering the high level of resilience provided by the SSP, the use of the CM is only envisaged for the processing of (very) critical payments in specific situations. These are:

- Unavailability or inaccessibility of the SSP components.
- The time needed for the activation of the alternate site/region lasts too long.

The CM ensures the processing of a limited number of (very) critical payments. The concept of (very) critical payments in TARGET2 defines payments which if processed with a delay could cause systemic risk.

16.19. Summary

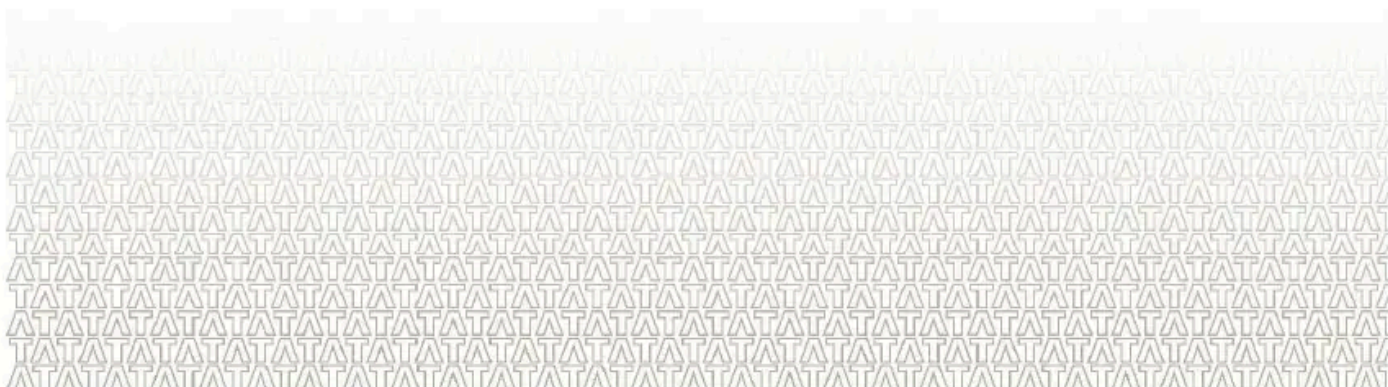
- The EU-wide real-time gross settlement (RTGS) system, The Trans-European Automated Real-time Gross settlement Express Transfer system (TARGET) is one of the leading among three largest wholesale payment systems in the world

- TARGET has a decentralized structure linking together 17 national real-time gross settlement (RTGS) systems (Some RTGS systems are linked through other RTGS systems) and the ECB payment mechanism.
- The logical and physical technical platform for the TARGET Interlinking is based on the SWIFTNet FIN network. S.W.I.F.T. has been selected as the preliminary network provider for the TARGET Interlinking.
- TARGET processes customer and inter-bank cross-border payments in euro. Data in TARGET payments are transmitted without any alteration of the content from end to end, although the order in which data appear may be altered slightly, because payment messages are mapped from domestic formats to TARGET Interlinking formats and vice versa.
- TARGET2 is the future real-time gross settlement system of the Euro system. It is based on a single platform infrastructure, i.e. the entire application is based on an integrated central technical infrastructure (so-called "Single Shared Platform" (SSP). Banca d'Italia, Banque de France and Deutsche Bundesbank are co-operating on the development of the new payment system.
- TARGET-2 has following features (some of them new and some old with improvement):
 - o Homogenous Technical Design/ A Single Technical Platform
 - o TARGET-Wide Flexible Liquidity Management Services
 - o Support For Payments With a Debit Time Indicator
 - o Pooling of Intraday Liquidity
 - o Interaction With Ancillary Systems
 - o Strengthened Business Continuity Measures
 - o Information & Control Module
 - o TARGET2 Directory
 - o Operational Day
- The following modules are implemented on SSP of TARGET2:
 - o Contingency Module (CM)
 - o Home Accounting Module (HAM)
 - o Information and Control Module (ICM)
 - o Payments Module (PM, including the interface for ancillary systems)
 - o Reserve Management (Module) (RM)
 - o Standing Facilities (Module) (SF)
 - o Static Data (Management) Module (SM)

1. TARGET is an essential instrument for the implementation of the monetary policy for the Eurosystem, and has contributed a lot to create a single money market within the euro area.
 2. TARGET has been developed to achieve three main objectives:
 - To provide a safe and reliable mechanism for the settlement of cross-border payments on an RTGS basis;
 - To increase the efficiency of intra-EU cross-border payments; and, most importantly,
 - To serve the needs of the ESCB's monetary policy.
 4. TARGET is a decentralised payment system, consisting of national RTGS systems and the ECB payment mechanism (EPM), which are connected to each other by the Interlinking system.
 5. Credit institutions access TARGET via RTGS systems participating in or connected to TARGET.
-







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Chapter-5 Process of MICR Check-clearing

This session discusses cheque and its clearing process details.

5.1 Learning Objective

- Discusses all the processes the cheque goes through to clear the payments
- Discuss the Importance of the numbers present on the cheque, how are they written & identified through the various channels.

5.2 Topics Covered

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5.3 Introduction

Paper based instruments like cheques, demand drafts are very important constituents of a nation's payment systems. Hence it is very important to know how the paper based instruments are cleared and settled. With the growth in trade and commerce the volume of the instruments like cheques and demand drafts have also increased astronomically over past couple of decades.

Did you know? For the year 2005 the value of total transactions done by cheques in US, UK, Japan and Canada were 302, 137, 105 and 269 times, respectively, of their respective GDPs (Gross Domestic Product - The market value of all final goods and services produced within a country in a given period of time, for example GDP of USA in the year 2005 was 12.98 trillion \$). Courtesy www.bis.org

However, with the advent of payment cards, paper-based instruments have seen a decline but still they are an integral part of the payment system. The introduction of MICR (Magnetic Ink Character Recognition) technology brought about a revolution in cheque clearance and settlement during the mid eighties and early nineties. Although other revolutionary techniques like E-check and Cheque Truncation are now in the phases of trial and testing, MICR forms the backbone of cheque clearing mechanisms in India and rest of the world.

This session aims at imparting knowledge about what are the basic concepts behind the settlement of cheques along with a discussion of MICR technology in Indian scenario.

5.4 The Basics of Cheque Clearing Process

- The customer (payee or holder) deposits the cheque that he has received from the Payer drawn in his favor with his Bank.
- The payee's bank processes the cheque and sends it to the clearing house along with other cheques that it has received for clearing purposes.
- At the clearing house the cheques are sorted bank-wise and all the banks (drawee banks) collect the cheques that are drawn on them (the cheques for which they have to make payment).
- The drawee banks process the cheques and validate them for details like signature, availability of amount in the customer's account and so on.
- The Drawee banks communicate the fate, (i.e. if the payer's account has sufficient funds to honour the payment or not, if the signatures matching or not), of the cheques to the clearing house within the time frame as specified by the clearing house.
- With this information, the payee banks either credit the customer's account with the cheque amount or return the unpaid cheque to the customer.

The Diagram below depicts the cheque clearing process:

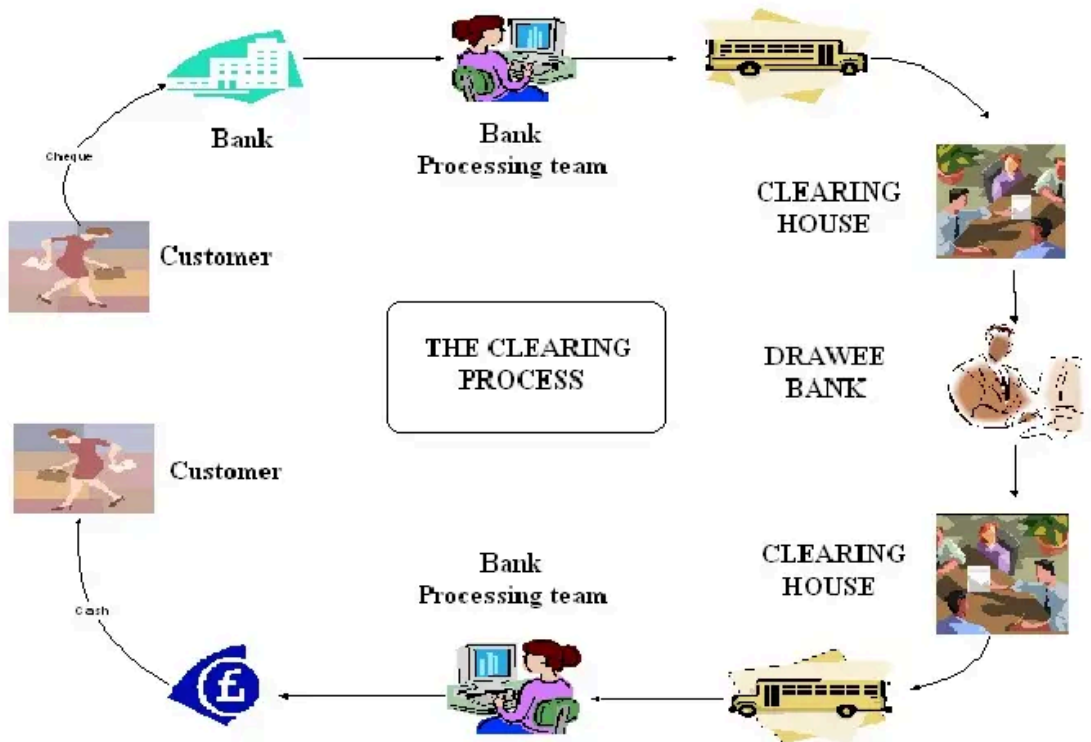


Figure 5.1 Process of Cheque Clearing

5.5 Mechanized Cheque Processing based on MICR- an introduction

It is defined as the machine recognition of numeric data printed with magnetically charged ink. It is used on bank cheques and deposit slips. MICR readers detect the characters and convert them into digital data. This technique is useful to accelerate the check routing process as also to route the check back to the location where the funds exist and to settle the transaction in minimum time.

Cheques were processed manually until the rapid increase in their use after the Second World War necessitated the introduction of a reliable automated method of cheque processing. This growth was most marked in the USA, and the American Bankers Association (ABA) in 1956 adopted a process of Magnetic Ink Character Recognition, or MICR, which was developed jointly by the Stanford Research Institute and General Electric Computing Laboratory. MICR is standardized by ISO 1004.

The MICR proofing systems have also developed since the early 1960s to incorporate scanning technology and advanced optical character recognition techniques to improve the automation of document processing. This technology, known as "Image Processing" provides the means for converting the image of documents into a digitized

format suitable for electronic processing and storage. The data contained in the MICR code line is still predominantly captured magnetically.

MICR cheque processing was introduced by the Reserve Bank of India first in Mumbai and Chennai in 1987 and was implemented in 1988 in Delhi to be followed by Kolkata in 1989. Following were the factors responsible for the adoption of MICR technology:

- Increasing clearing volumes
- Faster realization of cheques
- Better control over clearing reconciliation
- Use of state-of-the-art technology in cheque clearing
- The availability of cheque-wise data on instruments processed in clearing
- The facility to store, retrieve, and group data on clearing instruments with the aid of the computer systems used for the MICR processing
- The capability to provide data in a meaningful manner enabling further analysis.

Advantages and disadvantages of MICR

Advantages:

- MICR systems are secure against most common types of defacements like overprinting, dirt or writing across numbers
- A layer of transparent adhesive tape over the numbers does not prevent desirable results
- Imprinting with magnetic ink is highly durable and can withstand thousands of transits across the read-head with no impairment of the signal
- Documents are difficult to forge
- Reduces human errors.

Disadvantages:

- Highly stylized font required to ensure character discrimination
- Limitation to numbers and only about four other characters
- Difficulty in scanning with hand held device
- A very shallow depth of field
- MICR readers and encoders are expensive.

5.5.1 Equipments used in MICR Cheque Processing System's

Following are the instruments required:

5.5.1.1 MICR Encoder

The encoder is a tabletop machine, which can print the coded particulars of cheques and other payment instruments in magnetic ink on the 5/8" read band at specified position. The conventional encoder has a keyboard and a programmable journal

printer (i.e. lister). It endorses on the reverse of the instrument a fixed or variable stamp. The encoder has the facility to proof the pay-in-slip amount or control totals simultaneously by marking off successive amounts of encoded cheques thus arriving at a zero balance when all the cheques are encoded and bringing out discrepancies, if any, in the totals or errors during encoding. The figures are cumulated to enable encoding of the control documents viz., Batch and Block tickets. The encoders are also programmed to simultaneously affix/print the Clearing Endorsement Stamp on the reverse of the instrument, in the format prescribed. Encoders with compatibility to PCs are available, as also are power encoding machines and encoders with limited sorting facilities. Encoding work could either be decentralized at branches or centralized at the Service branch depending on the logistic in the bank. Clustering of encoding work at some branches to take care of smaller branches in the vicinity is another option available. Following are some images of popular encoders used by banks.



Figure 5.1 Encoders used by Banks

5.5.1.2 Reader/ Sorter

Previously two different methods for check processing were used. They were Sort-A-Matic and Top Tab Key Sort.

Sort-A-Matic method: It was made up of 100 dividers from 0-99 made up of leather or metal. It sorted checks through stepped or phased process. After reviewing the first two numbers of the account the checks were then placed into dividers which were then grouped by second two numbers and so on to make the check in numerical order of account numbers.

In Top Tab Key Sort holes are being punched at the top of each check indicating the ones, tens and hundred digits. For sorting the checks, a metal key was inserted into the checks separating them by the corresponding holes until they get arranged according to the account number. Both these methods were time consuming and outdated methods which eventually led to the emergence of the MICR encoding and sorting.

A Reader/Sorter is a device that reads the MICR encoded documents and sorts (direct) them to one of the many pockets as per the pre-determined sort pattern/program. The

characters are printed using special magnetic ink which contains iron oxide. As the document passes into the MICR reader, the ink is magnetized so that the shapes of the character can be recognized electronically. Documents are fed automatically from an input, which can handle documents of various sizes simultaneously. The documents travel past an electronic field, which magnetizes the characters, and symbols in the MICR read band and generates distinct wave patterns intelligible to the machine.

The physical sorting of cheques on the machine is carried out under the control of a computer program. This sort program, while directing the documents to the designated pockets, simultaneously captures and stores the information in the MICR code line on the cheques. The information captured from the documents is simultaneously stored on disk/tape, etc. and used for further processing. In case certain information is not read due to defective printing, encoding, etc., the cheque is directed to a 'Reject' pocket along with the control documents. These are taken out and the missing information is completed by manually keying in the data.

Following are the photographs of MICR readers one is an older version another is a current one.

An early MICR machine the "football field-long" machine from Recognition Equipment was used in the 1970s to process checks and credit card slips. *(Image courtesy of BancTec, Inc.)*

A current MICR machine this contemporary machine supports both MICR and OCR (optical character recognition) processing. *(Image courtesy of BancTec, Inc.)*

From Computer Desktop Encyclopedia
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© 1990 BancTec, Inc.



Figure 5.2 New MICR Reader

From Computer Desktop Encyclopedia
Reproduced with permission.
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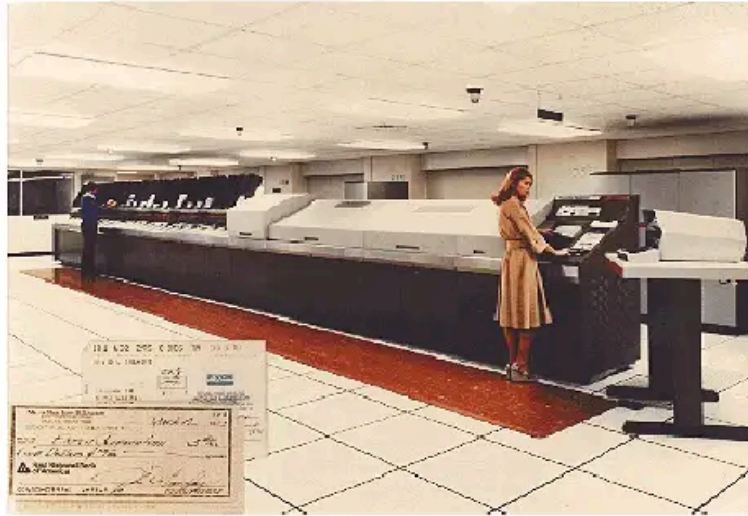


Figure 5.3 Old MICR Machine

5.5.1.3 Image Capture

Image capture and image processing technology is a recent development in document processing by which the image of a payment instrument is captured simultaneously when it is processed on reader/sorters by adding an image capture module and related software. The images so captured are stored on magnetic media for retrieval and processing. The images can be displayed on a screen and copies can be printed. It is also possible to transfer the image data to banks through magnetic media or through the communication backbone. The availability of image files enhances the processing quality and speeds up reject recapture, balancing, etc. The stored images could also be retrieved at a later date to facilitate quick reconciliation of clearing differences.

5.6 MICR Standards

Following are some character, fonts and printing standards followed.

5.6.1 MICR fonts

MICR fonts are printed at the bottom of every cheque or demand draft now –a-days. The MICR font looks like as under the following diagram.

⦿ 1 2 3 4 5 6 7 8 9 0 ⦿ ⦿ 1 2 3 4 5 6 7 8 9 0 ⦿ ⦿ 1 2 3 4 5 6 7 8 9 0 ⦿ ⦿ 1 2 3 4 5 6 7 8 9 0 ⦿

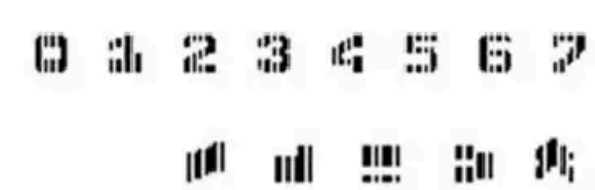
There are two different fonts that appear at the bottom of the checks or financial documents, which are used according to the established banking standards in the country. These are E-13B and CMC-7.

The E-13B fonts look like this:



This one is accepted in USA, UK, Japan, Australia, India, Canada, Mexico, Columbia and Turkey.

And the CMC-7 looks like:



and is accepted in the countries- France, Spain, Israel, South America and Mediterranean countries. (source: ANSI 9.27 specification)

The ABA (American Bankers Association) accepted the specification for the E-13B font and use of magnetic ink as a standard in 1958, and then in 1959 the first ABA publication for MICR was issued. Lithographic printing and impact ribbon were adapted to the process shortly afterwards. The American National Standard Institute (ANSI) adopted the ABA specification in 1963 as the American Standards, with local revisions.

In addition to their unique fonts, MICR characters are printed with a magnetic ink or toner so that, even when they have been overprinted with other marks such as cancellation stamps

5.6.2 Character Specification

The character set comprises 10 numerals and 4 special symbols. The ten numerals are:

0 1 2 3 4 5 6 7 8 9

The four special symbols are:

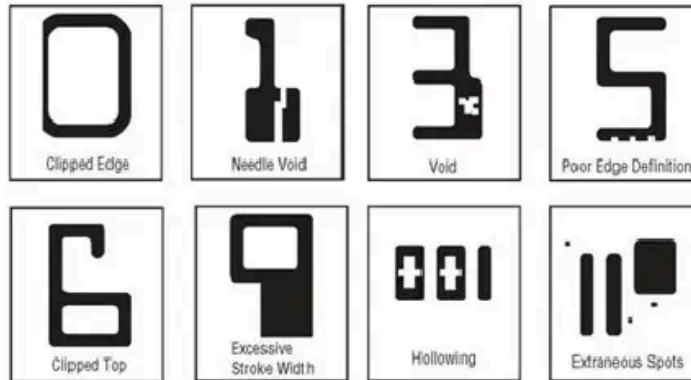
Amount	<
Domestic	=
BSB	;
Dash	>

But there could also be some problem with the MICR characters:

Common Problems with MICR Characters

Poor Quality MICR Line: Poor quality MICR characters can have problems with alignment, voids, ragged edges, extraneous spots, hollowed characters, clipped characters, etc. (see examples).

⑆0 234567890⑆0 23456789⑆0 23456789⑆0 23456789⑆



NOTE: For complete MICR specifications, refer to the ANSI X9.27 specification.

This way the character printing may lead to fallacies. So printing is the most important part of MICR enabling.

5.6.3 Print Specification

The instruments passing through clearing are required to be issued in standard format and defined size of 8" x 3 2/3". The instruments should be printed on MICR grade quality paper with a "read band" of 5/8" in width reserved at the bottom on which essential particulars occur in special MICR ink in the above mentioned E-13B Font. Approved security printers forming part of a panel, which is maintained by the Indian Banks' Association, print cheques. The numeral which we see at the bottom of any cheque these days is known as the MICR code line structure. This also follows a particular format, every sequence of number has some importance which depicts some information.

Pay.20.....
 or Bearer
 Rupees. Rs.
STATE BANK OF INDIA
 Jawaharlal Nehru University, New Delhi – 110067

⑆ 23456 ⑆ 234567890⑆ 345678 ⑆ 78

1 2 3 4 5

Figure 5.4 MICR Code Line Structure

1 - Cheque serial number of *six numeric digits* preceded and followed by a delimiter. The alphanumeric prefix to the serial number normally used by banks should be printed outside the code line in close proximity, just above the read band, in normal ink.

2 - Sort field or the city/bank/branch code number consisting of nine digits followed by a delimiter. *The first three digits represent the city; the next three indicate the bank and the last three digits signify the branch.* The nine-digit sort code is unique for any bank branch in the country.

The bank code is a three-digit code number allotted to the bank on an all-India basis. The branch code is the last three digits of the nine-digit sort code and is unique to a branch in a city. Allotment of branch codes is by the President of the Clearing House of which the Bank is a member; generally the service branch of a bank is allotted the branch code of '001'.

A sub-member will be treated as if it were a branch of the sponsoring bank. It would have the bank code number allotted to the sponsor bank to be followed by the branch code, which would normally commence from 251.

A full list of nine digit code number allotted to each bank/branch along with the three letter alphabetical abbreviations for the clearing stamp could be obtained from the President of the concerned Clearing House.

3 - Account number field consisting of *six digits* followed by a delimiter is an optional field. In the case of Government Cheques issued by CENTRAL BANK alone (the RBI in India), the account number is of seven digits. The Government Account number is 10 digits in length – 7 digits occurring in the Account number field and three in the transaction code field.

4 - Transaction code field comprising of *two digits* in all instruments except Government cheques drawn on CENTRAL BANK, which have a 3-digit transaction code. Control documents – batch and block tickets have a three digit representation in the transaction code field.

5 - The last field represents the amount field and consists of *13 digits* bounded on both sides by a delimiter. The amount is encoded in paise without the decimal point. It should be noted here that this field is empty while customer has his checkbook with him. Only after issuing a check to the Payee and when the check enters the clearing process in the Drawee's bank, then this amount field will be filled automatically by the system. Hence this implies that under normal circumstances, customers can never see this field filled!

5.7 The Process of check clearing at different levels

Now the in the current cheque clearing process the cheque has to travel through various levels before getting realized i.e. before the actual transfer of funds into the payee's account from the payer's account. All these steps comprise of the clearing process of the cheques.

Clearing is an arrangement through which a bank exchanges cheque drawn on other banks for those drawn on it, and the obligation of each party is determined. In the absence of such an arrangement, each bank will have to present cheques to each of the other banks for receiving payment of cheques over which they have a claim. The cheque clearing system provides a easy, systematic, efficient and cost-effective method of clearing cheques.

For Example Vinod Shah deposits a cheque into his account, at Dadar Branch of Bank of Baroda. This cheque has been given to him by Paresh Modi who has an account with Andheri Branch of Dena Bank. The processing of this cheque deposit transaction involves the debit of Paresh's account at Andheri Branch of Dena Bank, and credit Vinod's account at Dadar Branch of Bank of Baroda. The activities involved in carrying out this transaction are as follows:

- The cheque has to be sent to Andheri Branch of Dena Bank, so that they can carry out the validations, and debit Paresh's account in their branch.
- Andheri Branch of Dena Bank has to then debit the account in their branch and send the credit to Dadar Branch of Bank of Baroda.
- On getting the credit from Andheri Branch of Dena Bank, Dadar Branch of Bank of Baroda will in turn credit Vinod's account in their branch.

This activity becomes a complex task as every day individuals all over the country are sending cheques drawn on their account at one bank to people who bank with other banks. There will be a continual stream into each bank, of cheques drawn on each of the other banks. Such exchange of cheques therefore takes place in a *Clearing House*.

We will describe below the stages involved in cheque clearing. This is the process followed when the cheque that has been deposited is drawn on a bank that is in the same city. This is referred to as Local Clearing.

5.7.1 Local Clearing

The timings indicated are not uniformly applicable across all branches, but are quoted for better understanding. The various stages involved in the clearing of cheques are as follows:

1) Let us say that on any day around 500 cheques are deposited by the customers at Mahim Branch of Bank of Baroda. The branch would collect the cheques until a fixed

time say 3.00 p.m. Cheques deposited after 3.00 p.m. will be sent for clearing on the next working day only.

2) After collecting all the cheques, a clerk in the clearing department would enter the details of all these cheques in an Outward Clearing Register. The cheque amounts will then be totaled using an adding machine. The listing from the adding machine will be attached to the bundle of cheques, and sent to the Service Branch of the bank. The cheques are collected until 3.00 p.m., and are sent out of the branch to the bank's Service Branch at around 4.00 p.m.

3) As per central bank guidelines, each clearing cheque bundle or lot should contain at most 300 cheques. This activity at Bank of Baroda where the cheques drawn on different banks that have been deposited by the customers are collected and sent to Service branch is referred to as **Outward clearing**.

4) The Service branch of a bank receives cheques from all the branches of that bank in that city. These cheques would be of different banks. At the Service branch, the cheques are sorted bank-wise and encoded using the encoding machines. Encoding of the cheques involves the printing of the cheque amount, city code, bank code and branch code on the MICR line (at the bottom of the cheque). After encoding, the Service branch bundles the cheques bank-wise with control information on the number of cheques and the total amount for each bank.

5) The cheques from the Service Branch are then sent to the Clearing House which is generally managed by central bank or nominated bank.

6) Once the Clearing House receives the encoded cheques from the various banks (through their Service Branches), these cheques are processed during the night by the reader sorter machines. These machines read the encoded information from the MICR line, which contains details such as the bank branch where the cheque was issued, the bank branch where the cheque has been sent for clearing, and the cheque amount. The clearing of cheques involves transfer of funds from one bank to another, i.e. settlement. At the Clearing House, each bank has to maintain an account with the managing bank. The inter-bank transfer of funds required in the course of cheque clearing, is affected by debiting and crediting the accounts of the respective banks' accounts with central bank. Each bank in turn will credit or debit the account of the branch with the Head Office. Simultaneously, the cheques are sorted based on the bank branch where the cheque was issued.

7) Next day, the personnel from the Service branch of each bank, would collect the cheques issued from its own branches, from the Clearing House. The Service branch sorts the cheques branch-wise. Each branch will send a person to the Service branch of the bank early in the morning, to collect the cheques issued from their branches, which

have come through clearing. These cheques reach the branch at around 11.00 a.m. If there are any cheques which are to be returned due to insufficient balance or any other reason, then these have to be sent out of the branch by 3.00 p.m.

8) In case any cheque is returned, then the cheque is sent by the branch to the Service branch from where it is forwarded to the Clearing House who will return it to the bank where the cheque was deposited. If a bank branch does not receive any cheque return advice, then it treats the cheque as cleared. If a cheque return advice is received, then the account is debited.

9) Thus the Dena Bank (Andheri Branch) cheques deposited at Bank of Baroda (Mahim Branch) reaches Dena Bank (Andheri Branch) at around 11.00 a.m. on the following working day. These Cheques are posted to the respective customer accounts. In case of returns that might arise due to insufficient balance, signature mismatch, etc. the cheque has to be sent back from the branch at around 3.00 p.m. These return cheques will reach branch Bank of Baroda (Mahim Branch) early in the morning on the 3rd day. On the 3rd day morning all the cheques except the ones returned will be treated as cleared.

This activity at Dena Bank branch, where the cheques drawn on Dena Bank that have been deposited at various other banks are received from the Clearing House and processed is referred to as **Inward clearing**.

Following is a representation of the Clearing Cycle. Consider the Cheques of Dena Bank that has been deposited at Bank of Baroda:

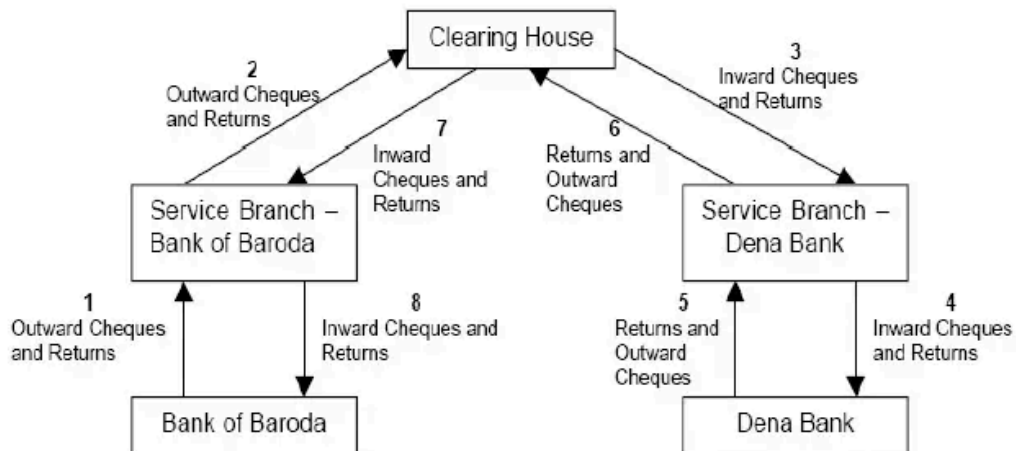


Figure 5.5 showing the Operation Cycle of Cheque Clearing

In this whole cycle the operations performed at the branch Bank of Baroda are referred to as the Outward Clearing operations, since the cheque is going out of this branch. The

operations at the branch of Dena Bank are referred to as Inward Clearing Operations. In the course of the cheque clearing the flow of funds will be as follows:

- The customer who has drawn the cheque gets debited
- Funds are transferred between the two banks involved
- The customer who deposits the cheques gets a credit in his account.

Now a branch can decide to credit the customer's account by the cheque amount on the day it was cleared or for the benefit of customer can decide to credit the account on the day it was deposited. In the latter case, the balance to the extent of cheque amount will remain uncleared till it gets cleared on the third day.

5.7.2 Outstation Cheque Clearing

If the cheque deposited has been drawn on a bank branch that is not in the same city, then this cheque is sent for Collection. A cheque of Bank of Baroda Chennai branch is deposited in Indian Bank at Nariman Point branch. This cheque will not go in the outward clearing of Indian Bank. Indian Bank – Nariman Point branch will send this cheque to Indian Bank branch in Chennai. From there it will follow the Local Clearing process and the Chennai branch of Indian Bank will receive the credit. Chennai branch will then forward the credit to the Nariman Point branch of Indian Bank.

5.8 Main Operators of the Clearing Cycle

5.8.1 Service Branch

Service Branch is a specialized branch of a bank for handling clearing of cheques, payment towards Demand Drafts, Pay Orders and collection of outstation cheques. Its basic objective is to facilitate the clearing process. Generally, banks have one Service branch in major cities.

Wherever it is not feasible to operate an exclusive service branch, an existing branch acts as a nodal branch and carries out functions of service branch.

5.8.2 Clearing House

The primary objective of the clearing house is to facilitate the speedy and economic way of collecting cheques, bills and other documents payable or deliverable at or through offices of the members and sub-members of the house situated in that town by a system of clearing.

In general, central bank undertakes the management of the clearing house. In the absence of this, one of the public sector banks in that centre, which may be specified by central bank, shall be managing the Clearing house. Clearing houses are established in all the cities, towns and even large villages where the volume of cheques is large and number of banks in the area is more than 2. This results in considerable saving of time

and cost. Clearing houses are autonomous institutions having uniform regulations and rules regarding the conduct of operations.

5.8.3 Settlement Agent

This is explained with the help of an example.

Central Bank of India presents cheques drawn on other banks which were deposited by its customers. The instruments are grouped by Central Bank of India bank-wise and the amount due from each bank is shown therein. The aggregate of these amounts represents the total amount to be received by Central Bank of India from all other banks. The instruments are presented to the respective bank along with a memo in a clearing house. Likewise Central Bank of India also received cheques presented by other banks. The aggregate of the totals of such instruments delivered by other banks represents the total amount of cheques drawn on Central Bank of India and payable by it. The difference between the two amounts is the net debit or credit to Central Bank of India. The Central Bank of India as well as other member banks keeps an account with the presiding bank-branch which manages the clearing house, through which these debits/credits are passed.

5.9 Reconciliation of clearing differences

Generally, there are two types of 'clearing differences' that may arise viz.,

- Inward Clearing Differences
- Outward Clearing Differences

5.9.1 Inward clearing differences

Inward Clearing Differences are of the following types:

- Cheques listed to the bank/branch but not received by it - the Clearing Receivables
- Cheques received by the bank/branch but not debited to it - the Clearing Payables;
- Cheques received by the bank/branch with the actual debit not tallying with the face value of the instrument.

For all the Clearing Receivable differences reported by a branch, the Service Branch should maintain full data and try to match these with the Clearing Payable differences reported by the other branches. If no such matching is possible, then the Service Branch could forward the Clearing Receivable Report to the presenting bank for further action as per the Clearing House Rules. For Clearing Payable differences too, the same procedure could be adopted and unresolved Clearing Payable Report sent to the cheque-processing centre as per the Guidelines. The CPC [Check processing Centre],

would, on the basis of the MICR data available at their end, find out the bank/branch to which the actual debit has been raised and inform the same to the bank/branch reporting the Clearing Payable difference that could issue a Pay order to the affected branch for settlement of this un-reconciled clearing difference.

In the case of a cheque listed for a value higher than the actual amount of the cheque, the branch may debit the drawee's account for the actual amount of the instrument, if it is otherwise in order, and the excess should be reported to the Service Branch for onward follow-up with the bank/branch concerned. In the case of a cheque which is listed i.e. the debit has been raised, for an amount lower than the value of the cheque, the branch may debit the drawee's account for the actual value of the instrument, if it is otherwise in order, and the difference passed on to the Service Branch for onward transmission to the presenting bank/branch after verifying the position from the cheque processing centers

5.9.2 Outward clearing differences

Outward Clearing Differences could arise in one of the following ways:

- No credit received in respect of an instrument presented.
- Short credit received in respect of an instrument presented.
- Excess credit received in respect of an instrument presented.
- Credit received in excess of the claim.

In respect of first three above, the differences should be reconciled promptly, if necessary by obtaining the additional information from the CPC or in consultation with the drawee bank/branch concerned. As regards to the last one where credit is received in excess of the claim, the Service Branch should be informed of the position by reporting the same in the daily Branch Clearing Control Report. The service branch should attend to such cases promptly and ascertain the reasons therefore by referring the matter to the CPC. Before making a reference, however, the Service Branch should verify from the Branch Clearing Statement and the branch wise claim figures whether the excess credit is adjusted against a corresponding short credit against another branch. Such inter-branch clearing differences should be reconciled internally without reporting the same to the CPC.

It must be recognized that reconciliation of clearing differences is an important activity that assumes greater significance since this is also an area that may give rise to frauds. The essence of reconciliation is the need for all banks and branches to follow the set rules and guidelines laid down in this regard. Prompt reporting and solving of the same should, therefore, be the prime concern of all players in the field. It would also be desirable that all un-reconciled clearing differences payable are transferred to the

Chapter-6 Electronic Funds Transfer and Electronic Clearing Systems

6.1. Introduction

This chapter focuses on electronic fund transfer and electronic clearing process. In this chapter the Indian scenario for both EFT and ECS has been highlighted to explain the concepts with better clarity.

6.2. Learning objectives

After reading this session, you will

- Learn about the aspects of electronic transfer of funds
- Basics of the ECS or electronic clearing service
- Know major players and their role in the EFT process

6.3. Topics Covered

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6.4. Electronic Funds Transfer – an introduction

Electronic Funds Transfer is a method of funds transfer by which the payer, (it could be an individual or a financial institution,) can give electronic (non-paper) instructions to the financial institution to debit or credit customer accounts and henceforth making the funds to move (transfer) from one account to another. The EFT mode of funds

transfer is faster and comparatively safer than funds transfer with the help of paper checks. Basically EFT is an alternative channel of payment, both for debit payments like mortgages, insurance premiums as well as for credit payments like payroll, dividends and other income. It has become more prevalent with more and more use of personal computers, arrival of cheap networks, internet, cryptography etc.

The history of Electronic funds transfer originated from the common funds transfer from the past. Since the 19th century with the help of telegraphic funds transfer were a usual thing in commercial transactions. Finally with the advent of computers transactions took place electronically and came to be known as 'Electronic funds transfer'

6.5. Advantages of Electronic Funds Transfer

EFT is similar to other direct deposit operations such as paycheck deposits, and it offers a safe modern alternative to paper checks. Providers who use EFT would realise the following benefits:

- The main advantage of this technology is time saving, since all transactions are done automatically and electronically.
- Reduction in paper-based transactions.
- Savings in time and other logistic operational costs normally incurred in presentation of paper checks and visit to Banks for this purpose.
- Elimination of the risk of paper checks being lost or stolen in the mail or in any form of transit. Also, the risk of mutilation and damage of checks is eliminated in this case.
- Faster access to funds as most banks credit direct deposits faster than paper checks (which go through the clearing process)
- Easier reconciliation of payments with bank statements.

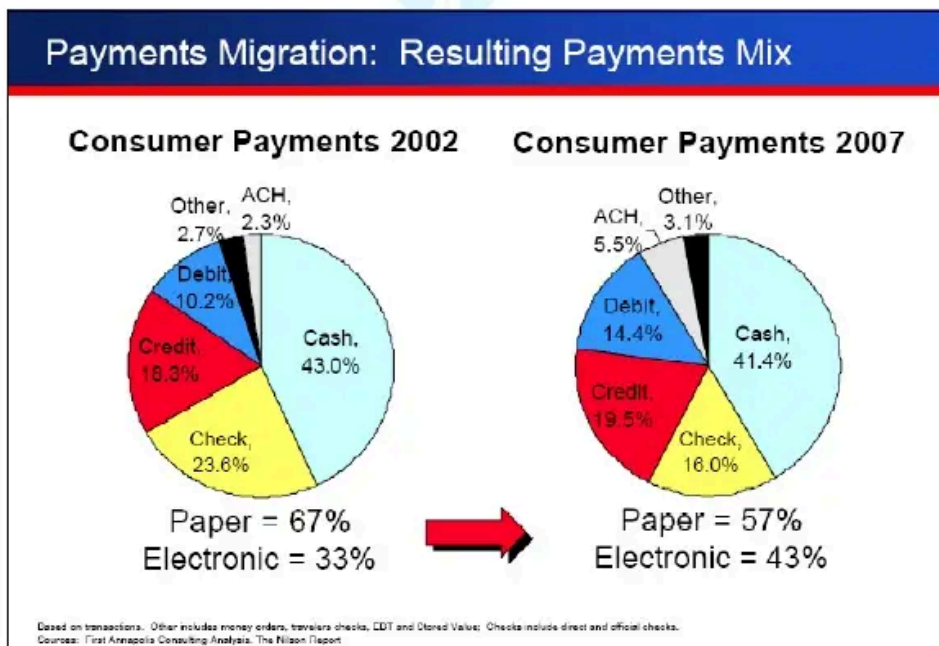
DID YOU KNOW?

In 1996, in the USA, the IRS introduced its free e-payment service, the Electronic Federal Tax Payment System. In 2004, 1.75 million people paid their taxes electronically. To sign up, all you need is your Social Security Number and checking account information. In addition to paying your tax bill online, you can access your payment history and schedule tax payments for next year

SOURCE: EFTPS (Electronic Federal Tax and Payment system)

Traditionally, the primary modes for Funds transfer are Demand Draft, Mail transfer and Telegraphic Transfer. The demand draft facility is paper based. The remitter, after purchasing demand draft from a bank branch, dispatches the same by post/courier to the beneficiary. The beneficiary, in turn, lodges the draft to his/her bank for collection and clearing. The time taken for completing the process is about 10 days. In the case of telegraphic transfer, fund reaches the beneficiary either on the same day or the next; but both the remitter and the beneficiary would have to be account holders of the same bank. If they are customers of different banks, a good deal of paper processing is required. On the other hand, the EFT system is an inter-bank payment system. The Central Bank of the nation acts as an intermediary between the Remitting bank and the Receiving bank and effects inter-bank funds transfer. The customers of these banks (i.e. the Payer or Remitter and the Payee or Beneficiary) can request their respective branches to remit funds to the designated customers' accounts irrespective of bank affiliation of the remitter and beneficiary. This makes the task of message reaching to the banks easier and quicker.

A recent survey and statistical analysis by First Annapolis came up with the results:



Here, it can be seen that the electronic transactions increased from 33% to 43% in last 5 years and there is a phenomenal drop down of the percentage of paper transaction occurrences. The check transactions significantly declined with time whereas those of electronic are showing a northwards move.

6.6. Payments Made By EFT

The Electronic Fund Transfer or EFT is deployed in several payment scenario. For example, when a company which is public i.e. has shares floated in the nation's stock exchange announces a dividend uses EFT, rather than sending the check of dividend to each shareholder. When you transfer funds from an account to another with the help of online banking is also known as EFT (rather it is NEFT- "N" for National). Other applications include payment of taxes and payment of salaries to the employees. Payroll disbursement is a very important usage of the EFT system. Hence, EFT is mainly employed for the bulk (high volume) but low value retail payment transactions.

6.7. The RBI EFT system

In this section of the session, we will learn about the Electronic fund transfer initiative of our nation's central bank, Reserve Bank of India. It is again a scheme initiated by the central bank so as to enable customers to transfer funds electronically.

The EFT system of RBI started its transactions on the BANKNET network. Where BANKNET was X.25 based packet switching network jointly owned by the Reserve Bank and the public sector banks. The computer system resources of the four IBM Mainframes (installed at the four metros for cheque processing operations) are used during the day time by BANKNET for data communication with additional equipment. This later evolved into the INFINET network.

The major applications and usage of INFINET are:

- Inter-bank fund transfers on banks' own account and on customers' account;
- Inter-branch funds transfers on banks' own account and on customers' account;
- Government transactions;
- Facilitates automated clearing services (similar to BACS);

Funds Settlement Time Frame in RBI EFT

- Till year 2001 from the time of its establishment in 1996 the RBI EFT used to settle funds on T+1 basis across other cities and on same day for metros
- Subsequent to year 2001, RBI EFT makes settlement throughout the country on a same day basis.

6.7.1 Process Flow in the RBI EFT

Step 1 - The customer fills up the EFT Application form in duplicate (similar to the draft application form) which contains the beneficiary's account information viz. A/c name, A/c no, A/c type, Bank /branch name and City

Step 2 - On receipt of the application from a Remitter, the branch checks Beneficiary details and Applicant details and sends it to its Service Branch along with following documents.

- EFT Scroll
- EFT application forms

Step 3 - At the Service Branch, the Beneficiary particulars are verified and ensured that all the details are correctly filled in.

Step 4 - The data is entered in an application which has a well defined template for all the fields.

Step 5 - Officer level user verifies the transactions and freezes them

Step 6 - At the cut-off time for settlement at RBI, the file is generated containing all the transactions for the day (incl. Acknowledgement of inward transactions).

Step 7 - RBI office merges files from various banks and generates centre-wise files. It transfers the files to other centers over the INFINET as per the pre-defined time schedules 12 noon, 2 pm and 4 pm.

Step 8 - At each of the RBI centers, files from all the centers are processed and uploaded to the respective service branches of the destination banks. Service branches process the file and print branch level report containing beneficiary details.

Step 9 - Destination branches credit the account to the beneficiary's account.

The following **figure 6.1** shows the process.

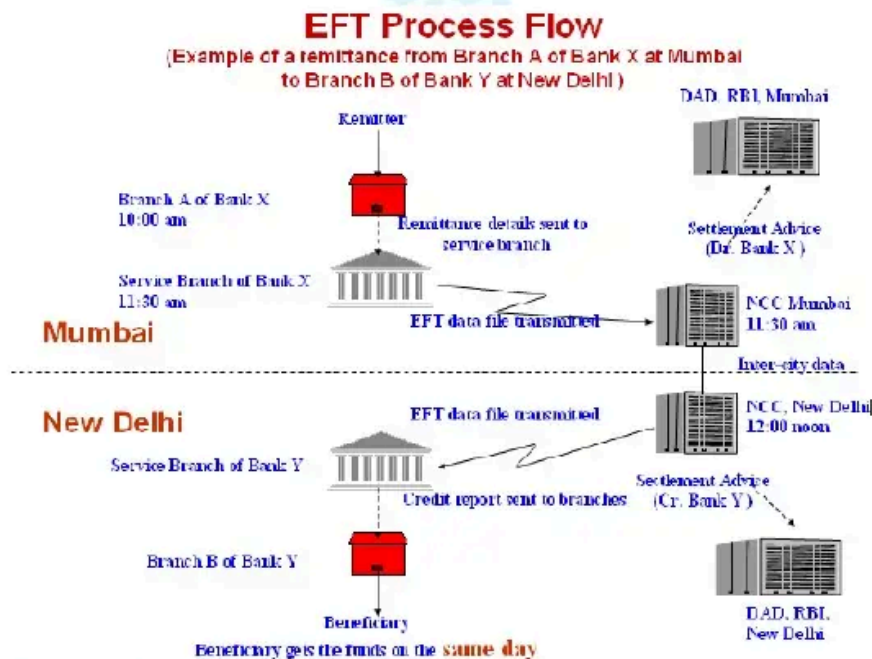


Figure 6.1 EFT Process flow

6.7.2 The RBI EFT Network

The EFT system presently covers all the branches of the 27 public sector banks and 55 scheduled commercial banks at the 15 centres (viz., Ahmedabad, Bangalore,

Bhubaneswar, Kolkata, Chandigarh, Chennai, Guwahati, Hyderabad, Jaipur, Kanpur, Mumbai, Nagpur, New Delhi, Patna and Thiruvananthapuram). Funds transfer is possible from any branch of these banks at these centres to other branch of any bank at these centres both inter-city and intra-city. The following diagram shows the total network. Centres which are connected with larger arrows are the centres where the mainframes are located and together they form the communication backbone.

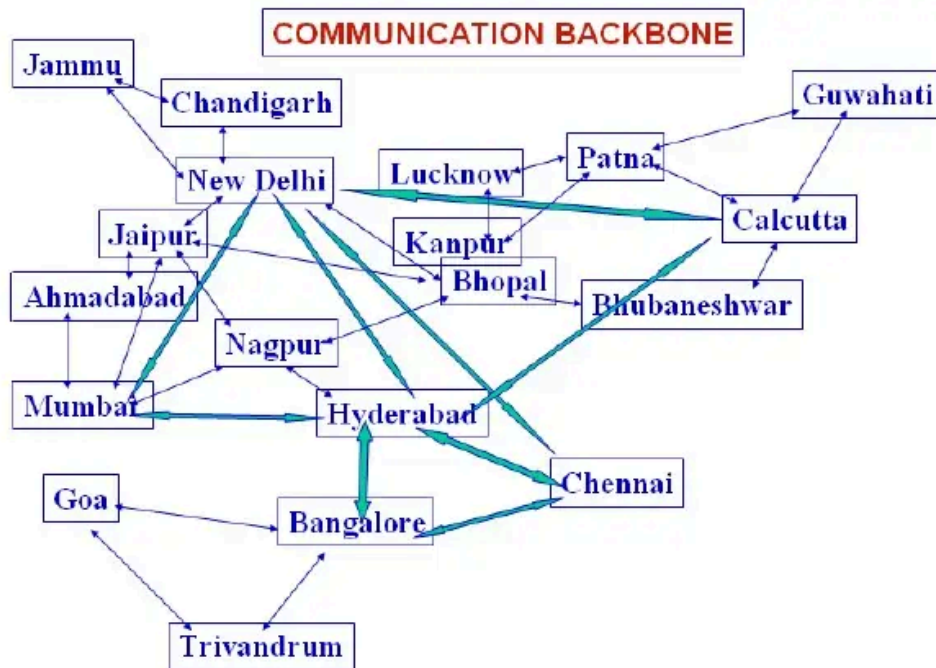


Figure 6.2 RBI EFT Network

6.7.3 Some Applications of EFT

- E- Cheque of ICICI Bank, Corporation Bank. Similar products of Bank of Punjab, Vysya Bank. Companies (including LIC) can pass on credits from their account after settling the case, from their office itself!!
- EFT for crediting merchant establishments of Credit Cards, Point of Sale, Smart Cards.
- Corporate crediting Suppliers through EFT. Railways and Defense Suppliers credits.
- Routing Foreign credits to beneficiaries in India.
- EFT interface on internet allows customers freedom to credit beneficiaries in other banks without visiting their bank.

- Banks having EDI interface with Govt. Departments say for example Customs, can credit payments like duty drawback etc., into accounts of other banks instead of giving pay orders.

6.8. Electronic Clearing Services (ECS) - an introduction

The above term, Electronic Clearing Service as the name clearly implies goes in tandem with the Electronic Fund Transfer. The objective behind its origination was to eliminate the paper based instruments for the bulk payments like payment of interest, dividend, refund, etc. which usually are disbursed by companies through paper-based instruments. This in turn will help immensely the banks and Companies/Corporations/Government Departments effecting the payments and hence resulting in faster and efficient customer service.

ECS is a mode of electronic funds transfer which uses the services of a Clearing house to enable transfers from one bank account to another account. This is normally used for bulk transfers. This can be used for both distribution and collection of amounts.

6.9. Importance of ECS

- Saves cost in bulk payments
- Paper based instruments put a lot of stress on the clearing and settlement systems.
- Moreover, studies done by RBI shows that volume wise paper checks arising from the periodic payments may be very high but value wise they are quite low. ECS scheme helps organizations such as Unit Trust of India, Life Insurance Corporation, banks and mutual funds and utility companies for their respective receipts and payments so that the ever increasing volume of paper instruments flowing to banks and clearing houses could be arrested.
- Chances of the paper based instruments getting lost in transit and exposure to fraud are very high
- In case of loss, instrument has to be re-issued which takes additional time
- To prevent fraud the validation procedures involves procedural delay
- The payment cycles become very long in paper based instruments.

6.10. Electronic Clearing Services (Credit Clearing)

In the case of ECS Credit, a series of electronic payment instructions are generated to replace paper-based instruments. The system works on the basis of one single debit transaction triggering a large number of credit entries. These credits or electronic payment instructions which possess details of the beneficiary's account number, amount and bank branch, are then communicated to the bank branches through their

respective service branches for crediting the accounts of the beneficiaries either through magnetic media duly encrypted or through hard copy.

e. g dividend, interest or salary payment come under ECS(credit clearing)

6.10.1 Participants of ECS (Credit Clearing)

- **User**

These are the companies/corporations/Government Departments or any other institutions effecting bulk payments to a large number of beneficiaries (For example a dividend paying firm or the firm making payroll payment).

- **Sponsor Bank**

It is the bank which would act as the agent of the User to submit the payment instructions prepared by the User to the National Clearing Cell (NCC) and which would give a mandate to Reserve Bank of India / Clearing Agency designated by RBI to debit its account.

- **National Clearing Cell (NCC)**

This is a functional unit of the local Bankers' Clearing House or such other agencies created by RBI which process the payment instructions received from the Sponsor Bank and generate relevant clearing reports for settlement of accounts of banks at RBI. The institutional arrangements made by RBI shall be final.

- **Destination Account Holder**

It refers to the beneficiaries who opt to receiving payments from the User directly by way of credit to bank accounts.

- **Destination Bank Branches**

It refers to the bank branches where the Destination Account Holders maintain their bank accounts into which ECS payments are credited.

6.10.2 How ECS Credit clearing works

The figure below (10.3) explains the working of the ECS Credit Clearing process. A payment is initiated by the corporate institution generating the payment instruction which will debit its (the corporate institution's) account and credit multiple user accounts. The corporate institution's bank will then submit the payment instructions on its behalf to the clearing house, which in turn clear the payments and credits the destination accounts of the beneficiaries.

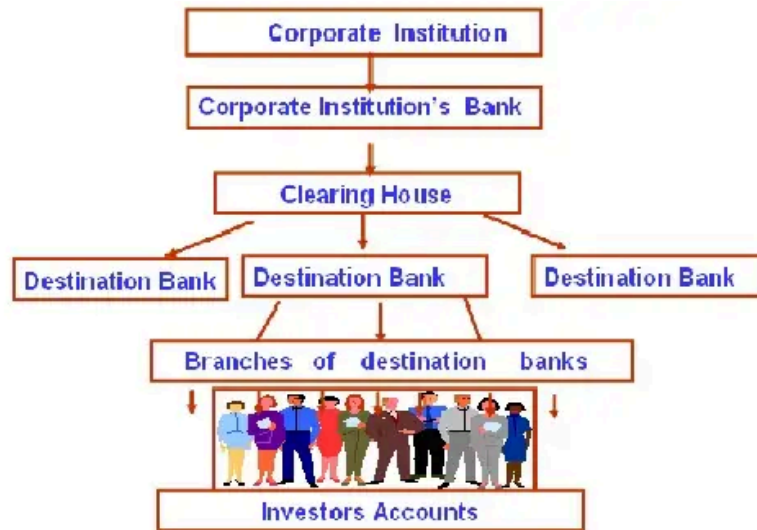


Figure 6.3 Working of ECS (credit clearing)

6.10.3 The ECS Process Cycle

Table 6.1 ECS Process Cycle

Day-1	Corporate Institution submits payment details on a data storage device to the Clearing House through its banker (Sponsor Bank)
Day-2	Clearing House sends the credit reports (on data storage device) to the Service Branches of the investors' banks (Destination Banks)
Day-3	Service Branches send the credit reports (mostly on paper) to investors' branches (Destination Branches)
Day-4	Destination Branches credit the investors' accounts on the due date of payment

Day-1

User → Sponsor Bank → Clearing House

Day-2

Clearing House → Service branches

Day-3

Service branches → destination branches

Day-4

destination branches → beneficiaries' accounts

Figure 6.4 ECS Process

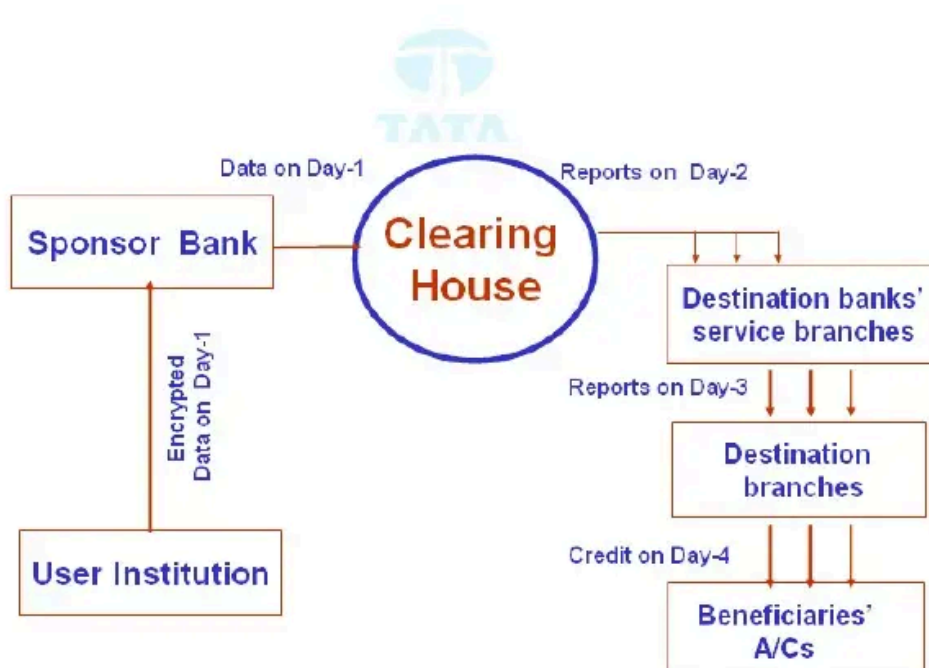


Figure 6.5 ECS Process Flow

The above figure 6.5 explains the same process of clearing in a schematic manner

6.11. ECS (Debit Clearing)

As opposed to the ECS credit clearing, the ECS debit clearing service is used when multiple debit entries from different accounts are required to be initiated and credited to a single account. The objective is to provide an alternative method of effecting payment transactions in respect of the utility-bill-payments such as telephone bills, electricity bills, insurance premium and loan repayments, etc. which would obviate the need for issuing and handling paper-based instruments and thereby facilitate improved customer service by the banks/companies/corporations government departments collecting / receiving the payments.

6.11.1 Participants of ECS Debit clearing

- **User**

User refer to the utility-companies, insurance/corporations and government departments or any other institution receiving / collecting repetitive payments from a large number of customers/beneficiaries, under the scheme. The individual transaction limit under the scheme is generally fixed at a certain amount.

- **Sponsor Bank**

It refers to the bank which has agreed to act as the agent of the user company and to submit the data storage device containing debit instructions prepared by the user to the RBI National Clearing Cell (NCC) (to be referred to as RBI hereinafter).

The Sponsor Bank also submits an undertaking to the effect that the standing instruction mandates to debit the accounts of the concerned users mentioned in the media device have been duly collected from the utility-users (consumers etc.) and have been forwarded to the respective destination bank branches for credit to the account of the Sponsor Bank with the sum mentioned therein.

This mandate will also authorize RBI to debit Sponsor Bank's current account maintained with them to the tune of un-debited returns.

- **National Clearing Cell (NCC)**

This is a functional unit of the local Bankers' Clearing House or such other agency created by RBI which processes instructions for debit payment received on magnetic media from the Sponsor Bank. The data is validated and then the NCC generates relevant ECS-Debit clearing settlement reports for debit/credit of the current account maintained at RBI. The instructional arrangements made by RBI is final.

- **Destination Account Holders**

This refer to the utility-consumers such as telephone and electricity users, insurance policy holders, debtors etc., under the ECS Debit Clearing scheme who opt for making payments to the User company directly by way of debit to their bank accounts as

indicated by them in the individual ECS mandate submitted by them to the utility company and also to their bank/branch.

- **Destination Bank Branches**

This refers to the bank branches where the Destination Account Holders maintain their bank accounts from which ECS utility payments are debited.

6.11.2 The process cycle for the ECS

The process cycle for the ECS debit is same as that of ECS credit with only difference that in the ECS Debit the initiators of the payment instructions being the beneficiaries of the ECS credit services.

6.12. Proportions of Retail Electronic Transactions in India

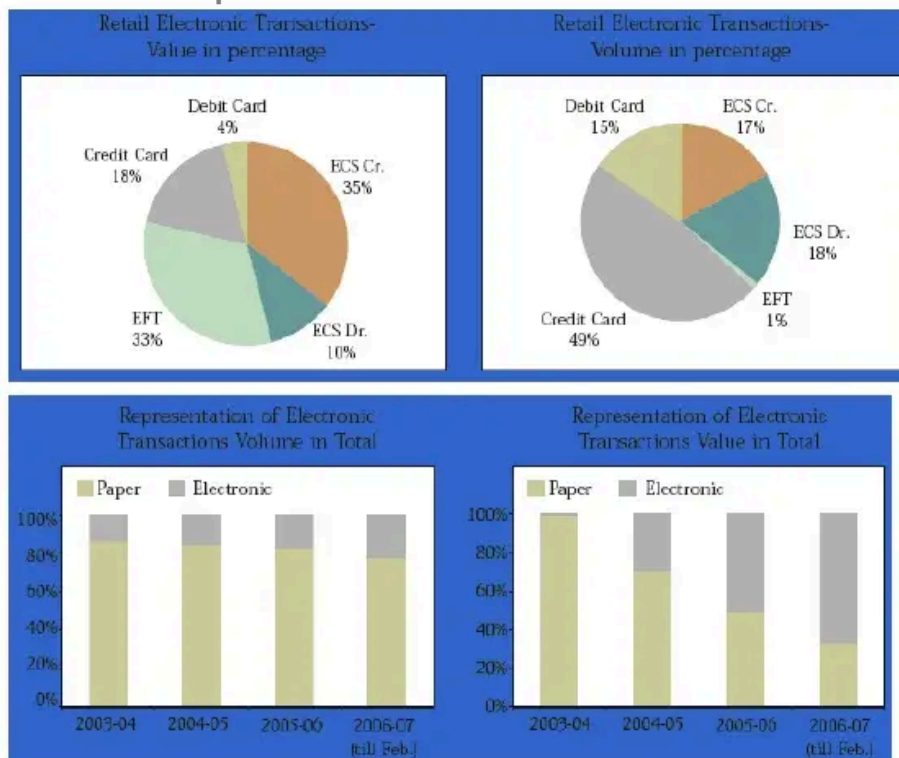


Figure 6.6 Retail electronic transactions (By value and volume)

6.13. Summary

- Electronic Funds Transfer is a method of funds transfer by a Payer who could be either an individual or a financial institution.
- Basically EFT is as an alternative channel for payments - both debit payments like mortgages, insurance premiums as well as for credit payments like payroll, dividend and other income disbursement.

7.4. Check 21 - an overview

In the year 2000, Federal Reserve Board (FRB) started working on improvement of the processing of checks in U.S in accordance with the changed scenario (to suit the needs of the 21st century). Fed mainly focused on promotion of check truncation, and electronic check presentation.

The final outcome of their endeavour was “Check clearing for the 21st Century Act” (popularly known as Check 21). The main idea was to come up with a machine readable copy of a check (a substitute check) for traditional check for forward collection or return. Substitute checks that meet the requirements of this act would be legal and practical equivalent of the original check. During this entire process, the FRB worked with the industry partners and other stake holders during the different versions of this Act.

On December 21, 2001 FRB sent a proposal to the members of the senate and House Banking Committees. Both the House and senate introduced a Bill in 107th congress. Many banking organizations monitored the legislative process and provided inputs. All sectors of the banking industry like small banks, large banks, credit unions, technology companies, FRB worked closely and supported the act in its passage strongly. Check 21 is short for “Check Clearing for the 21st Century Act”. Check 21 was signed into law on October 28, 2003 and became effective October 28, 2004. Check 21 allows banks to create and process a substitute check, also known as an **Image Replacement Document** (IRD) in lieu of an original check. Its purpose is to improve the speed and reliability of the check-clearing process by taking as much of the paper out of the process as possible.

It helped develop the **Substitute Check** that is the legal equivalent of the original check and it is the electronic copy or image of the original check.

Finally, the Accredited Standards committee focused on the development of industry standards for the financial services industry and developed the technical specifications for the substitute checks.

7.5. Why was Check 21 created?

While the current paper check clearing system in the U.S. is well established, the logistics of moving physical paper checks across America for processing and clearing has long been an impediment to further efficiencies. The check clearing for 21st century act was enacted into law on October 28th 2003 by 108th congress and is a United States Federal Law.

The Check 21 Act was enacted primarily in response to the September 11th, 2001 terrorists' attacks that grounded virtually all check payments and forced the U.S.

government to rethink how paper checks are processed. This new law will eliminate much of the risk associated with moving billions of checks, such as: transportation breakdowns (labor and mechanical), weather delays, fraud, theft and terrorism.

7.6. Highlights of the Check 21 act

Main purposes of Check 21 are

- To facilitate check truncation
- To develop innovation in the check payment without mandating the payment of check in electronic format
- Overall improvement of the payment system.

This act created a new negotiable instrument called “substitute check”

- If this substitutable check meets the criteria of the act then it is legally substitutable to the paper-written traditional check
- This substitute check can be processed similar to the paper check
- Involved Parties cannot refuse to accept these substitute checks if they follow all the requirements by the act “Parties” involve all the banks, Federal Reserve, consumers, customers and other financial services institutions.

The act provides legal equivalence only for substitute checks.

- The act provides check truncation and electronic image/ exchanges but does not provide legal equivalence for electronic check or image presentment
- Check clearing services where electronic checks are used, require the agreement of the parties who are accepting the electronic form of the instrument for value
- This act encourages the use of electronics to the empowering banks to truncate the original checks, and give paper checks where necessary.

All checks except foreign checks are eligible to become substitute checks, some of the following which can act as substitutable checks are

- i. Consumer checks
- ii. Business checks
- iii. Corporate checks
- iv. Government warrants
- v. U.S treasury checks
- vi. Money Orders
- vii. Control disbursement checks
- viii. Payable through drafts
- ix. Traveler’s checks.

- Usually the bank providing the substitute check to the subsequent parties should provide warranties and an indemnity to the subsequent parties involved in collection and return process
- Warranty involves that the substitute check meets the Act's legal requirements
- No party will be asked to make payment for the check which it has already paid (No double debit)
- The indemnity relates to the losses that incurred due to accepting this substitute check instead of the original check
- In the instance of warranty breach, the indemnity includes the damages proximately caused
- In the absence of warranty breach, the indemnity is for the amount of the substitute check and interest
- The indemnity bank may limit its liability if it can produce the original copy of the check.

Even this Act has an expedited re-credit facility for those who receive the substitute check.

- Consumer may claim expedited re-credit if the substitute check is improperly charged to the consumer account and even the consumer has the warranty claim if the consumer suffered any loss
- Banks are required to provide consumer awareness notices to the consumers whom they give substitute checks
- This will be applicable to both the existing as well as new customers.

7.7. What is Check Truncation?

Truncation is stopping the flow of the physical check issued by the issuer to the issuer's branch. Actually some where in the route the physical check is stopped from further travel and electronic image of the same is generated using some scanning device which will make sure that the front and back images of the physical check are captured properly with details like MICR and the content written on it, payee bank details, etc are added to the image.

Then this captured image is sent across the electronic channel that is then used for the further processing. This truncation is basically done to reduce the time required for the payment of checks and associated cost of transit and delay in processing. This process helps in speeding up the process of collection or realization of checks. Once a check has been truncated, banks, businesses can either work with the check's digital image or a print reproduction of it.

Substitute Check

A substitute check is an electronic or paper reproduction of the original check that contains the front and back information of the original check. The substitute check is in compliance with ANSI standard X9.100-140. All substitute checks are required to include the following statement, "This is a LEGAL COPY of your check. It can be used the same way you would use the original check."

Check 21 improves check processing without requiring customers to change the way they write checks. Check 21 allows the financial institution to make a decision to truncate all paper checks without agreement with any other parties. Check 21 authorizes the creation of the substitute check from an electronic record (image) of the check for those banks who have not agreed nor have the capability to accept the electronic record.

Substitute check is a paper reproduction of the original check.

- Contains the front and back images of the original check in electronic form
- Is suitable for the automated processing as the original check
- Conforms, in paper stock, dimensions as it is generally applicable to industry standards for substitute checks
- Contains a MICR line as the entire information that was available is printed in order for processing of the substitute checks, as accepted in the industry.
- Usually in the places where we find original check is found, a photocopy or image
- Bank customers may find substitute check with their periodic statement, when viewing checking images via online banking. Customers may request the copy of the paid check from the bank, or when a deposited check is returned unpaid.

Once a check is truncated, businesses, and banks can work with either the digital image or a print reproduction of it. Images can be exchanged between member banks, Saving & Loans institutions, credit unions, clearing houses, and the Federal Reserve.

The **figure 7.1** is the example of substitute check.

7.8. Uniqueness of the Image

The Images captured at the presenting bank level would be transferred to the clearing house then onto the drawee branches with the digital signatory of the presenting bank. Each image would have a digital signature apart from the endorsement from the presenting bank. In order to ensure that the images of the requisite quality reach the drawee bank, there will be quality check at the level of the Image capturing and the clearing house. In the entire process clearing house plays an important role of the interface and even the proper quality images with the presenting bank's signatory will

only sent to the drawee bank. Even the drawee bank may consider using the barcodes or holograms to maintain the uniqueness of the image.

DID YOU KNOW?

Research indicates that image exchange and other Check 21-driven changes can save the US banking industry about \$2 billion a year.

Source: www.carreker.com

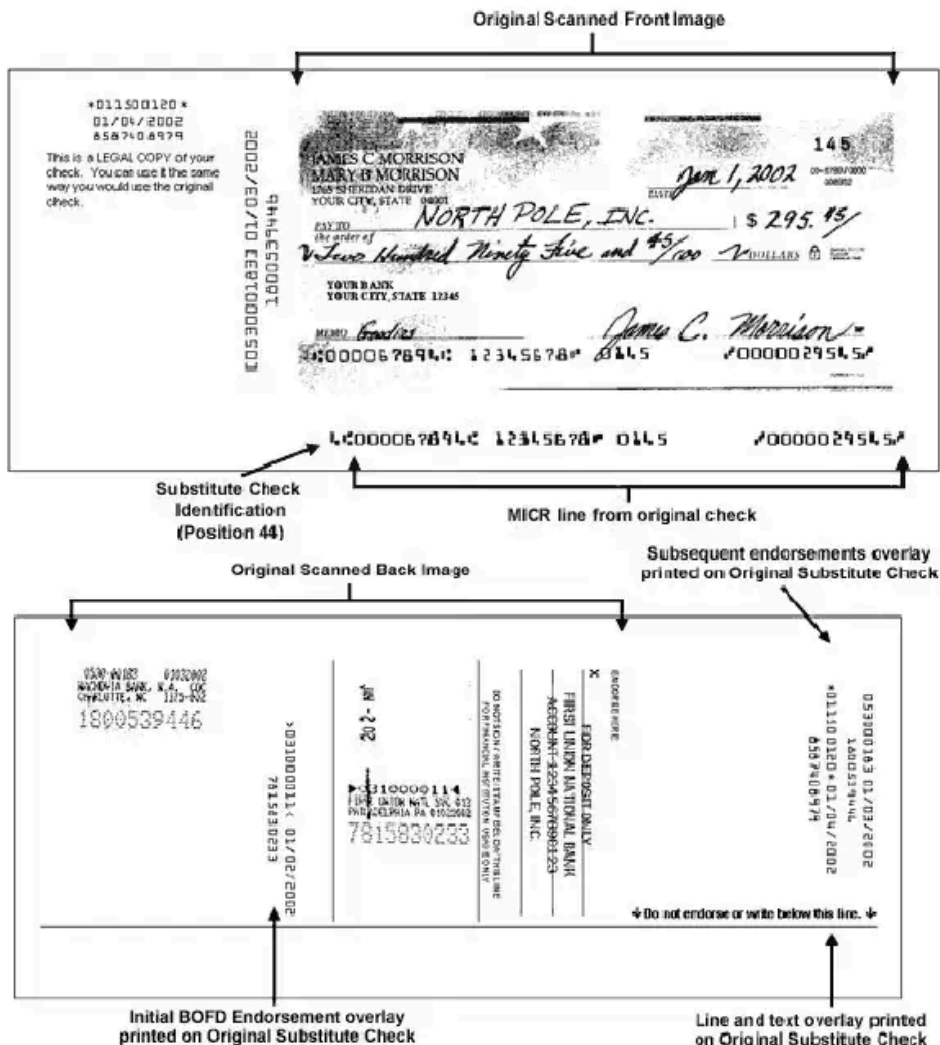


Figure 7.1 Substitute check

7.9. Check-21 Processing Cycle (Fig. 11.2)

Basically, the process of Check 21 starts with user or an account holder giving a check to a retailer in exchange for some service or some goods. Once the check changes

hands, retailer submits the check to his branch, and deposits in his account. This branch which receives the check sends the check for processing to the Bank service department, where the check is truncated and the image of substitute check is generated (usually after the substitute check is developed the original check is destroyed) and transmitted further for clearing to the Clearing house. The role of the clearing house is to process the check and transfer it to the Account holder's Bank. The account holder's bank deducts the amount of money from the account holder's

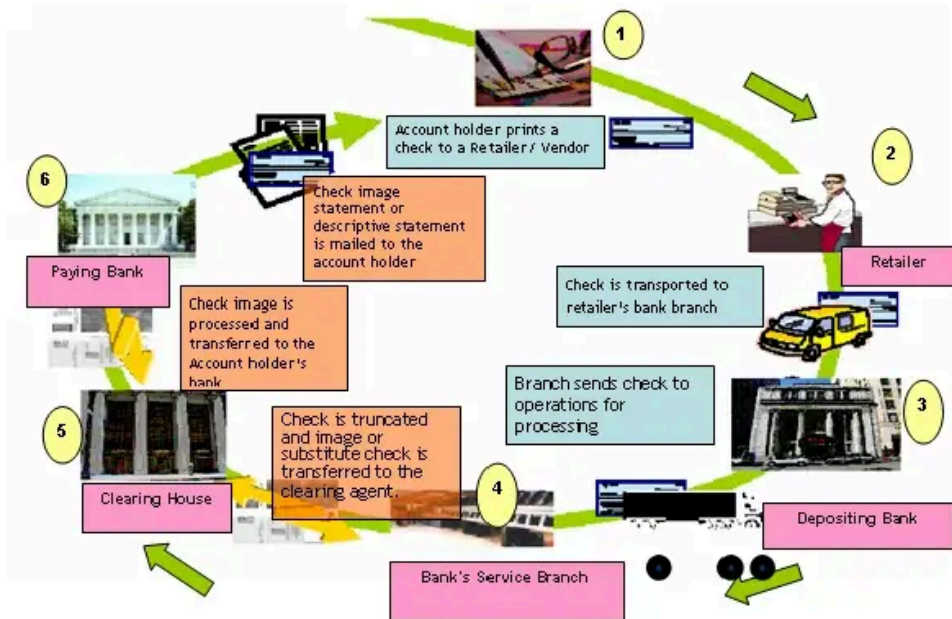


Figure 7.2 Post-Check 21 Image Check Processing Cycle

account and this particular image is saved for record purpose or same can be used for further payment and collection systems. And, usually this image of the check is sent to the payee or account holder who issued the check in his descriptive statement.

7.10. Check Conversion and Check 21

On March 16, 2007, a new standard for converting checks into electronic form for processing through the Automated Clearing House (ACH) system went into effect for U.S. businesses. Referred to as Back Office Conversion (BOC), this latest development represents one more step in the inexorable movement towards a paperless e-economy. The migration towards electronic payment began in the early 1970s with the introduction of direct deposit payroll and Social Security benefits via local private and government Automated Clearing House (ACH) networks. In 1978, these local ACH networks were linked into a nationwide system through the efforts of the Federal Reserve and NACHA (the National Automated Clearing House Association). Since the

mid-1990s the volume of paper checks has been gradually declining in favor of electronic payments.

The development of electronic payments was driven by the desire of government, businesses and consumers to make payment processes more secure and efficient. The advent of digital technology provided the means to achieve these goals. And, as they say, the rest is history. According to NACHA, the ACH network volume has doubled in the last five years to nearly 14 billion ACH payments in 2005. Much of this growth was due to check conversion applications.

Check conversion, sometimes called “electronic checks” or “e-checks”, is a process of converting paper checks into electronic debits. It was introduced in 1999 to utilize the same ACH network that had been used for direct deposit and direct payment transactions for more than 25 years.

As of March this year (2007), there are three types of ACH check conversion applications:

POP – Point of Purchase, available in September 2000

ARC – Accounts Receivable Entries, introduced in March 2002

BOC – Back Office Conversion, in effect as of March 16, 2007

All three were developed:

- To reduce handling and transport required for paper check processing
- To reduce the costs of processing payments
- To accelerate payment processing time
- To lessen the opportunity for loss and fraud.

All three check conversion systems apply to checks for less than \$25,000. All three are governed by provisions of the Electronic Fund Transfer Act of 1978 (implemented by the Federal Reserve’s Regulation E). Some of the consumer protection clauses include:

1. Payers must be notified, before issuing the check, that it will be converted into a one-time electronic debit
2. Payers can “opt out” of the check conversion
3. Payers have 60 days from the posting date to dispute the transaction. (Under check law, a bank or credit union has only until midnight of the business day following receipt of the check to return it through the system)
4. Bank statements listing converted checks must provide transaction details that include the check number, date, amount and payee’s name. (With paper checks, the payee’s name is not provided on the bank statement, although customers usually do have the option of receiving copies of their cancelled checks.)

The differences among the applications are primarily:

- (1) How the check is accepted by the business payee; and,
- (2) What is done with the paper check after conversion?

Two of the applications – POP and BOC – primarily relate to merchant sales to consumers. ARC has application for both consumer and B2B organizations.

7.11.1 POP (Point of Purchase) Check Conversion

POP is used for checks written at the point of sale, which are converted to a one-time electronic debit “at the register”, then voided and immediately returned to the customer. POP’s popularity was not as widespread as was hoped due to: Cost of the technology - scanning equipment had to be available at every payment station; and Training Issues – each checkout clerk had to be able to properly explain the transaction to the customer.

7.11.2 BOC (Back Office Conversion)

As with POP, Back Office Conversion relates to checks written either at the point of sale or at a manned payment location. However, with BOC, the seller retains the checks and converts them in a centralized “back office” location.

BOC can be considered an improvement over POP for a number of reasons:

- (1) Sellers only require equipment in their back office, not at every register;
- (2) Individual cashier’s do not require any special training, only the back-office personnel; and
- (3) the experience is more transparent to the customer, as no voided checks are returned to him at the point of sale.

7.11.3 ARC (Accounts Receivable Entries) Check Conversion

With ARC, checks are received by the biller through the mail, at a central location or drop box, and are converted into one-time electronic debits, which are batched and sent through the ACH network. The biller retains the check until the payment has been processed, at which time the check is destroyed.

Steps in the ARC Process:

- The biller mails a statement to its customer. (As customers must be notified that their checks will be converted and offered the option to “opt out”, the billing statement itself is a good format for notification)
- The payer pays the statement by check mailed either directly to a centralized billing location, or to a drop box.
- The payment check is processed
- The check MICR code is scanned

- It is determined if the check is eligible for conversion. It must be written for less than \$25,000 with no values in the auxiliary on-us line of the MICR. (“on-us” is a line of code sometimes utilized for routing information)
- One or more batches of ARC entries are electronically transmitted to the bank that then routes them through the ACH network
- The biller must ensure that the paper checks are securely stored so that there is no opportunity for double submission: once as the electronic debit, and again as a paper check. Once the payment has been processed, the checks are destroyed.

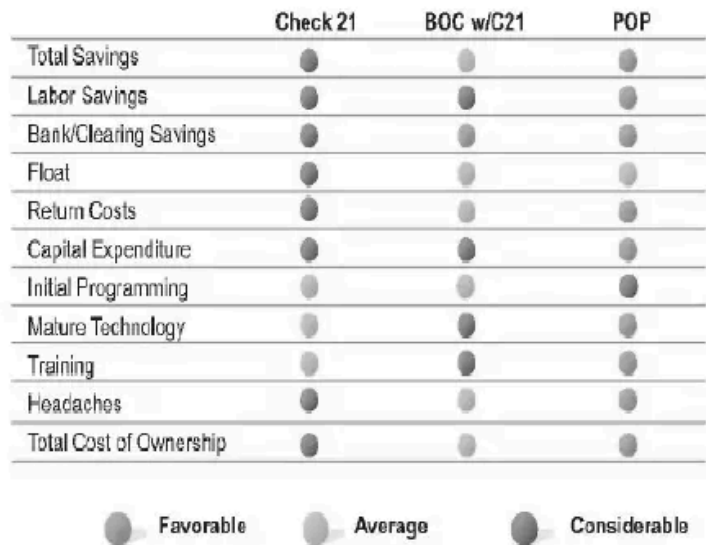
Any organization (business, not-for-profit, utility, etc) receiving volumes of check payments can take advantage of ARC.

Benefits include:

- Reduced processing times
- Reduced errors caused by manual processing
- Streamlined A/R reporting
- Decreased costs.

Interestingly, B2B companies are lagging behind B2C in utilization of e-checks. According to NACHA, in its March 5, 2007 “Check Conversion White Paper” in 2004, 80% of business-to-business transactions were [still] completed by paper check.

*Figure 4
Comparing the
costs of electronic
cheque processing
services*



Source: www.firstdata.com

7.11.4 Difference between Check Conversion and “Check 21”

Both check conversion and “Check 21” were developed for the same purposes: to increase the speed and efficiency of payment processing while decreasing the cost. Check conversion and the Check Clearing for the 21st Century Act (called Check 21), however, should not be confused, as the two are very different processes governed by entirely different rules and regulations.

With POP, ARC and BOC, the MICR code on the paper check is read by a piece of equipment and the data is used to create an electronic debit transaction which is then batch-processed and passed through the ACH system. With Check 21, scanners take pictures of the front and back of the check, and this digital image is used to create a substitute check that has the same legal weight as the original paper check. The substitute check is processed through the check clearing system, which has an entirely different set of regulations.

7.11.5 Impact of Check Conversion

Overall, Check 21, check conversion, and ACH processing have a positive impact on buyers, sellers and the U.S. economy as a whole. According to NACHA’s “White Paper” “the use of ACH in 2000 saved consumers, businesses, and the government approximately \$8.4 billion.”

There is, though, one drawback of check conversion that affects what is probably a good portion of consumers and business purchasers. That is the loss of the “float”. While check conversion does not result in an immediate draw from the payer’s account, as does, for instance, use of a debit card, converted checks will clear by the day after they are written. So ... the days of writing checks against future funds are rapidly waning. If a consumer writes a check, he should be sure that the cash is already in his account.

7.11.6 The Future of Check Conversion

The conversion of paper checks into electronic format, whether by conversion or Check 21, is definitely a growing business trend in the United States. A Federal Reserve Payment Study confirmed that electronic payment transactions in the United States exceeded check payments for the first time in 2004. The international consultancy firm for financial institutions, Celent, has projected that paper check processing will nearly disappear in the US by the end of the decade.

7.11 What is the difference between E-Check and Check Truncation?

An E-Check (Electronic check) is when a paper check is converted into an electronic debit at the point of sale. E-Checks are routed through the automated clearing house

(ACH) network. An E-Check is processed when a retailer scans check for financial information, and takes consumer's authorization before initiating the process of electronic transfer. Once the transaction is authorized the check is voided and is handed back to the consumer. A Substitute check is different from the E-Check in the way that it will appear as a check in consumer's statement not an ACH item.

7.12. Impact on Customers

By converting the original checks to the substitute checks banks can move the checks electronically through the payment systems. This will increase efficiency as well as reduce fraud in check collection and payment system.

Substitute checks usually take very less time to clear compared to traditional paper checks, meaning consumers have less float time. Once the substitute check is created the original check will be destroyed. It is assumed that over a period of time these substitute checks will replace the traditional paper checks.

7.13. Bank's Response to Check 21

There are many banking benefits of Check 21 system. First, it allows financial institutions to process checks faster, theoretically within minutes, thereby reducing account holder float. Secondly, it reduces the dependence on the transportation networks needed to move checks, including the associated costs and potential disruptions. The savings accrue from archiving checks (10%), electronic image exchange (65%) and distributed and fraud reductions (25%).

The legislation does not require institutions to present and receive electronic images for payment. The industry experts predict that mid-sized banks, those with assets of \$5 billion and more, are expected to follow suit and become image-capable by 2008. Half of the nation's small banks, with assets of \$100 million or less, are expected to use check truncation and become image-enabled by 2008.

7.14. Domino Effects of Check 21

- **Image Exchange:** Banks will have to invest in Image Exchanging infrastructure and software. With emergence of Image sharing and Image exchanges, it would still require the process in place to manage the transmission of check images and accept the returned ones.

Most of the US banks already have imaging capabilities for capturing and storing images. Today US banks spend approximately \$8 billion a year just on check processing and transportation. But with the adoption of check 21 roughly one quarter of these costs could be avoided

- **Image Application Cascade:** Once an image is captured and made available, the number of valuable uses to which it can be put is virtually unlimited. Application developers and product vendors are falling over themselves to make use of this huge opportunity.
Some of the uses like image statement delivery and positive pay have been in use for some time but applications like image lockbox and image self processing are still in nascent stage.
- **Distributed Capture:** The quicker the check is captured, the lower the expenses in terms of transportation, data entry and errors and frauds. Image capture at a branch is no longer a novelty, image capture at the ATMs and Point of Sale would soon be reality. Banks will have to be ready with applications to deal with the data and images coming in from 'n' different sources. By pushing capture out to its earliest point, banks can reduce their deposit processing expenses, achieve faster funds availability and share these benefits with the customers.
- **Payment Guarantee:** It refers to ability to guaranty check payments at the time the check is captured and imaged. With the wider acceptance of check imaging, banks will be in a better position to implement the much awaited shared utility and industry database that would enable them the guarantee of payments at the point of capture. It would go a long way in reducing the \$8 billion that the banks lose to check frauds annually. It could also lead to other types of frauds, like image tampering for which there would products required for validation, monitoring and fraud reporting.
- **Emergence of New players:** It's an opportunity for banks to enhance their market share through enhanced products and services to the customers. Banks who have missed out in the last race, especially the new ones, now have a renewed opportunity to define their niche and come out top. The same can be said about the technology vendors. Third party outsourcing providers are hugely benefited from the confusion in the market regarding the future of the check. It could be beneficial to company like ours which has completely missed out the last wave of check processing development. The current situation promises a huge opportunity for us to help banks reduce their investments.
- **Increased Outsourcing Potential:** The main back-office functions of exceptions, research, adjustments, and return items have been costly consumers of human resources. Image enablement of these back-office functions will remove the physical constraints and remove the dependence on local labor giving access to global labor supply. Outsourcing the back office payment processing would only be the first step in inviting the large third party

IT outsourcers to provide a comprehensive solution encompassing IT Development, Maintenance and Back office check processing.

DID YOU KNOW?

Carreker research indicates that, on average, U.S. banks can reduce approximately 10 percent of their of CIPC (CASH ITEMS IN PROCESS OF COLLECTION) as a result of Check 21 and image exchange initiatives.

7.15. Glossary of Terms for Check-21 and Check Conversion

Accounts Receivable Conversion (ARC): The conversion of consumer payments from checks to ACH transactions. For example, Wells Fargo Home Mortgage converts monthly payments received as checks to ACH, and destroys the original check. This creates a gap in the check sequence and means that the check would not be returned to customers in their monthly statement. ARC transactions currently appear in the "Other Withdrawals" section of the statement, along with other ACH payments.

Automated Clearing House (ACH): A nationwide electronic funds transfer system that provides for interbank clearing of electronic payments for participating financial institutions. Payroll direct deposit, Social Security and tax refunds, and direct payment of mortgages and utility bills are all examples of ACH payments.

Binary Image: A black and white image of a check where each pixel can be stored in memory by one bit of information since it is binary - either black or white.

Black and White Image: The print of the image of an original check or substitute check is only in black and white tones.

Bank of First Deposit: The bank where a customer deposits a check.

Check Conversion: The process by which a payment originating as a consumer check is turned into an electronic ACH debit.

Check Image: An electronic or digital image of an original check that is created by a bank or other participant in the check collection process. Check images can be exchanged electronically by banks through image exchange agreements, printed for customer statement purposes, displayed on Internet banking websites, and used to create substitute checks.

Check 21: The Check Clearing for the 21st Century Act —which was signed into law on October 28, 2003, and went into effect a year later — makes substitute checks the legal equivalent of original checks. A substitute check is a paper reproduction of an

electronic image of an original paper check. Under Check 21, the bank of first deposit (where the check is deposited) may present substitute checks instead of original checks to the paying bank (where the check is drawn). The law applies to virtually all check types. Check 21 seeks to:

- Increase the efficiency of check clearing in the United States
- Lower check processing costs
- Reduce the vulnerability of the check processing system to disruptions in air and ground transportation.

Check 21 does not require that banks send or receive checks electronically, it simply makes substitute checks the legal equivalent of original paper checks and requires that banks accept and process them.

Check Return: The service by which customers receive cancelled checks with their statements. Most customers choose the convenience of check safekeeping, where bank stores checks, and makes copies available on request. Many customers prefer to research and view their checks via, for example in case of Well Fargo bank, “WellsImage CD”, or online using the “Stops-Images-Search” service.

Check Truncation: The process, by which an original paper check is removed from the payment-processing stream, is archived as an image and replaced by a substitute check. The original check is usually destroyed after a short time.

Capture: Reading and storing data from the check MICR line to enable the funds represented by the check to move between banks and their customers.

Cash Letter: A group of checks packaged and sent by a Bank to another Bank, clearinghouse, or a Federal Reserve office. A cash letter is accompanied by a list containing the dollar amount of each check, the total amount of the checks and the number of checks in the cash letter.

Electronic Check Presentment (ECP): An electronic record governed by check law, created from the entire MICR line on a check and suitable for posting to a customer’s account. ECP transmissions may stand alone or may be followed by or accompanied by check images or paper checks. ECP transmissions are governed by check law.

Electronic Check (e-check): Used to refer to several types of electronic transactions that debit a checking account. An electronic check is not a substitute check.

Forward Collection: The transfer of a check by a Bank to a Paying Bank for payment. That is, the Bank forwards the check to another Bank directly or through an intermediary.

Float: The time elapsed between the time a check is deposited in a bank and the time the bank receives the funds.

Gray Scale Image: The print of the image of the original check or substitute check is in black or shades of gray.

Image: A digital representation of all or part of an original check or substitute check.

Image Exchange: An exchange of some or all of a digitized image (or images) of a check.

Image Capture: The act of reading or digitally recording a check or other document so it can be reproduced as an exact replication when needed. To create a substitute check, a bank first must image capture it.

Image Exchange: The electronic transfer of the MICR data and the front and back of an image-captured check between financial institutions for the purpose of clearing the check. Image exchange agreements must be in place between the exchanging parties.

Image Quality: The visual characteristics and readability of a check image, including the MICR line.

Image Statement: A statement that contains images of cancelled checks rather than the original cancelled paper checks.

Image Replacement Document (IRD): Another term for substitute check. The term under Check 21 is substitute check.

Magnetic Ink Character Recognition (MICR) Line: Numbers printed in magnetic ink near the bottom of the front of the check to facilitate automated processing. These numbers identify the bank the check is drawn on (the paying bank), the account number, the check number, and other information. The position and content of the MICR line are governed by industry standards.

Point of Purchase (POP) Conversion: This occurs when consumer check payments are converted to ACH transactions at the point of sale. For example, a cashier receives a check, then scans it through a MICR machine, creating an ACH or electronic transaction to directly debit the customer's checking account. The cashier voids the check, asks the customer to sign an authorization, and hands the check back to the customer. This process creates a gap in the check sequence, and such transactions have appeared in the "Other Withdrawals" section of the statement. Customers now see a clearer description of POP transactions on their statements.

Position 44: The legislation states that substitute checks will use position 44 of the MICR line to identify an item as a substitute check. A "4" in that position labels it as a substitute check for forward presentment; a "5" designates it as a returned item in the form of a substitute check.

Reconverting Bank: The bank that creates a substitute check; or if a substitute check is created by an entity other than a bank, the first bank that transfers, presents or returns that substitute check.

Chapter- 8 e-Check

8.1. Introduction

Electronic check is a technological innovation in the financial space. We shall try to understand the technology behind it, its structure and different operational models of e-check.

8.2. Learning objectives

After reading this session you will be conversant with

- Concept of e-checks
- Structure of e-check
- Flow of e-check
- Application of PKI in e-check.

8.3. Topics covered

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8.4. E-check – An Introduction

Checks are a very important part of the non-cash retail transactions. For many of the daily payments we need to write checks. But the processing of checks takes considerable amount of time. It is ironic that our own money gets trapped in the banking system as float due to considerable time taken for the clearing and settlement of these payment instruments.

With the advent of information technology and its profound impact on almost every business area today, the form of checks has undergone a transformation.

Electronic check is a payment instrument which comes with a two fold advantage, that it amalgamates the advantages of electronic transactions like speed, security and efficiency with the already existing legal infrastructure and business processes in place for the paper based checks. It is the first and the only electronic transaction mechanism chosen by the United States Treasury to make High Value Payments over public network.

E-check is an all electronic end-to-end payment instrument which works just like a normal paper check (electronic equivalent of paper check) and has the same legal treatment. It is a new payment instrument which combines speed and efficiency in processing of all electronic transactions. The e-check was developed as an initiative of the Financial Services Technology Consortium (FSTC). The FSTC is comprised of about 100 members, including most of the major banks, suppliers of technology to the financial industry, universities and research laboratories. *The technical work of the Electronic Check Project was carried out in a number of phases: generating the original concepts, performing preliminary research, building and demonstrating a prototype, formulating specifications for a pilot system, and implementing the pilot system.*

The vision behind the development of e-check is to substitute the paper based document with the electronic document by replacing the handwritten signatures with public key cryptographic signatures.

- The payer writes an e-check by structuring an electronic document with the information legally required to be present in a check and cryptographically signs it
- The e-check is received by the payee, he then verifies the payer's signature, the e-check is then endorsed by him, writes out a deposit, and signs the deposit
- The payee's bank verifies both payer's and payee's signatures, credits the payee's account and forwards the check for clearing and settlement
- The payer's bank verifies the payer's signature and debits the payer's account.

DID YOU KNOW?

According to NACHA (the electronic payments association)- More than 200 million e-check payments were made during first half of 2002, an increase of over 300 percent over the same period of 2001.

A recent study by Federal Reserve shows that check use in US is declining. But with billions of checks still being written every year at the point of sale and for bill payments, e-checks are poised for substantial growth and will contribute to the decline of paper check processing.

Here is a flow of e-check:

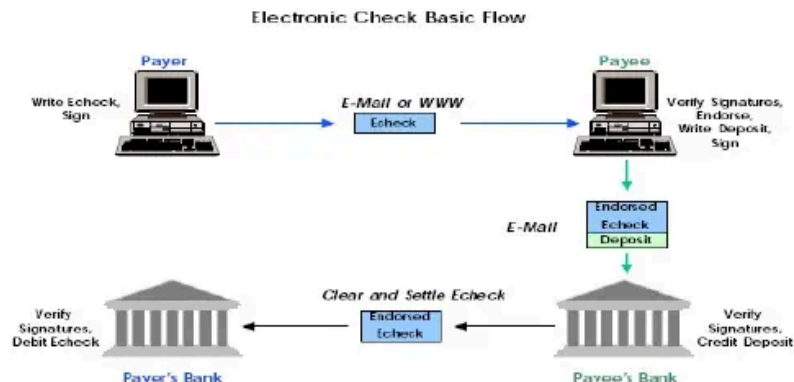


Figure 8.1 Flow of an e-check

Note: The advantage of e-checks is that the cryptographic signatures can be verified on every e-check, while handwritten signatures have no provision to be verified because it is very tedious.

8.5. Benefits of an e-Check

The obvious question arises here is that why do we need e-checks and what are its benefits. Following is a list of some important reasons for use of e-checks:

➤ **e-check saves cost** - A very surprising finding by the Federal Reserve says that currently the total costs of handling paper checks to the US economy is approx. \$44 bn per annum. While some other estimates also claim that it could be as much as 2 percent of the total GDP!

Electronic check will have a major benefit to drive this cost down. Certain elaborations below will explain this.

➤ **Time Savings with e-check:** Various time lags at various stages of paper check processing increases the total time for the clearing and settlement of checks. However the following are the times savings payer, payee and the financial institutions will have with e-checks.

- Waiting time while the paper checks are picked up from one bank to the clearing house for verification and further procedure by the transportation services or the postal services is totally eradicated with e-checks.
- Reduced/ eliminated mail float or lead time for bill payment instructions
- No paper envelopes to stuff or open. This may sound funny but the various support accessories like envelopes become mandatory while we transfer checks from one place to another. Hence this also wastes time but with the case of e-check this is also eliminated

- In the smaller cities checks are totalled manually hence with e-checks this additional step of counting manually is eliminated, which again saves number of man-hours
 - Additional requisites like printing check deposit slips are also eliminated
 - E-check facilitate improved document tracking and it's very easy to retrieve stored information too
 - Strong auditing and in the case of errors it's easy to track and rectify them
- **E-checks are well suited for clearing micro payments**, the conventional cryptography makes processing easier as compared to systems based on public key cryptography.
- **Electronic check technology connects the public networks** to the financial payments and bank clearing networks and hence leverages the accessibility of public networks with the existing financial payment infrastructure
- **Staff steps eliminated with e checks** with all payer, payee and financial institution's perspective. With the change of business process some steps taken by staff to process checks will change and hence it is found that man hours will be saved resulting into substantial saving.

Table 8.1 Steps saved by the business payer

Steps saved by the Business Payer	
Step	e-check
Management of the check stock for example order, storage, control, retrieval and destruction of checks.	Eliminates
Management of envelopes (order, store, retrieve, destroy)	Eliminates
Managing signature plates or font (order, lock up, retrieve)	Reduces; requires holder control of Electronic Checkbook
Loading checks in printer to print information.	Eliminates
Stuffing envelopes with checks and remittances.	Eliminates
Cash forecasting, the management of cash	Reduces and becomes easy; e checks will

Steps saved by the Business Payer	
Step	e-check
of working capital in the banks.	clear more consistently and have the better predictability for the cash requirements at the bank.
Reconciliation for outstanding checks.	Reduces; fewer remain outstanding for long periods
Record keeping and filing	Reduces; no paper checks to keep on file
Researching payments	Reduces; records on computer, no need for paper copies



Table 8.2 Steps saved by Business Payee

Steps saved by Business Payee	
Step	e-check
Mailroom (Receive envelopes in mail, sort envelopes by department, deliver)	Eliminates
Retrieve payments (retrieve mail, open envelopes)	Eliminates; system performs these functions manually or automatically
Verify payment information (match remittance, payment amount, conditions against accounts receivable)	Reduces; will eliminate with EDI {Electronic Data Interchange}
Verify check (appears legitimate, properly signed, paid to right party, right amount, dated, no hidden conditions, etc.)	Eliminates
Create payment record and file (photocopy check, store in paper company files)	Reduce/Eliminate; records stored automatically on computer
Prepare deposit (endorse checks, bundle, prepare deposit ticket, total)	Reduce/Eliminate; system automatically performs all steps except endorse
Make deposit (go to bank or night deposit box, give to teller, get receipt)	Eliminates; system automatically sends deposit to bank
Statement Review	Reduced. More descriptive statement information available

Table 8.3 Steps reduced by Financial Institutions

Steps reduced by Financial Institutions	
Step	e-Check
Receive check deposits in branch or night deposit (teller inspection, deposit ticket proofing, bundling, balancing, bagging)	Eliminates
Branch pickup	Eliminates
Item Preparation and Proof and Transit (delivery, unbagging, power encoding, balancing, error correction, batching,	Eliminates

Steps reduced by Financial Institutions	
Step	e-Check
pickup)	
Suspicious item inspection (outsourcing, delivery, manual inspection, signature card lookup, paper tests, etc., pickup)	Eliminates manual effort. Every eCheck is automatically validated for signatures, dates, etc.
MICR Capture (may include image capture)	Eliminates
Reject repair	Eliminates; e-checks never need be "repaired"
Automated Sorting	Eliminates physical sorter passes. All sorting done electronically. Fine sorts can be done in a single pass.
Bundling, cash letter preparation, bagging	Eliminates manual steps. Cash letters automatically prepared for both.
Clearing (bag pickup, handoff to transportation company/group, delivery to clearing house, Fed, or other Financial Institution)	Eliminates manual effort. All clearing is electronic.
Inclearing (bag delivery, MICR capture, etc. as above)	Eliminates manual effort. All in clearing is electronic.
Returns (item pull, storage, bundling, cash lettering, etc., transportation to returns clearing house or collecting bank)	Eliminates manual effort. All returns are electronically processed.
Filing (item filing, microfilming, statement retrieval)	Reduces; all items available online through data base inquiry. No physical check to include in statement.
Statement Rendering (retrieve checks, fine sort, insert checks into statements, re-image for image statements, stuff envelopes, etc.)	Reduces; no physical items to handle
Research (item pull, finding on microfiche, photocopying, deciphering the back, etc.)	Reduces

➤ **The reduction of Risks and Frauds by e-check-** The table below shows various risks or frauds and parties exposed to them and how they can be eliminated by the e-check.

Table 8.4 Risk and Fraud Reduction by E-Check

Risk Description	Parties with greatest exposure to loss	e-check impact
Check to be stolen	Primarily banks and payee	Reduced through PIN-protected hardware
Unauthorized issuing of Checks	Payer	Reduced through personal assignment of signing tokens. Also reduced through support for dual signatures.
Forgery	Payee, Paying Bank	Virtually eliminated through digital signatures, automatic verification, and PIN-protected hardware signing keys.
Forgery where Facsimile signatures are used on account	Payee, Account owner	Eliminated
Counterfeiting	Paying Bank, Payee	Virtually eliminated through digital signatures, automatic verification, and PIN protected hardware signing keys.
Duplication	Innocent Payees, Depositing bank for fraudulent payee, Paying bank	Reduced through rigorous duplicate detection functionality, use of payee public key in addition to name in pay to field.
Fraudulent unsigned demand draft	Payee, Depositing Bank, Paying bank	Eliminated.
Alteration of payee name	Depositing bank, fraudulent payee, payer	Effectively Eliminated
Alteration of amount (no	Depositing bank,	Effectively Eliminated

Risk Description	Parties with greatest exposure to loss	e-check impact
positive pay)	payee	
Alteration of amount (with positive pay)	Depositing bank, payee	Effectively Eliminated
Amount encoding error	Depositing bank, payee inconvenience, small errors often written off by banks	Eliminated
Non-sufficient funds, account closed	Payee	Reduced due to faster, more consistent clearing times.
Stop Payment	Payee	Reduced due to faster clearing.
Check drawn on non-existent bank or account	Payee	Eliminated due to digital certificates issued by banks to account holders

Other Handling Costs saved: While there are some obvious costs which will be saved by the implementation of e-checks. The following table shows the costs saved in dollar terms for the paper check handling.

Table 8.5 Paper check handling cost and saving by e-check

Item	Typical Cost	Saved by
Paper check stock	\$.02-\$.25	Payer
Paper remittance forms	\$.02-\$.15	Payee
Envelopes	\$.02-\$.10	Payer or Payee
Postage	\$.22-\$.33	Payer, Bank on statement
Photocopies of checks	\$.02-\$.05	Payee, Bank on research items
Filing cabinets, storage space	Varies	Payee, Payer, Bank

While the above savings may seem small, the total cost per check turns out to be \$.5 which is significant savings if e-checks are used.

➤ **Enhancement in Customer Service:** With e-checks banks can save their costs and as well enhance their customer service manifold.

Table 8.6 Customer Service Enhancement

Wider Choice	eCheck enables financial institutions to provide their customers an increased array of cost-effective electronic payment alternatives. Its design, which provides true end-to-end security, makes it suitable for use by any bank customer, even in insecure environments like the Internet.
Better Information	eChecks provide better, more accessible, information about the transaction than paper checks. For example, on the payer's bank account statement, eCheck transactions will show the payee name in addition to the check number, date, and amount.
Pay Anyone	eChecks are designed to allow payment between any two parties, for most types of transactions. Anyone with a bank account can receive e-checks. As with paper checks, we expect services to be developed to enable anyone to receive an eCheck, even if they don't have a bank account.
Pay and Receive	eChecks are designed to enable the same payment instrument to be used for both paying and receiving payments. Many other instruments, such as debit cards, are designed primarily as retail payment vehicles, and can therefore be used by consumers only to make payments.
Do business directly	eChecks are designed for direct exchange between transacting parties, rather than requiring an intermediary in the middle of the transaction. They support direct exchange of transaction information, and payment over the Internet.
Familiarity	Since e-checks are modeled after paper checks, they are familiar, and don't require a significant effort to learn new terminology or processes.
Accepted Legal Basis	eChecks build on the most familiar payments legal infrastructure in the US today, check law. Banks and customers understand the risks of this system, and know which risks are acceptable on a daily basis. Users of the system can be assured the system is stable and relatively unchanging.
Payer Control	Through the use of strong digital signatures, e-checks enable only authorized transactions to be posted to a bank account, and the authorization is checked every time. Payer's control and

	banks can enforce who can make payments from their account, and how much they can do. Business practices, such as dual signatures and transaction limits, are part of the system design, and automatically enforced.
Payee Control	Payees can receive e-checks directly, or can use electronic lockbox services to simplify their processing. In addition, payees have control over which payments to accept or reject, which account to deposit into, how much risk they are willing to take, and the timing of depositing the transaction (which is particularly important for cash basis businesses managing quarterly results).
Co-exist with paper. Helps transition to electronics	eChecks are designed to coexist with paper, reducing the confusion and complexity of accepting eCheck transactions into businesses that currently receive paper checks. They help businesses migrate in a cost-effective, non-disruptive manner from paper to electronic payments.

8.6. Architectural Specifications for e-checks

In this section we will look into the various technological aspects defining the format of an e-check and the work process flow of e-checks. Paper based checks have information both handwritten and printed on it. Amongst all of them payer's signatures is one of the most important information. It clearly is the mode to validate the check, but in electronic checks this issue of certification and generation of unique identification such as a signature is resolved differently. An alternative of checkbook in the e check systems is discussed. The markup language used to specify body of check and other issues are handled below.

8.6.1 Financial Services Markup Language (FSML) - *The language to define e-checks*

FSML is a mark up language designed to allow the creation of electronic financial documents. An e-check must have cryptographic information for its unique identification and other fields like MICR numbers, amount etc need to be stored in a check.

Hence an e-check is written in FSML. It's a markup language defined in SGML (Standard Generalized Markup Language). The document structure and data items for e-checks are delimited by "tags" similar to those used in HTML (HyperText Markup Language), another language defined using SGML. FSML is designed in a manner to support the

data structures and the cryptographic information required in an e-check. The e-check written in FSML will contain all of the information that is normally found in paper checks, including that which is handwritten, preprinted and in the magnetic ink character line at the bottom of the check.

- FSML includes signature blocks designed to support the addition and deletion of FSML document blocks and to support rich signature semantics to enable signing, co-signing, and endorsing of e-checks by the several parties who may sequentially process an e-check. FSML also provides the ability to encapsulate and cryptographically attach other documents, such as an advice of payment, invoice, or remittance information to enable the payee to correctly post the payment in accounts receivable.
- To ensure the widest possible compatibility with electronic mail, HTTP (HyperText Transport Protocol) and other types of transport, FSML specifies a limited line length and a restricted ASCII character set for encoding all e-check data.
- Each FSML Document is a sequence of blocks. The start and end of a block is specified by a start and end tag. The start tag and end tags identify the type, and hence the contents, of each block. Each block starts with block name tag, criticality and version tag. We shall see contents of check block and account block.

8.6.2 Check Block

In check block, the first part of the block content will be a sequence of data items which are logged by the electronic checkbook (will be discussed later). This gives the signer a secure, portable record of the most important items which have been signed by the private signing key in the electronic checkbook.

The number of the electronic checkbook and the check number are automatically inserted by the electronic checkbook to ensure uniqueness of each e-check. The check block requires the date that the e-check becomes valid and the date that the e-check was issued. The writer may enter the country that the e-check was written in order to comply with local laws. The writer must specify the amount and type of currency, and the "pay to the order of" data item must be present. However, in order to unambiguously identify the payee, the payer may specify the payer's bank and account or customer number at the bank. Alternatively the payer may specify the payee's public key. A payer may place restrictions on the e-check, such as to

Table 8.7 Check Block

Tag	Data	Purpose
<check>		Start tag of the check block
<bllname>	string	Name by which the block is referenced
<crit>	"true"	Check is a critical block
<vers>	decimal	Version number of this block
<checkdata>		Start tag of data elements (shaded area below) to be logged in electronic checkbook register
<checknum>	integer	Check number automatically incremented by electronic checkbook
<dateissued>	ISO date	Year, month, day that check was written
<datevalid>	ISO date	Year, month, day that check is payable
<country>	ISO country	Optional, check is subject to this country's law
<amount>	decimal	Numerical amount of currency
<currency>	ISO currency string	3 character currency code, e.g. USD for US dollars
<payto>	string	Name of person to whose order the check is payable, may repeat for joint or several payees
<paytoabank>	string	Optional, bank routing code of payee's bank
<paytoaccount>	string	Optional, payee's account at payee's bank
<paytocustno>	string	Optional, payee's unique identifier at payee's bank
</checkdata>		End tag of check data elements (shaded area above) to be logged in electronic checkbook register
<checkbook>	integer	Manufacture, model, serial number of payer's electronic checkbook
<restrictions>	string	Optional, "duration pynummd", "for deposit only" or "all payees must endorse" (pynummd is an ISO standard format for representing a period of time as n years, n months and n days).
<paytokey>	hexstring	Optional, public key to be used to verify payee's endorsement
<payeracct>	string	Optional, payer's account with payee's business
<memo>	string	Optional
<conditions>	string	
<legalnotice>	string	
</check>		End tag of check block

distinguish between writing an e-check to "John and Mary Smith", which both must endorse, or an e-check to "John or Mary Smith" which either can endorse. Just as on paper checks, the payer may enter memo information as a reminder of the purpose of the e-check. A specific memo field is provided which the payer may use to enter the payer's account number at the payee's business to assist the payee in correctly posting the payment in the payee's accounts receivables. Lastly, the check block carries a legal notice that the e-check is subject to check law.

Table 8.8 The FSML Account Block

Tag*	Data	Purpose
<account>		Start tag of the account block
<blkname>	string	Name by which the block is referenced
<crit>	"true"	Account is a critical block
<vers>	decimal	Version number of this block
<bankcode>	string	Bank routing code of the issuing bank
<bankacct>	string	Account number of this account at the issuing bank
<bankser>	integer	Serial number of this account block, assigned by the issuing bank
<expdate>	ISO date	Expiration date of this account block
<accttitle>	string	Optional, account title
<accttype>	string	Optional, account type
<bankname>	string	Optional, name of the issuing bank
<bankaddr>	string	Optional, address of the issuing bank
<acctrest>	string	Optional, a string containing "minimum amount nnnnn.nn ccc", "maximum amount nnnnn.nn ccc", "n signatures required", "n signatures required above amount nnnnn.nn ccc", "special processing", or "currency ccc".
<certissuer>	string	The unique distinguished name of the issuer of the certificate that can be used to verify signatures on FSML documents containing this account block.
<certserial>	integer	The unique certificate serial number of the certificate.
</account>		End tag of account block

8.6.3 Account Block

The account block shown in Table 12.8 contains the bank code and account number that corresponds to the bank code and account number that would be found on the MICR line of a check, and it contains a reference by issuer and serial number to a certificate containing a public key that can be used to verify signatures on e-checks drawn against the account. It may also contain restrictions on the value, the maximum time interval before the e-check becomes stale, and requirements on the number of signatures required for e-checks above certain values. A bank customer may have more than one account block corresponding to the several accounts over which the customer has signature authority. There also will be multiple account blocks issued on a single account when the account is held jointly or in the case of commercial accounts where several persons have signature privileges.

The following **Table 8.9** gives a comprehensive overview of the total number of blocks defined for an e-check.

8.7. Cryptographic Signatures in E-checks

Till now we discussed about FSML, about the various blocks an e-check has and how it carries information. Now we shall see how the digital signatures of a payer are made and put on an e-check.

Figure 8.2 which follows this description will help you to understand the process. The figure explains the process of creating a digital signature and putting up that signature on the e-check. The hash code generated will be the unique identification of the payer. For the transmission purposes it will be encrypted with the private key of the sender and public key of the recipient. At the receiving end the private key of the recipient and the public key of the sender will come into use to decrypt the document.

The signature block in FSML is shown in **Table 8.10**

When an e-check is originated, a minimum set of information is written and signed. As it is processed, more information and more signatures will be added as it is passed from party to party. For example, an e-check may be:

- originated by a payer,
- co-signed by a co-payer,
- certified by a bank,
- endorsed by a payee,
- co-endorsed by a co-payee,
- deposited, and
- paid.

This requires a flexible document structure and signature mechanism. The principle characteristics of the FSML signature mechanisms are:

- The document consists of a sequence of blocks, and blocks may be nested.
- Signatures implement cryptographic signatures and/or hashes on other blocks referred to by name.
- Signatures may sign other signature blocks.
- Signature blocks refer by block name or by issuer and serial number to the certificate block which contains the corresponding public signature verification key.

These characteristics provide enough flexibility to handle the signature requirements of electronic checks and the signature requirements of a broad range of financial documents.

The signature block contains the names of the other blocks being signed, and the corresponding hashes computed on those blocks. The <blockref> and <hash> data elements are repeated in pairs (as shaded) when multiple blocks are being signed. The random nonce is generated by the electronic checkbook prior to hashing any of the blocks, it is prepended to each block, and it is also included in the scope of the hash calculation on the signature block itself. This defeats attacks where the attacker fabricates two messages that have the same hash value, persuades the victim to sign one, and then substitutes the other.

The signature block also contains a reference to the public key to be used in verifying the signature, either by naming another FSML block or by referencing the X.509 certificate by the Certificate Authority's distinguished name and the certificate serial number.

The signer can choose to include other personal data, such as name, address, phone number, email address, etc. These data items are stored in the electronic checkbook when it is initialized by the bank, and are changeable only after the electronic checkbook has been unlocked using the bank's administrative PIN. This method of providing personal information is not as secure as including the information in the X.509 certificate or the account block. However, it is expected to be secure enough for purposes such as analysis and decision making by check guarantee services, while providing the bank's customer with complete control over which personal information is released to which payee.

The electronic checkbook signature is actually a signature on a hash of hashes. The advantage of this design is that the signature can be verified for any subset of the blocks which were originally signed. However, it also implies that if any blocks are absent, the application can determine whether the subset of blocks still present constitute a valid e-check. For example, attachments which might contain remittance or other business information are deemed not to affect the validity of the e-check, and attachment blocks can be removed by the payee or discarded by the bank of first deposit with impunity.

Figure 8.2 shows how signatures are applied to the e-checks. The right hand set of blocks are the check as made out by the payer. The payer's signature signs the action, check, account, attachment and invoice blocks. The payer's signature block references the payer's account block which references the payer's certificate. The payer's certificate contains the payer's public key, which verifiers use to verify the payer's signature. The payer's bank has signed both the payer's account and certificate blocks, and the payer's bank's certificate contains the public key needed to verify the bank's signature.

The endorser has signed the action and endorsement blocks of the endorsement, plus the check and the payer's signature on the e-check being endorsed.

The depositor has signed the action and deposit blocks of the deposit, plus the endorsement and endorser's signature of the e-check being deposited. Note that since

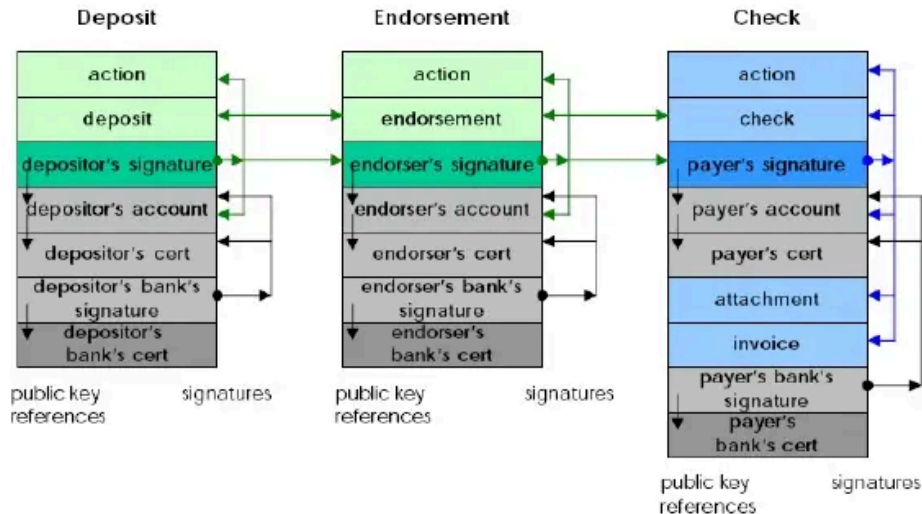


Figure 8.2 E-check signature and endorsement

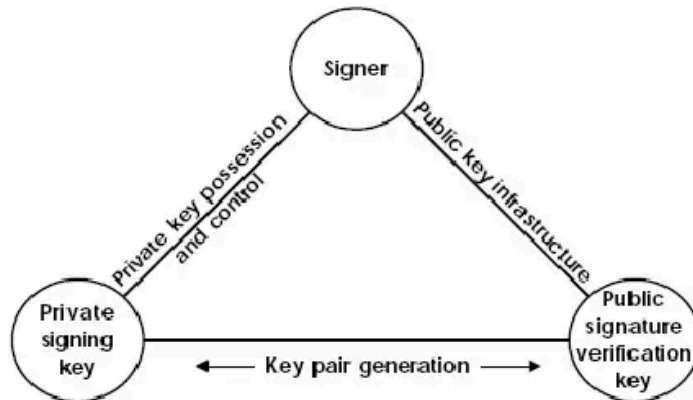
the payer's signature block only contains the hashes of the attachment and invoice blocks, when these blocks are removed by the depositor, the payer's signature can still be verified on the remaining check blocks by the paying bank.

8.8. Electronic Checkbooks

A handwritten signature captures the reflexive movement of the signer's muscles and is partly a biometric characteristic of the signer. This makes it difficult for a forger to create a perfect forgery, even if the forger has a sample of a handwritten signature.

However, a perfect forgery of a cryptographic signature can be made by anyone who has the signer's private signature key.

Therefore, it is critical to establishing an e-check system based on public key signatures that payees and banks are able to trust that payers can maintain possession and control over their private signing keys at all times. Electronic checkbook smart cards or other cryptographic hardware are used to help ensure that signatures are made only by legitimate signers, so that check forgery is made difficult. Electronic checkbook cards also standardize and simplify key generation, distribution and use, so that a high and uniform level of trust can be established without depending on the skill and diligence of every e-check customer.



Source: www.e-check.org

Fig8.3: Public Key Signature Security Fundamentals

Verifiers must believe three types of things about the signer and the signer's private and public signing keys:

1. Private key possession and control – The signature verifier must believe that the signer has exclusive possession of his signing key. If an attacker can get possession of the signer's private key, then the attacker can forge signatures using his own system from any location on the network. If the attacker can gain control of the signer's computer, then the attacker can forge signatures without the signer's knowledge.

Table 8.9 The Blocks defined for the E-check

Block Type	Usage
<account>	A block containing information about the payer's or payee's account, such as the bank code, account code, account title, signer's name and address, and the issuer and serial number of the signer's certificate that may be used to verify signatures on echecks drawn against the account. The account block is signed by the bank holding the account.
<action>	A block describing the action to be performed by the recipient.
<attachment>	A block containing information sent from the payer to payee, and which is to be associated with the echeck.
<bankstamp>	A block containing processing status information.
<bundle>	A block containing totals for a bundle of echecks being sent between banks.
<cashletter>	A block containing cash letter totals for echecks being sent between banks.
<cert>	A block containing an X.509 certificate. Certificates are needed for bank customers, bank tellers, bank information systems, and certificate management purposes.
<certification>	A block containing information provided by a bank, along with the bank's signature, to create a certified echeck.
<check>	A block containing echeck information such as the amount, pay-to-the-order-of, and the date on which the echeck becomes payable.
<deposit>	A block containing information about a set of echecks being deposited as a batch.
<endorsement>	A block containing information supplied by the endorser, such as date, "for deposit only", etc.
<invoice>	An invoice/remittance document containing payment information sent between payer and payee.
<signature>	A block containing a list of the names of other blocks being signed, a list of their hashes, a reference to the public key certificate needed to verify the signature, and a cryptographic signature on the above.

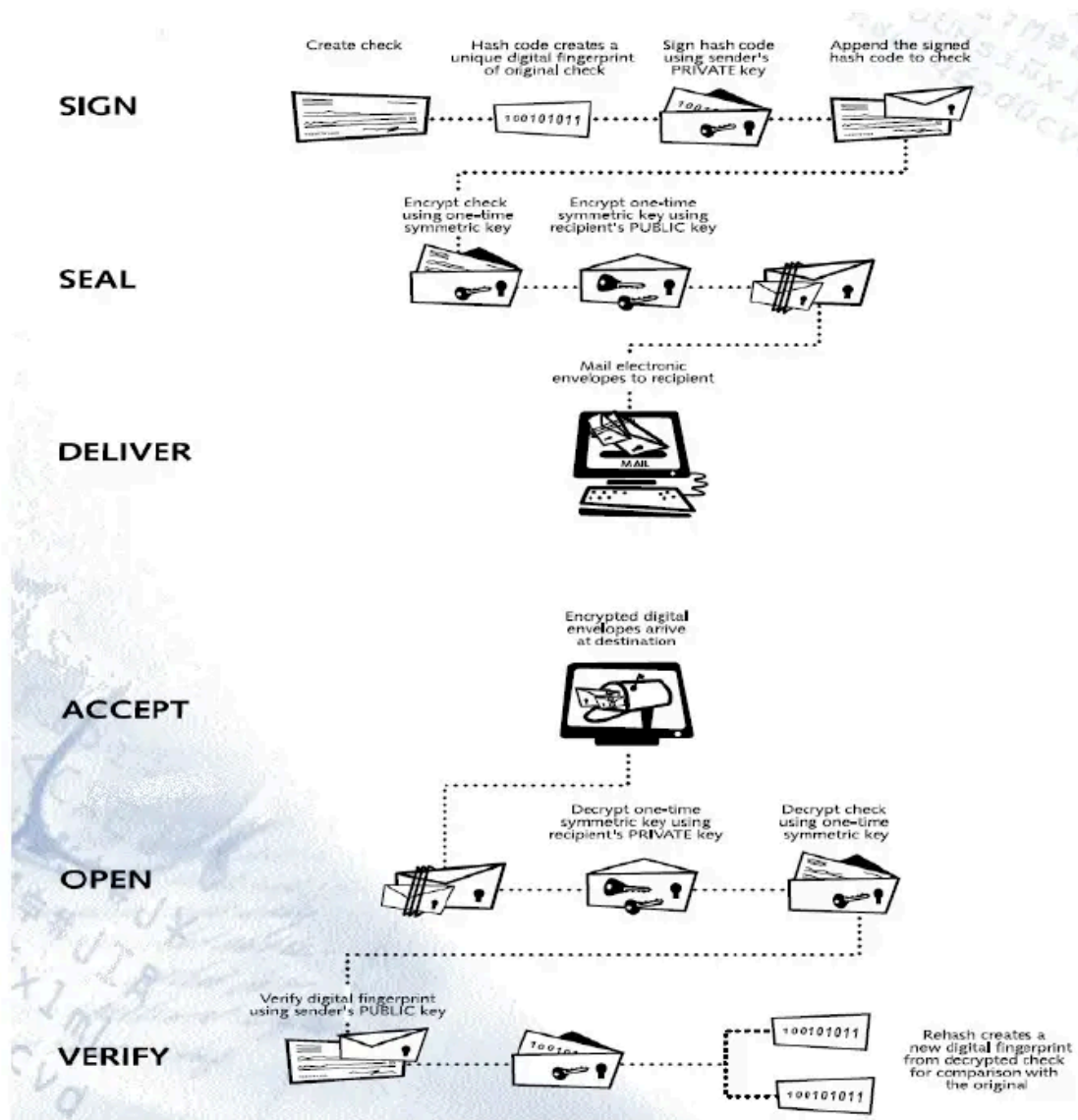


Figure 8.4 Signing a e-check (source: www.entrust.com)

The electronic checkbook, in the form of a PIN-activated tamper-resistant smart card or similar cryptographic hardware, performs the signing algorithm so that the private signing key is always kept inside the trusted hardware and is never read into the signer's much more vulnerable networked personal computer or server. The electronic checkbook is aware of e-check syntax and logs critical data from e-checks to provide the signer with a trusted log of signing actions.

2. Key pair generation – The signature verifier must believe that the private/public key pair was generated such that the private key cannot be calculated or guessed by an attacker based on knowledge of the public key.

The electronic checkbook performs key generation within the tamper-resistant hardware using algorithms that have been properly tested and certified by the

Table 8.10 Signature Block

Tag*	Data	Purpose
<signature>		Start tag of the signature block
<blkname>	string	Name by which the block is referenced
<crit>	"true"	Signature is a critical block
<vers>	decimal	Version number of this block
<sigdata>		Start tag of data in this block which is hashed and signed by the cryptographic signature at the end of the block
<blockref>	string	Name of the block whose hash is next
<hash alg="TYPE">	hexstring	Hash of the block just named The TYPE attribute identifies the hash algorithm used to hash the block, e.g. "sha" for NIST Secure Hash Algorithm
<nonce>	string	A random string generated by the electronic checkbook and used to render the blocks unpredictable to attackers before they are hashed
<sigref>	string	Optional, the name of an account block or cert block containing a reference to the certificate or the certificate needed to verify this signature
<certissuer>	string	Optional, the unique distinguished name of the issuer of the certificate that can be used to verify signatures on FSML documents containing this account block.
<certserial>	integer	Optional, the unique certificate serial number of the certificate.
<algorithm>	string	Hashing and signing algorithms used to compute the signature on <sigdata>, either "md5/rsa", "sha/dsa", or "sha/rsa"
<username>	string	Optional, the signer's name
<userphone>	string	Optional, the signer's phone number
</sigdata>		End tag of data in this block which is hashed and signed by the cryptographic signature at the end of the block
<sig>	hexstring	Cryptographic signature computed by the electronic checkbook using the signer's private key
</signature>		End tag of signature block

manufacturer. Only the public key is exported from the hardware, and the private key is never revealed to anyone.

3. Public key infrastructure – The signature verifier must be able to trust that the public key provided for use in verifying the signature really belongs to the signer and is the other half of the signer's public key pair.

The electronic checkbook is initialized using procedures based on bank card issuing processes. The public key exported from the card is included in an X.509 certificate

signed by the bank's Certification Authority, and associated with an account block also signed by the bank's Certification Authority. The bank e-check servers also keep an independent database of the bank's signer's public keys, such that they always know the most current relationships of keys to accounts and signers.

Besides key management and logging functions just described, electronic checkbooks also:

1. Include the checkbook's unique number (manufacturer, model and serial number) in each e-check,
2. Consecutively number each e-check as it is signed, in order to ensure that each e-check is unique,
3. Generate random numbers that are prefixed to blocks to increase security of the hashing functions,
4. Contain signer personal data which the signer can selectively apply to e-checks,
5. Separately unlock e-check writing, checkbook administration and bank administration functions using PINs,
6. Deactivate if PIN hacking is detected.

Furthermore, the design of the electronic checkbook must be such that the private signing key cannot be extracted via its electrical connector and any successful attempt to extract the private key will visibly damage the electronic checkbook and render it inoperable. The security concern with the e-check is that whoever has the private key of the signer can forge perfectly the cryptographic signature of the signer. For the prevention of these attacks the tasks of key generation and management, logging important information is done by a PIN protected and tamper resistant smart card known as an electronic checkbook.

8.9. Flow of E-checks

The following **figure 8.5** shows the flow of e-check in the system

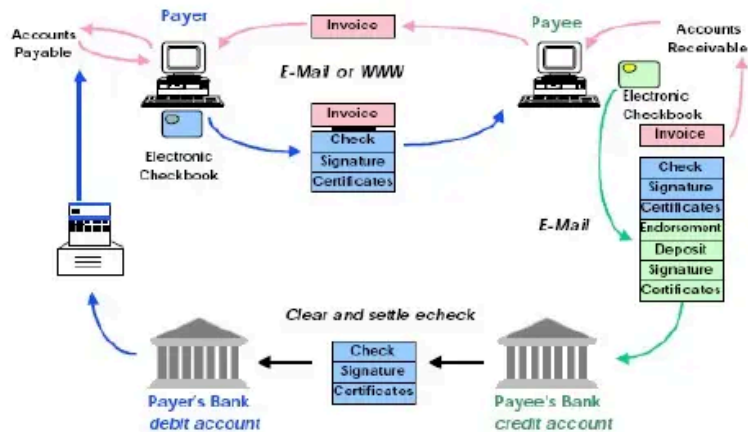


Figure 8.5 Flow of e-check source: www.e-check.org.

e-checks are currently cleared and settled between the banks using the ANSI X9.46 and X9.37 standards.

The Electronic Check Clearing House Organization (ECCHO) has adopted rules for inter-bank clearing of electronic checks which consider the e-check to have the status of a "negotiable instrument" for the purposes of the Uniform Commercial Code and a "check" for purposes of Regulation CC. This is reinforced by including a legal notice within the e-check which states "This instrument subject to check law", as well as by uniform customer agreements for use between the banks and their customers, which agree that e-checks are subject to check law. If a consumer, rather than a business, writes the electronic check, then Regulation E also applies. Regulation E provides consumers with additional rights and protections, and it supersedes Regulation CC where they overlap.

In the normal check flow described here, the banks providing e-checking accounts must be members of ECCHO, members of an equivalent organization or clearing house providing clearing and settlement rules, or they can agree to clear and settle e-checks bilaterally. An exception to this is the alternative e-check flow described in **Figure 8.5**, 8.6, and 8.7 below.

The business transaction begins with the payee sending an invoice or bill to the payer, which is processed by the payer's accounts payable system. When the time comes to pay the invoice, the invoice information is retrieved from the accounts payable system, and the invoice data is used to create an e-check. The e-check includes familiar check information such as the payee's name, the amount, and the date and the account information. To sign the e-check, the payer enters a PIN to unlock an electronic checkbook card in the form of a smart card. This card is a secure container for the

payer's private signature key, and assures a degree of non-repudiation. The signature on the e-check may also cryptographically bind a copy of the invoice to the e-check, so that an attacker cannot substitute a different invoice in order to commit fraud. The invoice format is not fixed, but it can be flexible with respect to length, format and data content, so that the payer can return the document received from the payee. This provides the payee with the complete information needed to correctly post the payment.

The signed e-check and invoice is sent to the payee by email or a web transaction. The payee verifies the payer's signature on the e-check and invoice, detaches the invoice information, and posts the payment to accounts receivable. The payee enters his PIN to unlock his electronic checkbook and uses the electronic checkbook to endorse the e-check and to sign an electronic deposit slip to deposit a batch of e-checks.

The endorsed e-check is forwarded to the payee's bank for deposit and subsequent clearing. The clearing process can be done by integrating e-check into existing Electronic Check Presentment systems or other clearing and settlement systems. Both the payee's bank and payer's bank verify all signatures on the e-check and endorsement using a two layer certificate system which links the signature verification keys to the signer and signer's bank account. The paying bank verifies that this transmission of the e-check is not a duplicate, that the payer's certificate and account are currently valid, and posts the e-check to the payer's Demand Deposit Account (DDA).

Finally, the payer receives a line item on his statement, which may now carry a full description of the transaction, since the entire contents of the e-check are machine-readable.

8.10. Alternative Operational Models

The e-check flow shown in **Figure 8.5** is typical, but e-checks can be used in other ways, just like paper checks.

Figure 8.6 shows how a lockbox operator can process an e-check on behalf of the payee. In this case the lockbox operator does the cryptographic processing to verify the payer's signature. The invoice or advice of payment information can be converted by the lockbox operator to the same format used for paper checks. This allows the payee, typically a biller, to receive e-check payments without implementing new, e-check-specific software and hardware. The lockbox operator endorses and deposits the e-check on behalf of the payee. If the lockbox is operated by the payee's bank, then the lockbox functions can be interfaced directly to the bank's e-check server, saving the separate endorsement and deposit signing steps.

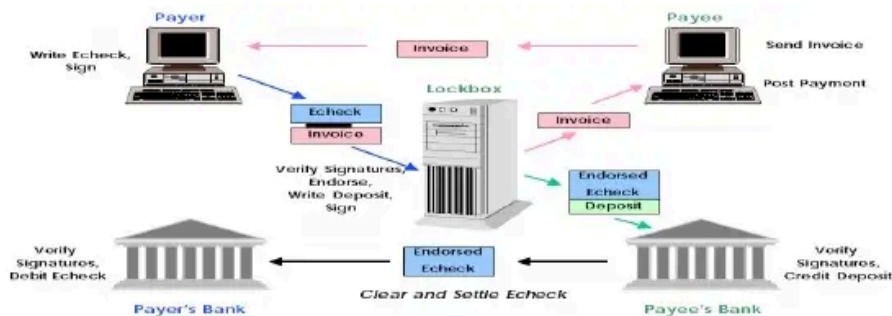
Figure 8.6 E-check lockbox flows

Figure 8.7 shows how a payee can endorse and cash the e-check at the payer's bank. The payer's bank transfers the proceeds to the payee's bank by electronic funds transfer. This flow is particularly useful if a payer wishes to pay payees whose bank does not yet offer e-check accounts. The payer's bank can provide the payee with a "check cashing" electronic checkbook valid for cashing the payer's checks. The electronic funds transfer can be done over a variety of networks, including international networks. In this case the payer could write the e-check in the payee's currency, and the payer's bank could implement the funds conversion from the payer's currency to the payee's currency when making the electronic funds transfer.

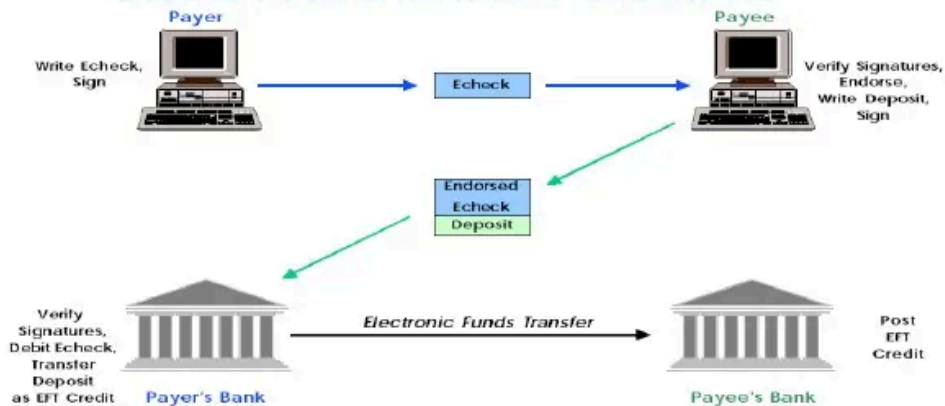
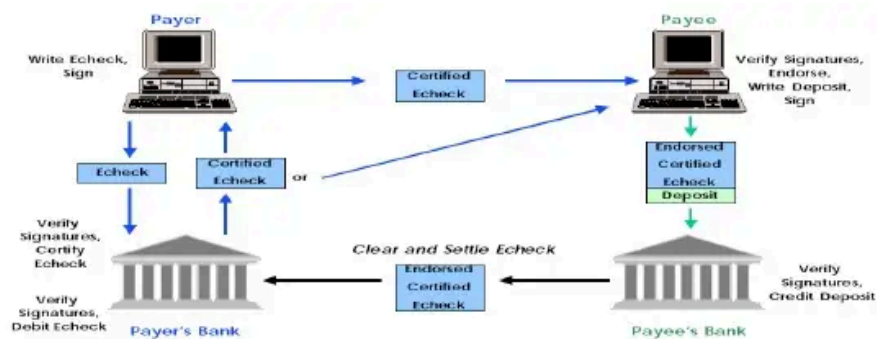
**Figure 8.7** E-check cash and transfer flow

Figure 8.8 shows a certified electronic check flow. In this case, the payer sends the e-check to the payer's bank. The payer's bank verifies the payer's signature, determines that there are funds available in the payer's account, and places a hold on the funds. The payer's bank then countersigns the check to certify it, and send the check back to the payer. Alternatively, the bank can send the certified check directly to the payee, possibly over an encrypted channel to provide the payee with the greatest degree of security and confidentiality.

Figure 8.8 Certified e-check flow

8.11. Summary

- Electronic check is a payment instrument which comes with a two fold advantage - it amalgamates the advantages of electronic transactions like speed, security and efficiency with the pre-existing legal infrastructure and business processes in place for paper based checks
- The e-check was developed as an initiative of the Financial Services Technology Consortium (FSTC)
- The vision behind the development of e-check is to substitute the electronic document with the paper based document by replacing the handwritten signatures with public key cryptographic signatures
- With e-checks banks can save their costs and as well enhance their customer service manifold.
- An e-check must have cryptographic information for its unique identification from others.
- It's a markup language defined in SGML (Standard Generalized Markup Language).